Can the Public Sector Deliver Adaptive Learning Technology at Scale? Experimental Evidence from India*

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Abstract

The hardest part of any new educational innovation is scaling it through a public school system. The Government of Andhra Pradesh has scaled a tablet-based personalized adaptive learning (PAL) software through its 444 government middle schools. The software combines video lessons, practice, and assessments at the enrolled grade, with adaptive remediation reaching up to two grade levels below. We embed a cluster-randomized trial in 120 additional schools (60 treatment, 60 control). Two years of exposure raise endline math ability by 0.39 SD, at a per-hour causal return of 0.011 SD. A Year-3 follow-up shows the trial treatment-control gap halves where PAL delivery fell, while the per-hour return reproduces in 40 newly scaled residential schools where delivery is sustained. Sustaining the per-hour return at the program's full footprint depends on the integrity of the delivery infrastructure that produces engaged hours per student. Earlier studies have focused on tailoring content to each student's level. This program produced similar gains by keeping most content at the enrolled grade.

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I. Introduction

Public school systems in low- and middle-income countries operate well below the productivity frontier of education. In Andhra Pradesh, the setting of this paper, 48% of grade-8 students cannot perform grade-3 arithmetic against a curriculum that has moved on to algebra (ASER Centre, 2024). The literature documents large learning gains in such settings from targeted instruction (Banerjee et al., 2007, 2017), personalized adaptive learning (Muralidharan and Ganimian, 2019; Muralidharan and Singh, 2025; Oreopoulos et al., 2026), structured pedagogies (Gray-Lobe et al., 2022), and reforms to teacher accountability (Muralidharan and Sundararaman, 2011; Duflo et al., 2012) and school information (Andrabi et al., 2017). Translating that evidence into a state's classrooms is a chain of requirements—delivery capacity, training, maintenance, political attention through leadership transitions, and social acceptance from teachers, parents, and students—each capable of failing on its own. Each link runs through the state's existing administration: the program is taught by teachers already carrying their regular load, supervised through the standard school inspection chain, and resourced from the same budgets that fund the rest of the school system.

Within the governance frame, one design choice in the literature warrants attention. Targeted-instruction and personalized-adaptive-learning programs typically go deep—routing students to remediation content several grade levels below their enrolled grade. Going deep can run into social-acceptance pressures in public school systems: teachers under curriculum-completion pressure may resist, parents may read below-grade routing as stigma, and students may experience it as a setback. A shallower design that holds most content at the enrolled grade may be easier to sustain under state adoption. Whether such a design can produce gains comparable to deep-targeting designs is an open empirical question.

This paper studies one such adoption. The Government of Andhra Pradesh has run a tablet-based personalized adaptive learning (PAL) program developed by ConveGenius in its government middle schools since 2019, now reaching about 1,200 schools (about 524 regular middle schools plus 700 residential schools added in January 2025). We embed a stratified-randomized trial in 120 of these schools (60 treatment, 60 control) across two academic years (AY 2023–24 and 2024–25). A Year-3 endline (February–March 2026) covers three rollout configurations: an 80-school trial subset, 40 schools from the original 444-school regular rollout, and 40 from the roughly 700-school residential expansion. The paper addresses three research questions. First, what is the causal effect of two years of program access on math learning? Second, what is the per-hour causal return to PAL usage—instrumented by treatment$\times$section-enrollment variation under the 30-tablet supply constraint—and what does this cost per unit of learning? Third, as the state scales the program, does the effect persist, and which state actions hold it up?

The intervention had several state-built elements. The software is a tablet-based math platform developed by ConveGenius and aligned to the state curriculum, with most content at the enrolled grade and adaptive remediation reaching up to two grade levels below. Each treatment school received a 30-tablet computer lab procured by the state, dedicating the lab to math and replacing two of its eight weekly math periods with PAL sessions supervised by the regular math teacher across two academic years (AY 2023–24 and 2024–25), with continued exposure in a Year-3 follow-up (AY 2025–26). State implementation was supported by structured teacher and headmaster training and a school–district–state monitoring chain (from the State PAL Coordinator at the Commissionerate to district nodal officers to headmasters). The chain operated through three channels: weekly WhatsApp usage rankings, periodic in-person review meetings, and real-time per-school dashboards. Technical support sat outside this chain: a Field Management Staff (FMS) workforce contracted through ConveGenius, deployed at roughly one staffer per 12 schools. Implementation was funded through state and central (Samagra Shiksha) education budgets.

Our empirical strategy has three components. The endline is a 72-item math assessment scored using item response theory, placing Year-2 and Year-3 ability estimates on a single scale. Treatment$\times$section enrollment under the 30-tablet cap instruments the per-hour causal return. Compliance is the fraction of platform-prescribed activities a student completes. The prescription is individualized: a per-chapter diagnostic gate routes students to below-grade content when prerequisite questions fail, then to at-grade content.

Two years of access to the program produced large gains in math learning. The intent-to-treat effect on endline math ability is $+0.39$ SD ($p < 0.001$). That is roughly 1.8 additional years of learning compared to how AP control-school students progress from one grade to the next. On a foundational-skill metric, the share of grade 7–8 students with full grade-3 arithmetic mastery (subtraction and division) rises from 54.8% in control schools to 61.5% in treatment schools—a 6.7 percentage-point gain.

Effects vary across student characteristics. Girls gain $+0.46$ SD against $+0.33$ SD for boys ($p = 0.001$), surviving multiple-testing corrections on the primary subgroup interaction; the gender gap is driven by usage rather than per-hour return, with girls accumulating roughly two additional cumulative hours of PAL over the two years, concentrated in active practice (questions and assessments) rather than passive content (videos). Effects decline across the baseline-ability distribution, with students in the lowest baseline quintile gaining $+0.48$ SD and those in the highest gaining $+0.31$ SD. Within every baseline quintile, students in the top tercile of compliance outperform those in the bottom tercile by $+0.3$ to $+0.5$ SD, with the largest within-quintile gap at Q1. Completing the prescribed work, in other words, mattered most where students started weakest.

Effects also decline across grades—from $+0.51$ SD in grade 7 to $+0.33$ SD in grade 9 ($p = 0.091$). The gradient is concentrated in larger schools where section enrollment exceeds the 30-tablet cap; in smaller schools where every student has a device, effects are uniformly large (0.58–0.65 SD across all grades). The grade-by-school-size pattern is consistent with tablet congestion as the mechanism: in larger schools, older grades have larger sections and less PAL time per student, which the within-school dose-response translates into smaller effects.

Each hour of PAL produces measurable gains in math learning. Two-stage least squares (2SLS) estimation, using treatment$\times$section enrollment under the 30-tablet cap as the instrument and anchoring the linear projection at zero usage, yields a per-hour causal return of $+0.011$ SD ($p < 0.001$), close to the ordinary least squares (OLS) estimate of $+0.009$ SD per hour. The within-school dose-response is flat below 10 hours of two-year usage, steep between 10 and 30, and flat above 30; trial students average 35.3 cumulative hours over two years and sit at the upper end of the active region.

The trial produced these gains at \$52 per student over two years. That translates to 0.94 LAYS per \$100—within the range reported for similar adaptive-learning interventions (0.24–2.66 LAYS per \$100). This is the cost-effectiveness of the trial configuration: 35.3 hours of math PAL per student over two years, delivered through state prescription and active monitoring. Cost per LAYS depends on dose, and dose depends on implementation—a point we develop below.

In Year 3 the delivery infrastructure that produced trial-arm dose contracted across multiple layers, and dose attenuated. The same small state team that supervised PAL for 444 schools at trial start was responsible for roughly 1,200 schools by Year 3—a near-threefold increase against an unchanged team—including a new residential rollout launched in January 2025; central monitoring contracted, with weekly state-coordinator WhatsApp messages dropping to about a quarter of the Year-1–Year-2 rate. The contracted FMS workforce experienced a contract gap at the close of AY 2024–25 and remained roughly a quarter below plan through December 2025. Tablet replacements in the original 524 schools, including the RCT treatment arm, did not begin until December 2025. Across the 60 RCT treatment schools, 6 stopped PAL entirely in Year 3 and 28 more dropped below 5 hours per student per year—a continuation of variation present from the start, with chronic under-implementers running at about half the pace of the high-implementing arm. In Year-3 surveys, 52% of RCT treatment headmasters reported a tablet shortage, and 29% of teachers reported having reduced or canceled PAL sessions during the year, citing competing school activities (75%), syllabus pressure (40%), and hardware or electricity issues (36%). The Year-2 treatment–control gap halves where delivery dropped.

The same per-hour return reproduces across the state's other rollout configurations. In 40 KGBV residential girls' schools, students average 13.6 hours of cumulative PAL with a descriptive within-school dose-response slope of $+0.015$ SD per hour, in the active range of the trial's curve. In the 444-school program rollout—where the 30-tablet lab is split across math, English, and Telugu, and supervision is lighter—students average just 0.64 PAL hours and post outcomes indistinguishable from controls. What varies is dose.

This paper contributes to two bodies of work: state delivery in low- and middle-income countries, and in-school adaptive learning. On state delivery, two strands are relevant here: technology and political economy. Tech-enabled state delivery documents how infrastructure extends state administrative reach: Muralidharan et al. (2016) on biometric smartcards in Andhra Pradesh, Muralidharan et al. (2021) on phone-based monitoring of agricultural transfers, Okunogbe and Pouliquen (2022) on electronic tax filing in Tajikistan, and Beg (2022) on digitized land records in Pakistan. We extend that strand to classroom delivery: the AP government built the technology stack required to deliver an adaptive curriculum to each instructional period. The political-economy strand has names for the Year-3 pattern we have just documented. Andrews et al. (2017) call it premature load-bearing: an organization asked to carry weight before its underlying capacity has been built up. Glaeser and Poterba (2021) call it ribbon-cutting bias: the political incentive to inaugurate visible new infrastructure that short-changes upgrade and maintenance of existing infrastructure. Our trial puts a price on the pattern's consequence: each hour of dose lost is worth 0.011 SD of math learning.

The second is the in-school adaptive-learning literature, which establishes that implementation structure—typically a hired lab-in-charge per school—is what makes the platform produce learning at scale. Muralidharan and Singh (2025) adapt Mindspark for in-school delivery in Rajasthan public schools through a hired lab-in-charge (LIC) per school responsible for hardware and student login support. In a sample twenty times the original Mindspark trial (Muralidharan and Ganimian, 2019), they find a 0.22 SD math gain over 18 months. Their Year-3 LIC-reduction test halved usage; usage recovered when monitoring resumed. Oreopoulos et al. (2026) add a part-time LIC to Uttar Pradesh boarding schools that already had Khan Academy access—tasked with scheduling weekly sessions, troubleshooting, and monitoring student progress—and raise math achievement by 0.44 SD in 31 weeks, isolating implementation structure as the binding margin. On implementation architecture more broadly, Bold et al. (2016) show that NGO-implemented contract teachers in Kenya raised math and English scores while the same intervention implemented through the government's regular bureaucracy produced no effect, establishing implementation architecture as a binding margin in scaled education delivery.

The same literature increasingly recognises a related point: depth of targeting is one component of these programs, not the singular engine. de Barros and Ganimian (2024) hold the Mindspark platform constant and vary only whether students receive level-targeted or grade-level content: students gain only 0.05 SD on average ($p > 0.10$), with the effect concentrated in the bottom-quartile baseline cohort ($+0.22$ SD)—evidence that deep personalization helps low performers but is not what drives gains for the average student. Ardington et al. (2026) synthesise twenty-five significant and twenty-one null estimates across TaRL evaluations in India and sub-Saharan Africa and attribute cross-study variation to who delivers, how much extra time is allocated, the intensity of monitoring and coaching, and program duration—rather than to depth of level-matching. Kaffenberger et al. (2026) decompose the TaRL approach into nine core components spanning pedagogy, pedagogical support, and the authorising environment, of which 'aligning instruction to current learning levels' is one. On both axes—implementation structure and depth of remediation—no single component is what drives the gains; the picture is of a multi-component bundle.

We extend this literature on three dimensions: who delivers, how deep the remediation goes, and how the program performs as the state scales. On delivery, the implementation chain—scheduling, monitoring, and content integration—runs through the state's regular school hierarchy alongside a thinner contracted technical-support workforce (FMS at $\sim$1:12) than the per-school LIC configurations of the comparison studies. On content depth, our shallower design contrasts sharply with the Mindspark Rajasthan configuration, where over 97% of math items presented to Grade-8 students were below grade level (Muralidharan and Singh, 2025); in our trial, by contrast, $\sim$ 82% of analogous Grade-8 math items sit at the enrolled grade. On performance at scale, the trial identifies a per-hour return present at every baseline quintile but declining toward the top; Year-3 evidence shows the return reproduces wherever delivery is sustained, while the trial–control gap halves where delivery drops. Across these settings, the per-hour return to in-school adaptive learning is robust across platforms and content depths; whether it translates into learning at state scale depends on the integrity of the multi-layer delivery infrastructure that produces engaged hours per student.

The remainder of the paper is organized as follows. Section II describes the AP education system and the delivery infrastructure the state has built around its PAL adoption. Section III describes the design, instrument, and estimation. Section IV reports state implementation cadence and student PAL usage. Section V reports the intent-to-treat effect, heterogeneity, the per-hour IV return, and the within-school dose-response. Section VI reports cost-effectiveness. Section VII reports the Year-3 follow-up and external validity to the state's broader rollout configurations. Section VIII discusses. Section IX concludes.

II. Background

A. Program rollout and study sample

The Government of Andhra Pradesh deployed Personalized Adaptive Learning (PAL), a tablet-based math program, in partnership with the educational-technology firm ConveGenius. Launched in 2019, suspended during the COVID-19 pandemic, and restarted in 2022, the program reached 444 government middle schools across 17 of the state's 26 districts by 2022 and has since scaled to roughly 1,200 schools statewide. We partnered with the state to evaluate the program in 120 randomly selected schools added to the rollout—60 assigned to receive PAL and 60 retained as controls, stratified across 8 districts and 21 mandal-cluster groups—with implementation in the treatment arm beginning in academic year 2023–24. Figure 1 situates the AP program rollout in national context (Panel A) and locates the 120 RCT schools within the broader 444-school AP rollout (Panel B), with the 21 randomization strata shown as colored polygons. The main results in this paper come from these 60 evaluation schools; we discuss external validity to the broader 444-school program rollout (the subset with ConveGenius usage records) in Section VII.

B. State implementation infrastructure

Each treatment school sat at the bottom of a four-tier chain that decided whether the PAL lab actually ran each week (Figure 2). At the top, the State Project Director (SPD) and the Additional SPD (ASPD) at the Commissionerate of School Education set policy; under them, a State PAL Coordinator handled day-to-day operational traffic via the Joint Director for State Academic Monitoring (JD-SAMO) wing. Each district had a District Nodal Officer (DNO) for PAL who cascaded state instructions to the schools in their district and coordinated laterally with the District Educational Officer (DEO). At the school, the Headmaster (HM) oversaw delivery and the math teacher ran the lab session itself.

The chain operated through three concrete channels. *First*, a WhatsApp broadcast group from the State PAL Coordinator to all DNOs—used to share weekly school- and district-level usage rankings, to issue direct nudges to increase usage, and for operational pushes such as data syncing, orientation sessions, tablet maintenance guidance, and field-visit logistics. *Second*, scheduled review meetings: state-coordinator-led district-by-district sessions with HMs, math teachers, and DNOs in the program's first weeks; subsequent

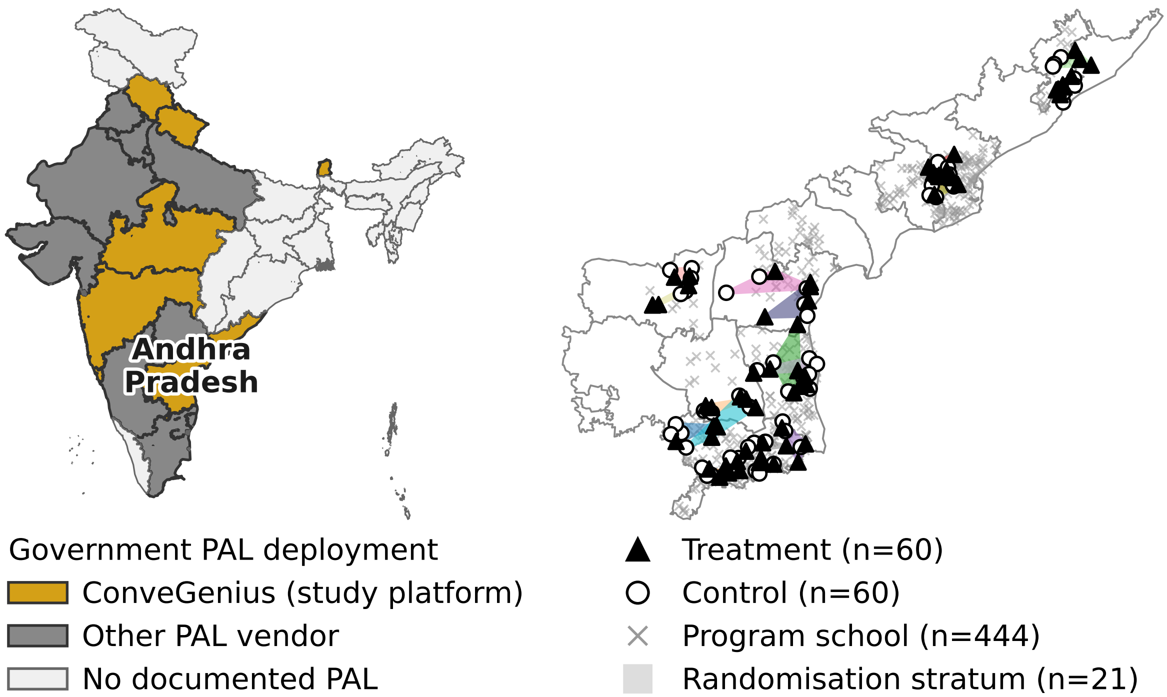

FIGURE 1. RCT SCHOOLS WITHIN THE ANDHRA PRADESH PAL ROLLOUT

Notes: Panel A: Indian states color-coded by PAL deployment status. Gold = states where ConveGenius operates (the PAL platform of this study). Dark gray = states with another documented PAL vendor. Light gray = no documented government PAL deployment. *Panel B:* Andhra Pradesh with all district boundaries. The 120 RCT schools (60 treatment = filled triangles; 60 control = open circles) sit within the 444-school government PAL rollout (gray × markers) covering 17 of AP's 26 districts. The 21 randomization strata, formed by grouping adjacent mandals within each of the 8 RCT districts, are shown as translucent polygons in distinct colors.

in-person summons of low-performing DNOs to the Commissionerate; and SPD-led review meetings with the broader DNO group. *Third*, real-time monitoring dashboards: per-school usage metrics that the State PAL Coordinator and DNOs could query directly, with the underlying telemetry coming from the ConveGenius platform itself. All 28 RCT treatment HMs in our Year-3 survey report receiving these state communications periodically, and 71 % report having seen the DIL school-level usage report card.

Alongside the government chain, a Field Management Staff (FMS) workforce contracted through ConveGenius operated at the school level. One FMS member was assigned to a cluster of approximately 12 schools, visiting each treatment school regularly to handle technical issues with tablets and software, sync data from the offline platform, and in some cases flag low usage to the HM. The FMS is not a tier of the government chain—it is a parallel technical-support workforce that interfaces only at the school. Its existence is what allows the rest of the infrastructure to assume that tablets work and that data reaches the dashboard.

This steady-state arrangement broke in Year 3. The state's contract with ConveGenius lapsed at the close

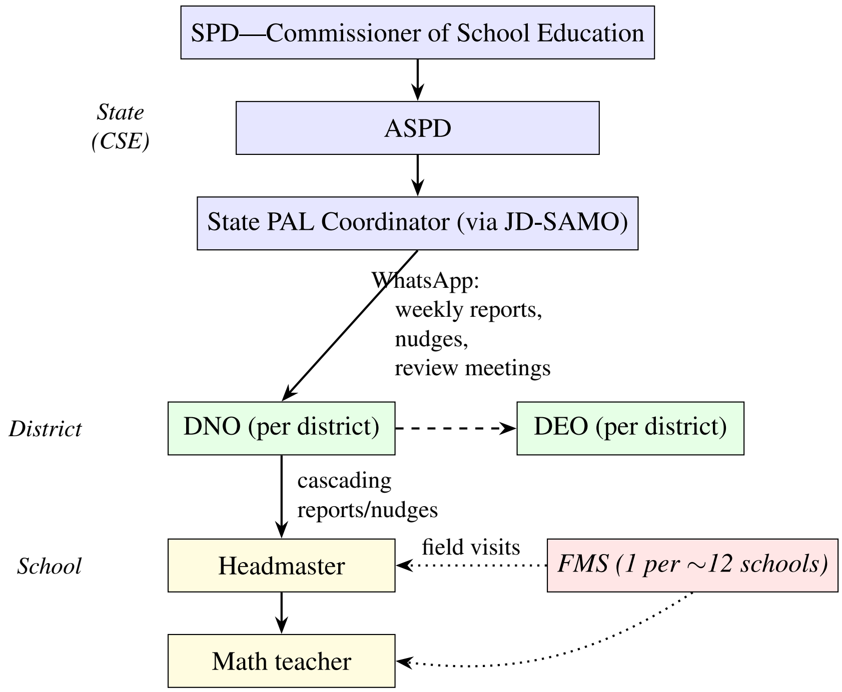

FIGURE 2. THE SCHOOL–DISTRICT–STATE ESCALATION CHAIN FOR AP PAL

Notes: Solid arrows: direct line of authority. Dashed: lateral district-level coordination. Dotted: technical-support contractor interface. SPD = State Project Director. ASPD = Additional SPD. JD-SAMO = Joint Director, State Academic Monitoring. DNO = District Nodal Officer. DEO = District Educational Officer. HM = Headmaster. FMS = Field Management Staff (contracted through ConveGenius). Funded through state and central (Samagra Shiksha) education budgets and operated by the AP government's regular school administration.

of AY 2024–25 and was not renewed until late July 2025; the existing FMS workforce was let go in the interim. Hiring resumed in batches from late July, with substantial attrition during the two-week induction training. Based on state-team correspondence, FMS staffing remained roughly a quarter below plan through December 2025[1] and reached the planned level only in January 2026. Effective field presence was near zero from the start of AY 2025–26 through August, partial from September through December, and at the planned staffing level only from January 2026 onward.

C. School-level delivery and the PAL software

In each treatment school, the government established a PAL lab: a designated classroom equipped with 30 tablets, charging stations, and earphones. The tablet count was fixed at 30 regardless of school or section enrollment. In the status quo, students had eight 40-minute periods of math instruction per week. Schools were instructed to convert two of these periods to PAL sessions, targeting approximately 80 minutes of

[1]Approximately 61 FMS in place by mid-October 2025, 65–67 by late December, 97 by January 2026, against a planned approximately 80 FMS for the 1,224-school Year-3 footprint (524 regular middle, 352 KGBV, 348 additional residential schools added July 2025), rising to 1,502 schools after the December 2025 addition of 278 PMShri schools. Year-3 contract target: 24 school visits per month per FMS. Source: state-team correspondence; preliminary, pending CG confirmation.

software use per week. During PAL periods, the math teacher took students to the lab and supervised while they worked on the software independently. Math teachers reported spending less time on remedial instruction and practice problems during the remaining six regular math periods, instead focusing on teaching key concepts in the classroom and directing students to use PAL for practice on each chapter. In sections with more than 30 students, tablets had to be shared: teachers were asked to split the section into batches that alternated between PAL and regular instruction, though in practice we also observed students being paired on a single tablet or taking turns. Across the 60 treatment schools, 44.7% of the 300 class sections exceeded the 30-tablet count; the median oversized section held 37 students (a 1.23:1 student-to-tablet ratio), and 59.6% of treatment students were in a sharing-required section (Appendix Figure D1).

Personalized Adaptive Learning (PAL) is a tablet-based math software developed by ConveGenius, an Indian education technology company. The software delivers curriculum-aligned content organized into chapters that mirror the state mathematics textbook; remediation extends up to two grade levels below the student's enrolled grade. Across 8.78 million module-level events in the 60 treatment schools, 27% of events are below the student's enrolled grade, with an average depth of 1.43 grade levels below. The lowest content grade delivered is grade 4, seen only by grade-6 students (for whom it is 2 grades below enrolled); for grade-9 students, the deepest remediation available is grade 7. We test whether this two-grade limit binds in a pre-registered within-school experiment reported in a companion paper (Cupito et al., 2026, in preparation). Per the vendor's intended use model, students work individually on tablets, watching instructional videos, answering practice problems, and completing assessments. The software operates offline; teachers sync usage data to the server approximately once per week.

Using these module-level data, we reconstruct the instructional pipeline that students experience. Each chapter follows a standardized sequence of six stages (Figure D2 in Appendix A): (1) a diagnostic test of 2–3 questions on prerequisite content; (2) below-grade instructional videos; (3) below-grade remedial practice; (4) a baseline assessment at enrolled-grade level; (5) at-grade videos paired with formative knowledge checks; and (6) a chapter endline assessment. Within any given chapter, the module sequence is largely uniform: the top sequence covers 13–36% of students in high-N chapters and the top five cover 38–54%. In the program's standard configuration in AP, students proceed through chapters in a fixed curriculum order (chapter locking). For the RCT schools, chapter locking was disabled, allowing the math teacher or the student to choose any chapter at any time; the cross-chapter order variation we observe is therefore an RCT-specific feature, not a software default.

When a student opens a new chapter, the platform delivers content in three phases (Figure 3, Panel A): up to three short diagnostic stages testing prerequisite content from earlier grades; below-grade remediation for any diagnostic stage the student fails; and the at-grade chapter content shared by every student. The diagnostic stages—each two to five questions—are completed before any chapter content begins. The first stage tests material one grade level below the student's enrolled grade. Depending on the chapter's prerequisite depth, the platform then runs one or two additional stages, each testing material two grade levels below. The set of stages the student fails (scores below 80% on) determines the remedial content the platform delivers: pass every stage and the student moves directly to the at-grade content for the chapter; fail the one-grade-below stage and a one-grade-below remedial module is inserted before the at-grade content; fail any of the two-grade-below stages and two-grade-below remediation is added on top. In the platform's reference chapter (Basic Geometric Concepts, Grade 6), which has two diagnostic stages, this assigns one of three bundles: 54 activities (at-grade only), 61 (with one-grade-below remediation), or 67 (with both grade levels of remediation). Each "activity" is either a short instructional video—a *learning objective*, or LO—or a brief multiple-choice quiz testing the videos in a module (a *knowledge check*, or KC). Across 121,000 student-chapter attempts in our 60 treatment schools, approximately 3% of students pass the first stage and skip remedial content; the remaining 97% receive one or two grade levels of remediation, with the deeper bundles assigned to students who fail more stages.

This per-chapter routing is student-specific—different diagnostic outcomes deliver different bundles of activities to different students. But students do not necessarily complete all of what they are asked to do (Figure 3, Panel B). Across $N = 6{,}052$ treatment students, the mean fraction of prescribed learning objectives a student opens—which we term *compliance*—is 0.61 (median 0.65).[2] Compliance varies meaningfully within each baseline-ability quintile, but is essentially flat across quintiles: students in the bottom quintile open 60% of their prescribed activities on average, students in the top quintile 62%. Time on the platform and compliance are correlated ($r = 0.58$) but distinct: hours measure raw exposure during PAL periods, while compliance measures the fraction of the prescribed sequence the student walks.

[2]Compliance is defined only for students with at least one chapter started on the platform: a student who logs in but never opens any chapter has no prescription assigned and therefore no denominator. About 690 treatment students with no module-level data are dropped here but appear in the Table 2 hours sample (where they are coded as zero hours / zero activities). The remaining gap between $N = 6{,}052$ here and the full treatment sample is driven by missing baseline-ability data, which compliance-by-quintile statistics require.

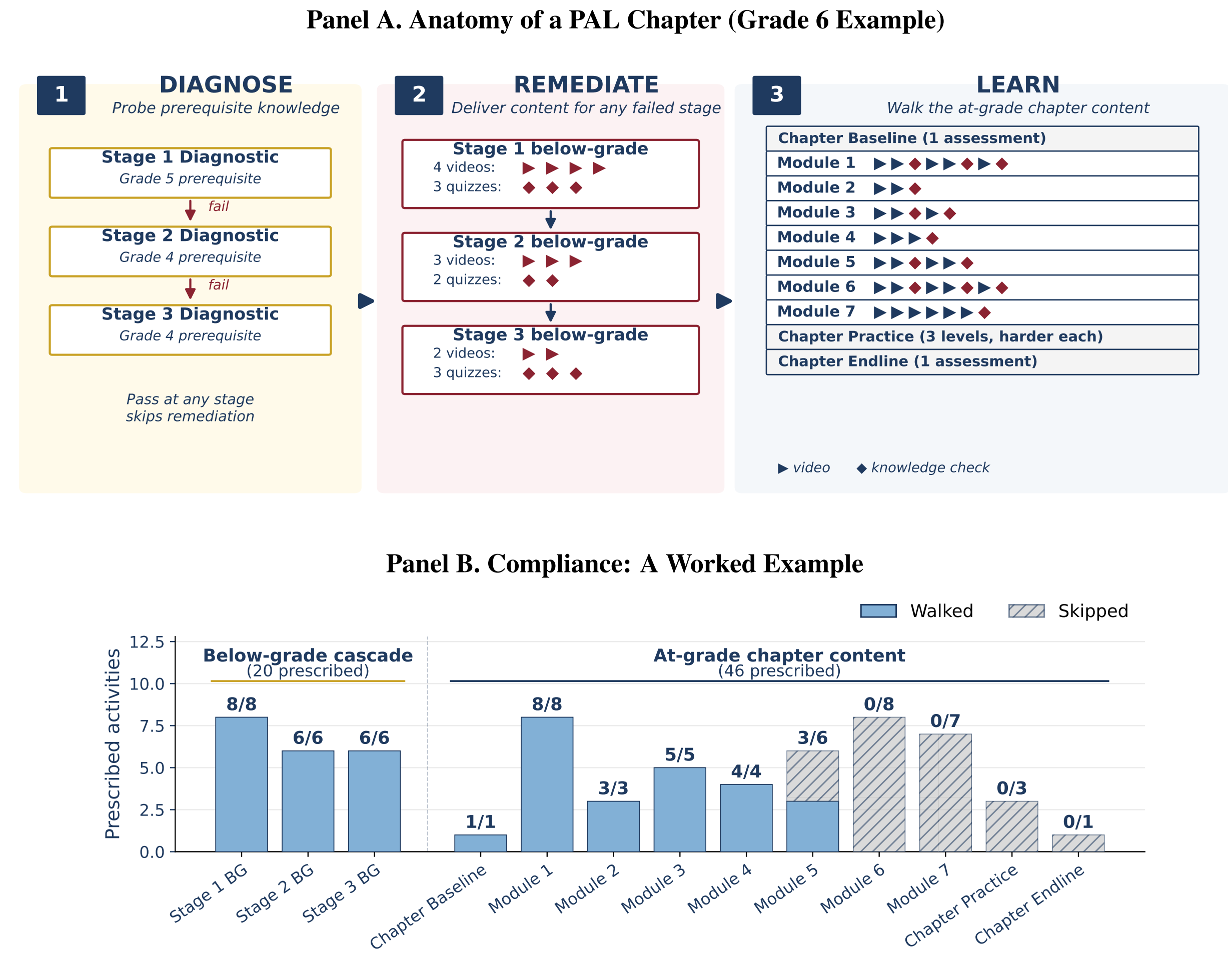

FIGURE 3. THE PAL CHAPTER AND STUDENT COMPLIANCE

Notes: Panel A: the chapter unfolds in three phases. *Phase 1 (Diagnose):* up to three short diagnostic stages testing prerequisite content from earlier grades, all completed before any chapter content begins. *Phase 2 (Remediate):* below-grade videos and knowledge checks for any failed diagnostic stage. *Phase 3 (Learn):* the at-grade chapter content (a chapter baseline assessment; modules each containing instructional videos and knowledge checks; chapter practice; a chapter endline assessment), identical for every student. *Module* = a topic block within a chapter; *learning objective* (LO) = one short instructional video; *knowledge check* (KC) = a quiz testing the LOs in a module. The figure depicts a Grade 6 chapter with three diagnostic stages and seven at-grade modules; chapters with shallower prerequisites have fewer diagnostic stages. *Panel B:* stylized worked example showing compliance for a single student in a single chapter. Compliance = activities walked $\div$ activities prescribed. The illustrative student walks the full below-grade cascade (all three diagnostic-failed stages, with their associated remedial videos and quizzes) and Modules 1–4 of the at-grade content, gets halfway through Module 5, then stops. Activities walked = 44; activities prescribed = 66; compliance ≈ 0.67.

III. Methods

A. *Experimental Design and Study Sample*

Our study is a cluster-randomized controlled trial conducted in government schools in the Indian state of Andhra Pradesh. The government of Andhra Pradesh identified eight districts in which to implement the PAL program. Within these districts, schools were grouped into 21 geographic strata based on mandal clusters within each district. From a pool of eligible co-educational government schools serving grades

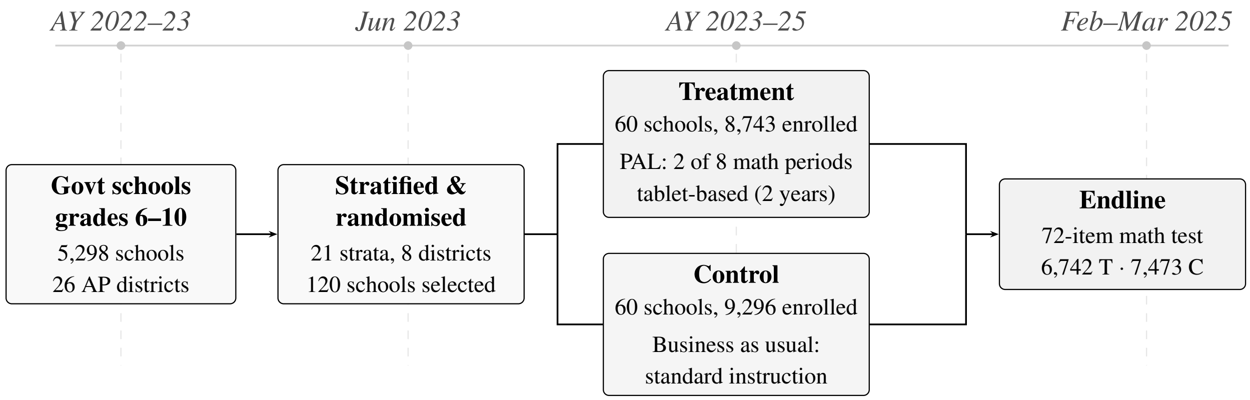

FIGURE 4. CONSORT FLOW DIAGRAM

Notes: Government schools with grades 6–10 in Andhra Pradesh (5,298 schools across 26 districts). Eight districts were selected as study sites and stratified into 21 geographic strata based on mandal clusters within each district. From this pool, 120 schools were randomly selected within strata and assigned to treatment or control. In treatment schools, PAL (Personalized Adaptive Learning) replaced 2 of 8 weekly math periods with tablet-based adaptive learning for 2 academic years (2023–24 and 2024–25). Enrollment counts are from government administrative records (AY 2024–25).

6–10, 120 schools were selected and stratified-randomly assigned to treatment (60 schools) or control (60 schools).[3] A separate 20-school within-school randomized experiment testing deeper remediation, conducted in program schools outside our 120 evaluation schools, is reported in a companion paper (Cupito et al., 2026, in preparation). Figure 1 maps the 120 RCT schools within the broader government PAL rollout. Table 1 presents balance on school and student characteristics. Treatment and control schools are statistically balanced across 15 school-level characteristics in Panel A (joint $F = 1.14$, $p = 0.336$), spanning enrollment, staffing, and ICT infrastructure, and across nine student-level characteristics in Panel B (joint $F = 1.07$, $p = 0.387$), including grade composition, gender, age, parental education, and socioeconomic status (SES). The one exception is the pre-intervention government math score (Panel B, last row), where treatment students score 2.6 points lower—but this test is underpowered and compositionally biased because data coverage is only 61% and asymmetric across arms (see §B and Appendix E).

Treatment schools received the PAL intervention, which replaced two of eight weekly math periods with tablet-based instruction. Control schools continued with their regular math curriculum. The intervention ran for two academic years: AY 2023–24 and AY 2024–25.

A total of 18,039 students enrolled in grades 6–9 across the 120 study schools in AY 2024–25 comprise our study sample. Students' exposure to PAL varied by cohort: those in grades 7–9 at endline received

[3] Three size-based filters were applied to the eligible pool before randomization, ensuring feasibility of the 30-tablet-per-school intervention: (i) schools with any grade 6–9 section exceeding 60 students were excluded (at most a 2:1 student-tablet ratio at maximum congestion); (ii) schools with more than 20 total sections across grades 6–9 were excluded; (iii) schools in the top decile of the pupil-teacher ratio were excluded. These filters also truncate the support of the congestion instrument used in Section C.

two full years of the program, while grade 6 students (a new cohort entering in AY 2024–25) and grade 10 students (who aged out after AY 2023–24) received one year. We collected baseline data from government assessments administered in AY 2022–23, prior to the start of the intervention. Endline assessments took place in February and March 2025. Figure 4 presents the CONSORT flow diagram.

Of the 18,039 enrolled students, 14,215 (78.8%) completed the endline assessment—6,742 in treatment schools and 7,473 in control schools. Attrition was 3.9 percentage points higher in treatment schools (23.7% versus 19.8%). However, the majority of students who did not take the endline assessment remained enrolled in school: 81% appeared in government assessment records for 2024–25, and 70% had non-zero math scores. We present formal attrition analysis and Lee bounds in Appendix G.

B. Data

Our main sources of data come from independent assessments at endline conducted by a private survey firm; administrative data from the state government; and student usage and learning data collected through the software by ConveGenius.

We conducted math assessments at endline to capture grade-level and earlier-grade learning. The instrument comprised 72 multiple-choice items spanning five broad math topics—algebra, geometry, measurement, statistics, and number systems—with questions drawn from the grade 2 through grade 9 curriculum. Each student answered items from grade 2 up to their enrolled grade, yielding between 50 and 72 items depending on grade level (50 for grade 6, 72 for grade 9). Using item response theory (IRT), we transform each student's performance into a single ability parameter, which we standardize to the control group mean and standard deviation. Following other work using this method to generate comparable scores across children and grades, we call this our IRT score (Muralidharan and Ganimian, 2019; Muralidharan and Singh, 2025). We describe the development and piloting of the instrument in Appendix B.

Baseline math ability, used in robustness checks and for heterogeneity by baseline ability, comes from pre-intervention government assessments. The source is cohort-specific: grade 7–9 endline cohorts use the AY 2022–23 SA1 (the year immediately before PAL began); the grade 6 endline cohort, who entered the study schools in AY 2024–25, uses the AY 2023–24 SA1 from their prior (grade 5) school. Overall coverage is 61%, but missingness is sharply asymmetric across arms (8% treatment versus 45% control), driven by differential government data extracts rather than student characteristics; for analyses that condition on baseline (Appendix Table G2 robustness column and the baseline-quintile heterogeneity in Table 4) we include a missing-baseline flag and interpret the resulting estimates with caution. Appendix H validates these government scores against our independent endline IRT and documents why they serve only as within-school ranked controls, not as a cross-school ability measure.

We administered the assessment on tablets using SurveyCTO, conducted by enumerators independent of the schools and intervention. The tablet platform served several data-quality objectives. First, mandatory responses combined with an explicit "Don't know" option eliminated missing values entirely. Second, randomized presentation of both topic order and question order across three test forms reduced opportunities for copying. Third, the platform recorded per-item response times, enabling ex post detection of guessing or rushing. Fourth, same-day data syncing allowed us to monitor school-level completion rates in real time and schedule efficient return visits to high-absence schools. Separately, to address the concern that treatment students' greater familiarity with tablets could bias performance, we conducted a pre-pilot confirming similar response times and accuracy on the device across treatment and control, and a Year-3 within-student sub-study that formally tests tablet-vs-paper measurement equivalence on Grade 8 students; both find no differential mode advantage for treatment students (see Appendix C).

C. Empirical Specification

Our main estimating equation is the intent-to-treat (ITT) specification:

$$Y_{isg} = \alpha + \beta\, T_s + \mathbf{X}_i' \gamma + \delta_g + \mu_m + \varepsilon_{isg} \tag{1}$$

where Y_{isg} is the endline math score (IRT ability parameter, standardized to control group mean and SD) for student i in school s and grade g; T_s is a treatment indicator (1 = PAL school); $\mathbf{X}_i$ is a vector of student-level controls (SES index,[4] parental education, age, gender, and missing-value flags); δ_g are grade fixed effects; and μ_m are randomization strata (mandal cluster) fixed effects. Standard errors are clustered at the school level (120 clusters). The coefficient β is the average ITT effect of PAL on math learning. Baseline math ability (described in Section B) is not included in $\mathbf{X}_i$ for the headline specification because the missingness is sharply imbalanced across arms (8% in treatment versus 45% in control), driven by differential government data extracts; a missing-flag dummy is therefore itself treatment-correlated and would absorb causal variation. We report a robustness column adding baseline in Appendix Table G2, where the pooled ITT moves only

[4]First principal component of nine household-asset indicators collected at endline: refrigerator, laptop, father's mobile phone, mother's mobile phone, air conditioner, washing machine, car, scooter, and bicycle.

TABLE 1—SAMPLE CHARACTERISTICS AND BALANCE

	AP (1)	Control (2)	Treatment (3)	Difference (4)	(SE) (5)	p-value (6)	N (7)
Panel A: School Characteristics [AP: $N = 5,298$ schools]							
Enrollment (grades 6–9)	217.1	160.0	155.1	-4.333	(17.848)	0.808	120
Total enrollment	309.1	234.6	223.2	-11.020	(22.825)	0.629	120
Urban	0.129	0.100	0.117	0.027	(0.055)	0.627	120
Has primary section	0.493	0.683	0.583	-0.111	(0.087)	0.203	120
Math teachers	2.1	2.0	1.9	-0.098	(0.153)	0.525	120
Physical ed. teachers	1.7	1.5	1.3	-0.205*	(0.109)	0.060	120
Total teachers	11.8	10.8	10.5	-0.286	(0.706)	0.685	120
Classrooms	10.8	9.4	8.9	-0.512	(0.629)	0.416	120
Functional computers	1.9	1.3	1.6	0.212	(0.610)	0.728	120
Projector	0.614	0.600	0.550	-0.051	(0.092)	0.584	120
Smart TV	0.681	0.767	0.650	-0.108	(0.076)	0.158	120
Digital board	0.286	0.267	0.283	0.020	(0.070)	0.773	120
Uses ICT tools	0.794	0.817	0.833	0.020	(0.073)	0.782	120
Digital content available	0.695	0.717	0.733	0.037	(0.079)	0.638	120
Generator/UPS	0.133	0.133	0.117	-0.017	(0.060)	0.780	120
Joint F-statistic		1.14		[$p = 0.336$]			120
Panel B: Student Characteristics [AP: $N = 1,150,232$ students]							
Grade 6	0.223	0.237	0.228	-0.011	(0.008)	0.168	14,215
Grade 7	0.241	0.235	0.237	0.006	(0.007)	0.414	14,215
Grade 8	0.255	0.248	0.262	0.013	(0.008)	0.109	14,215
Grade 9	0.281	0.280	0.272	-0.007	(0.009)	0.407	14,215
Female		0.492	0.479	-0.020	(0.019)	0.301	14,215
Age		12.30	12.35	0.039	(0.037)	0.293	14,124
Father education		2.14	2.07	-0.057	(0.059)	0.336	12,744
Mother education		2.10	2.07	-0.037	(0.045)	0.412	13,135
SES index (PCA)		-0.02	0.02	-0.030	(0.040)	0.454	14,215
Joint F-statistic		1.07		[$p = 0.387$]			12,267
Baseline math score (govt SA1)		35.23	32.10	-2.585**	(1.299)	0.047	10,238

Notes: Column (1) shows means for all 5,298 government schools with grades 6–10 in Andhra Pradesh (1,150,232 students in grades 6–9). School characteristics from government administrative records (AY 2023–24); ICT variables from UDISE+ 2024–25. Columns (2)–(3) show unconditional group means for the 120 randomized study schools. Column (4) and standard errors are from OLS of each variable on a treatment indicator and 21 randomization strata (mandal cluster) fixed effects. Panel A reports school-level characteristics (including enrollment in grades 6–9, the study grades) with HC1 robust standard errors ($N = 120$ schools). Panel B reports student-level characteristics for the grades 6–9 study sample, with standard errors clustered at the school level. Baseline math score (government SA1, 2022–23) is excluded from the joint F-test due to differential administrative data coverage (62%; see text). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

from 0.390 to 0.393, indicating the headline is robust to including baseline despite this contamination.

To estimate the causal return to PAL usage, we exploit exogenous variation in dosage arising from tablet congestion. All treatment schools received 30 tablets regardless of enrollment, so students in larger sections share fewer tablets per student and receive less PAL time. We instrument for usage hours using the interaction of treatment status and section enrollment:

$$\text{First stage:} \quad H_{isg} = \pi_0 + \pi_1(T_s \times S_{sg}) + \pi_2 T_s + \pi_3 S_{sg} + \mathbf{X}_i' \pi_4 + \delta_g + \mu_m + \nu_{isg} \tag{2}$$

$$\text{Second stage:} \quad Y_{isg} = \alpha + \lambda \hat{H}_{isg} + \phi T_s + \psi S_{sg} + \mathbf{X}_i' \gamma + \delta_g + \mu_m + \varepsilon_{isg} \tag{3}$$

where H_{isg} is total PAL usage hours and S_{sg} is section enrollment (the number of students in the student's class section). The exclusion restriction requires that the difference in treatment effects between larger and smaller sections is driven entirely by the difference in PAL usage between these sections—equivalently, if usage were equalised across section sizes, treatment-effect magnitudes would be the same in larger and smaller sections.

For heterogeneity analyses, we estimate equation (1) separately by subgroup or include interaction terms with treatment status.

IV. Implementation and Usage

A. State implementation cadence over time

Figure 5 traces the operating cadence of the state implementation infrastructure described in §B. The top lane shows state-to-DNO communication traffic (WhatsApp messages classified by content + scheduled review meetings); the bottom lane shows mean monthly Field Management Staff visits per RCT treatment school. The state apparatus produced approximately one WhatsApp DNO message per week across both Years 1 and 2 (46 messages each year), with FMS visits cadencing at roughly 1–4 visits per RCT treatment school per month over AY 2024–25. In the partial Year 3 window for which transcribed monitoring data are available (through 27 October 2025), the WhatsApp channel produced 4 messages across 4 months—roughly one-quarter the Years 1–2 rate. The State PAL Coordinator's agenda over this period also took on a $\sim$700-school KGBV residential expansion that launched in January 2025. We return to both shifts in §VIII.

B. Student PAL usage

Table 2 summarises PAL usage across both academic years. Schools were allocated one 80-minute PAL period per week, though the effective instructional window was approximately 60 minutes after accounting for student transit to the computer lab and device setup. PAL sessions were suspended during holidays,

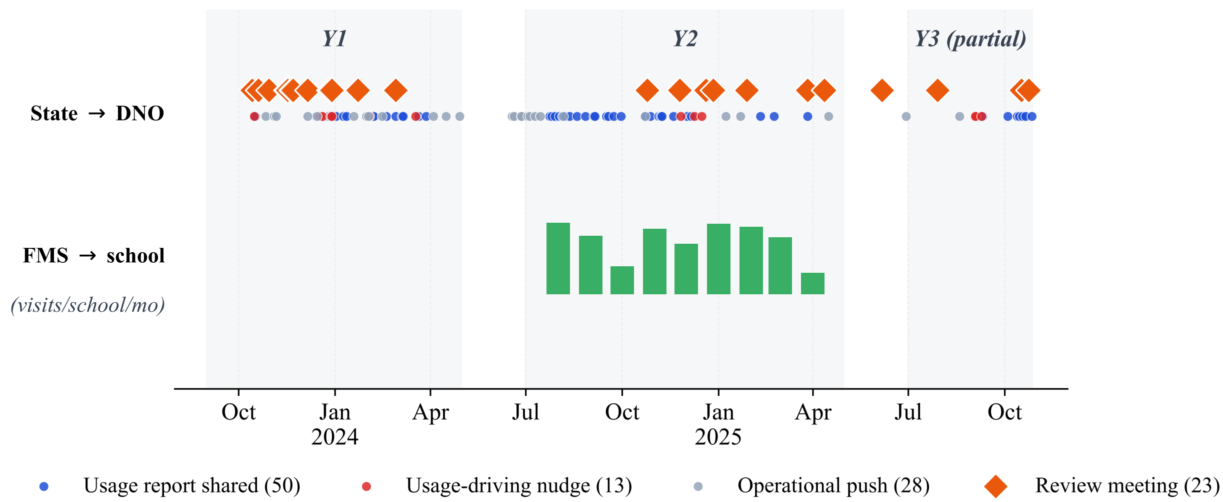

FIGURE 5. STATE IMPLEMENTATION INFRASTRUCTURE OPERATING ACROSS TIME, OCTOBER 2023 – OCTOBER 2025

Notes: Top lane: state-to-DNO communication traffic from the State PAL Coordinator's WhatsApp log and a parallel review-meetings record, classified by content as usage report shared (blue), usage-driving nudge (red), operational push such as data sync, training, tablet maintenance, or field-visit logistics (gray), and scheduled review meeting (orange diamond). *Bottom lane:* mean monthly Field Management Staff (FMS, the technical-support contractor) field visits per RCT-T school, from daily-grid visit logs available for AY 2024–25 only (no FMS data outside that window). Mean visits per school per month, by month: Jul 2024 1.5; Aug 3.8; Sep 3.1; Oct 3.5; Nov 3.7; Dec 3.6; Jan 2025 2.7; Feb 3.0. Light shading marks AY-active windows. The Y3 column is partial through 27 October 2025.

formative and summative assessment (FA/SA) weeks, and pre-exam revision periods—leaving 22 eligible weeks in Year 1 and 21 in Year 2 (Appendix Table I5). Against this calendar-adjusted effective target of 43 hours over two years, students in grades 7–9 averaged 38.2 hours (89%). Compliance was higher in Year 2 (92%) than Year 1 (86%), consistent with implementation ramp-up during the program's first year. Lab setup was staggered across schools, with 49 of 60 schools beginning in September 2023 and the remaining 11 in October. Figure 6 traces the resulting monthly usage profile across both academic years and the corresponding content grade-level gap.

Baseline math test scores and age are the strongest predictors of PAL usage, while SES is not (Appendix Table I4). Students spend slightly more time watching videos (54%) than answering questions (46%; Appendix Table D1). Eighty percent of treatment students report enjoying PAL "a lot," though 37% report language comprehension and video loading as implementation challenges (Appendix Table D2). Appendix A characterizes the platform's instructional pipeline using module-level data, showing that the student experience is largely uniform across the baseline-test-score distribution.

TABLE 2—PAL USAGE STATISTICS (TREATMENT GROUP)

	Mean	Median	SD	P25	P75	*N*
Panel A: Overall Usage						
Total usage (hours, 2 years)	34.2	30.2	23.3	17.4	47.3	6,742
Year 1 usage (hours)	14.7	12.6	15.5	0.0	23.8	6,742
Year 2 usage (hours)	19.5	17.8	12.5	10.7	26.2	6,742
Video time (hours)	18.9	16.5	14.2	9.6	25.2	6,742
Question time (hours)	15.3	13.5	10.5	7.3	21.3	6,742
Months active (Y2)	6.6	7.0	1.9	6.0	8.0	6,742
Chapters attempted	23.0	24	12.1	14	31	6,742
Chapters completed	15.4	14	11.2	6	23	6,742
Compliance (overall)	0.60	0.66	0.21	0.49	0.76	6,742
Compliance — remedial (BG)	0.74	0.78	0.20	0.66	0.86	6,742
Compliance — at-grade (AG)	0.56	0.62	0.22	0.44	0.73	6,742
Panel B: Usage by Grade						
	Y1 hours	Y2 hours	Total	Months (Y2)	Compliance	*N*
Grade 6	—	20.4	20.4	6.0	0.53	1,540
Grade 7	18.9	20.5	39.4	6.8	0.58	1,599
Grade 8	19.7	20.0	39.7	6.9	0.63	1,769
Grade 9	18.4	17.5	35.9	6.8	0.65	1,834

Notes: Treatment group only ($N = 6{,}742$ students in 60 schools). Panel A reports individual-level usage aggregated across both academic years (AY 2023–24 and 2024–25). Video time is passive instruction; question time is active practice. Chapters are curriculum units within the PAL platform. *Compliance* is the student-level mean across chapters of unique learning objectives (LOs) completed divided by the LOs prescribed by PAL's diagnostic for the student's (chapter $\times$ cascade signature $\times$ prior-incomplete history) cell, capped at 1; the BG (below-grade) and AG (at-grade) rows apply the same ratio separately to the remedial and at-grade portions of each prescription. Treatment students with no module-level data are coded as zero across hours, chapters, and compliance to keep the Panel A sample at $N = 6{,}742$. Panel B shows usage by endline grade. Grade 6 students entered the study in Year 2 only. Usage data from ConveGenius platform logs, capped at the endline assessment start date (February 17, 2025). Each school was allocated one 80-minute PAL period per week; accounting for school closures and setup time, two-year students achieved 89% of the calendar-adjusted effective target (Appendix Table I5). Topic-level usage breakdown is reported in the appendix.

V. Impact on Learning

A. ITT Estimates

Figure 7 plots kernel density estimates of endline math scores by treatment status and grade. The treatment distribution shifts visibly rightward for all four grades, with the largest shift at grade 7 and the smallest at grade 9. The shift is not confined to one tail of the distribution: treatment widens the distribution as well as shifting its mean, indicating that PAL generates variance in math-test-score gains across students.

Table 3 reports the corresponding regression estimates. The pooled ITT effect is 0.39 SD ($p < 0.001$). By grade, effects range from 0.33 SD in grade 9 to 0.51 SD in grade 7. The null hypothesis that effects are

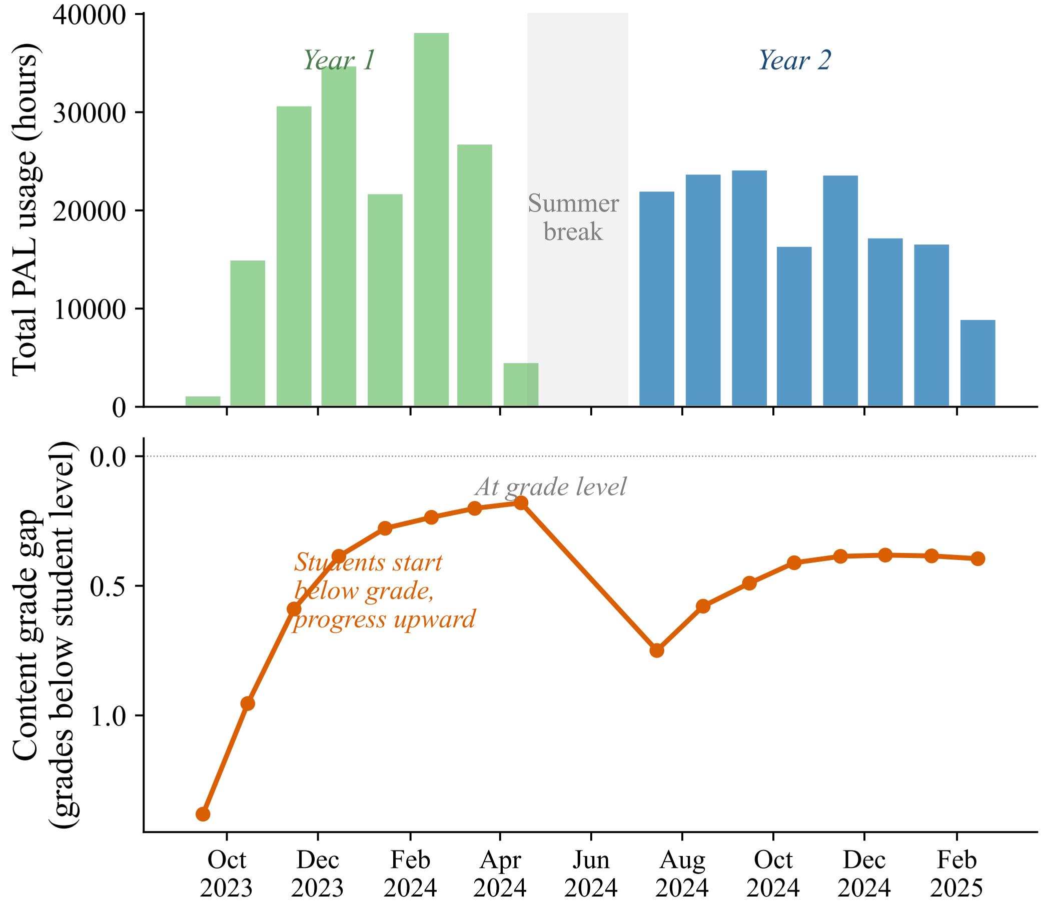

FIGURE 6. MONTHLY PAL USAGE AND CONTENT PROGRESSION

Notes: Top panel: total PAL usage hours per month across all treatment schools (green = AY 2023–24, blue = AY 2024–25). Usage ramps up from September, peaks in December–February, and drops during summer break (May–June). Bottom panel: average content grade gap (student grade minus content grade being studied). The system starts students approximately 1.3 grade levels below their enrolled grade, progressively moves them toward grade-level content, then resets at the start of the new academic year. Data from ConveGenius platform logs for 60 treatment schools.

equal across grades is marginally rejected ($p = 0.091$), with the pattern suggesting a declining gradient from younger to older cohorts. We investigate this gradient—and its explanation through tablet congestion—in Section D.

These results are robust to alternative specifications and inference procedures. The treatment effect is stable across three control specifications (0.38–0.39 SD; Appendix Table G2). Randomization inference, which does not rely on asymptotic assumptions, confirms the main result (RI $p < 0.001$; Appendix Figure G1). Lee (2009) bounds, which trim the sample to address differential attrition between treatment and control (23.7% vs. 19.8%), remain large and positive (0.27–0.48 SD; Appendix Table G1). Finally, a within-student tablet-vs-paper sub-study conducted on Grade 8 students in Year 3 (Appendix C) finds no significant

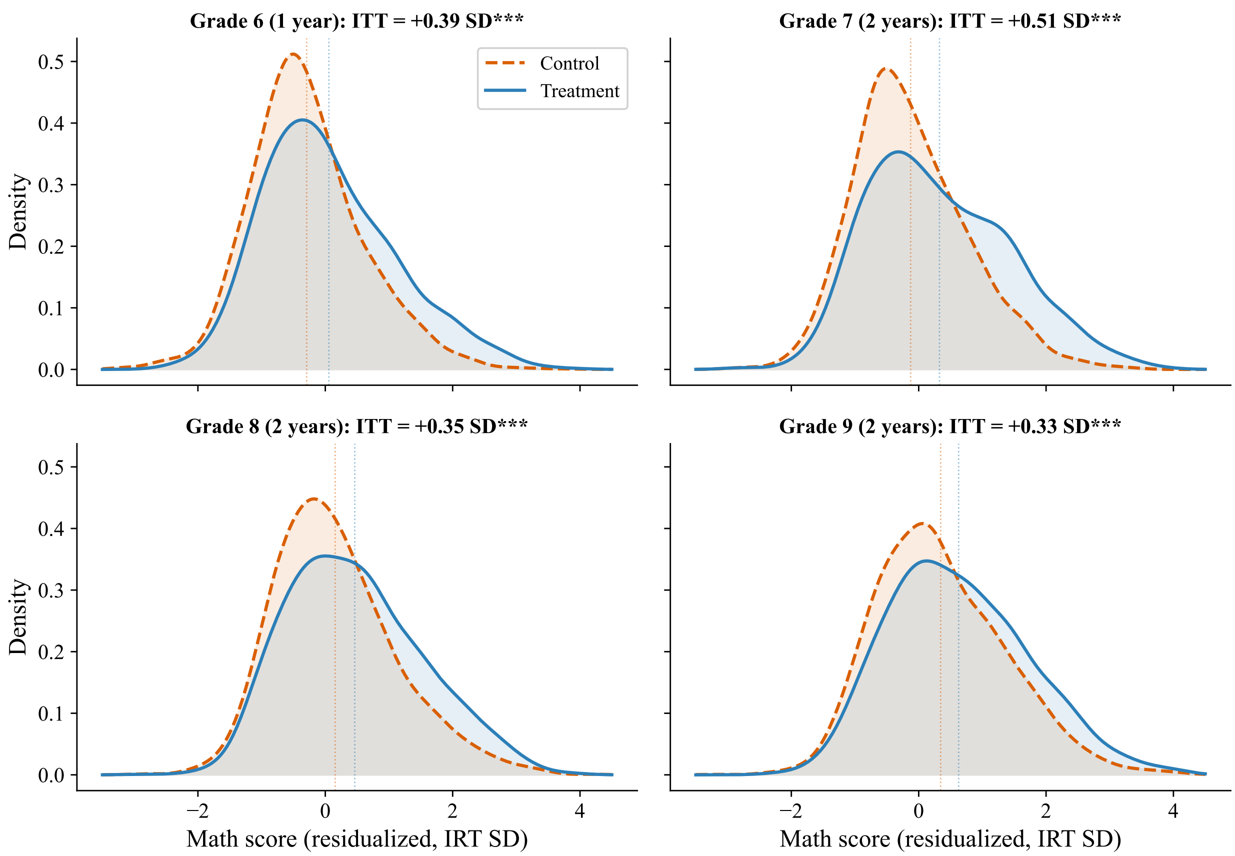

FIGURE 7. SCORE DISTRIBUTIONS BY GRADE: TREATMENT VS. CONTROL

Notes: Kernel density estimates of endline math scores by treatment status and grade. Scores are residualized for student covariates (SES index, father/mother education dummies, age, female, with missing-flag indicators) and mandal-cluster strata fixed effects, then re-centered at the grade-level mean to remain on the standardized IRT scale. Dotted vertical lines mark group means. Panel titles report the regression-adjusted ITT (matching Table 3) from OLS of *mathscore_z* on treatment, the same covariates, and strata fixed effects, with school-clustered standard errors. By Frisch–Waugh–Lovell, the T–C mean shift in each panel equals the panel-title ITT up to the residual correlation between treatment and the partialled-out covariates, which is approximately zero under random assignment. Grade 6 students had 1 year of PAL exposure; grades 7–9 had 2 years. Bandwidth = 0.25.

mode $\times$ treatment interaction, ruling out tablet familiarity as a driver of the headline effect.

B. Heterogeneous Treatment Effects

Table 4 examines treatment effect heterogeneity along several dimensions.

Gender. Girls gain significantly more than boys (0.46 vs. 0.33 SD; $p = 0.001$). This gender interaction survives multiple hypothesis testing corrections (Bonferroni, Holm, and Benjamini–Hochberg; Appendix Table G3). The advantage is not explained by pre-existing differences in baseline ability or attendance, which are balanced across gender within treatment status. Rather, girls accumulate approximately 2 more hours of PAL usage than boys, driven by more time on active practice (questions) rather than passive content (videos;

TABLE 3—INTENT-TO-TREAT ESTIMATES OF THE EFFECT OF PAL ON MATH LEARNING

	Pooled (1)	Grade 6 (2)	Grade 7 (3)	Grade 8 (4)	Grade 9 (5)
Treatment	0.390^{***}	0.393^{***}	0.509^{***}	0.346^{***}	0.327^{***}
	(0.054)	(0.070)	(0.069)	(0.068)	(0.062)
Controls	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Grade FE	Yes				
Control mean	0.000	-0.338	-0.157	0.116	0.316
Observations	14,215	3,313	3,354	3,623	3,925
R^2	0.137	0.153	0.125	0.097	0.086
p-value (equality across grades)			0.091		

Notes: Dependent variable: IRT math ability parameter standardized to control group mean and SD. Each column reports OLS estimates of the treatment effect with 21 randomization strata (mandal cluster) fixed effects. Column (1) includes grade fixed effects (grade 6 = base). Controls: SES index, father education, mother education, age, gender, and missing flags for each imputed variable. Standard errors clustered at the school level (120 clusters) in parentheses. The p-value tests the null hypothesis that treatment effects are equal across grades (from a pooled interaction model). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Figure E2). The gender-stratified IV analysis confirms that boys and girls have similar per-hour returns to PAL (Appendix Table F3), indicating the gap reflects differences in usage quantity rather than in the productivity of each hour. Appendix Table E1 provides the full regression results for this diagnosis.

Baseline math test scores. Treatment effects decline with baseline achievement: 0.48 SD for the lowest quintile and 0.31 SD for the highest ($p = 0.08$ for Q1 vs. Q5). PAL benefits students across the entire baseline-test-score range, but the lowest-baseline students gain the most. A conditional quantile analysis (Appendix Figure I9) reveals that within each baseline quintile, effects increase from the 10th to the 75th percentile of the endline distribution before flattening—suggesting that PAL creates within-group variance in outcomes in addition to the across-quintile gradient. The large treatment effect for the "missing baseline" subgroup (0.50 SD) reflects a data extraction artifact rather than a true subgroup effect: the government provided broader baseline data for treatment schools than for control schools in grades 6–7, creating compositional imbalance in this subgroup (Appendix Tables E3–E4 and Figure E3).

SES and school characteristics. Effects are uniform across SES terciles ($p = 0.42$ for Low vs. High; see also Appendix Table E2). Students in small sections (at or below 33 students, the approximate tablet supply) gain 0.46 SD compared with 0.34 SD in large sections ($p = 0.05$). Rural and urban schools show similar

TABLE 4—HETEROGENEOUS TREATMENT EFFECTS

	Treatment	(SE)	Control mean	N
Pooled	0.390***	(0.054)	0.000	14,215
Panel A: Gender				
Boys	0.329***	(0.056)	-0.063	7,289
Girls	0.464***	(0.059)	0.068	6,877
p (Boys = Girls)		0.001		
Panel B: Baseline Achievement				
Q1 (lowest)	0.483***	(0.071)	-0.410	2,483
Q2	0.353***	(0.063)	-0.207	1,897
Q3	0.330***	(0.075)	0.057	1,969
Q4	0.339***	(0.079)	0.341	1,936
Q5 (highest)	0.305***	(0.086)	0.929	2,020
Missing baseline	0.498***	(0.077)	-0.155	3,910
p (Q1 = Q5)		0.083		
p (joint)		0.126		
Panel C: SES				
Low SES (T1)	0.350***	(0.059)	-0.057	4,746
Mid SES (T2)	0.422***	(0.066)	0.039	5,182
High SES (T3)	0.381***	(0.060)	0.018	4,287
p (Low = High)		0.418		
Panel D: School Characteristics				
Small section ($\leq$33)	0.463***	(0.059)	-0.014	7,220
Large section (>33)	0.336***	(0.075)	0.014	6,995
p (Small = Large)		0.050		
Rural	0.456***	(0.057)	-0.018	11,615
Urban	0.520***	(0.151)	0.095	2,600
p (Rural = Urban)		0.075		

Notes: Dependent variable: IRT math ability parameter standardized to control group mean and SD. Each row reports the treatment coefficient from a separate OLS regression on the indicated subgroup, with 21 strata fixed effects, grade fixed effects, and controls (SES, parental education, age, gender, missing flags). Standard errors clustered at the school level in parentheses. Baseline achievement quintiles are within school $\times$ grade percentile ranks of government math SA1 (AY 2022–23). "Missing baseline" includes students without baseline data (28.8%). Baseline coverage differs by treatment arm for grades 6–7 due to a government data extraction artifact (Appendix Table E3); the treatment effect for this subgroup is not interpretable as a clean subgroup estimate. SES terciles are based on a PCA index of 9 household assets. Section enrollment is the number of students in the student's class section. p-values test equality of treatment effects between the indicated groups (from pooled interaction models). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

effects. The school-level heterogeneity by section size foreshadows the tablet congestion mechanism in Section D. Appendix Figure E1 provides a visual summary of heterogeneity across all four dimensions.

Topic-level effects. Treatment effects are uniform across all five math topics (0.31–0.38 SD; Appendix Table I7 and Figure I5), indicating that PAL improves foundational math skills broadly rather than producing topic-specific gains. Effects are slightly larger on harder items (Appendix Figure I6), suggesting that PAL helps students reach questions they would otherwise get wrong. Treatment students also outperform control students at every content grade level, with the largest gains on grade 4–6 content (Appendix Figure I7). PAL increases math enjoyment by 0.19 SD, with positive spillovers to other subjects, and reduces self-reported absences by 0.22 days over two weeks (Appendix Table I6).

C. Compliance and the Engagement Margin

The heterogeneity results above describe how the school-level treatment effect varies by student characteristic. We turn next to a different cut: how the within-treatment dose-response varies across baseline ability, using *compliance*—the fraction of the prescribed activity sequence the student walks (defined in §C, summarised alongside hours in Table 2)—as the engagement anchor. This complements the pooled IV LATE on hours in §E.

Figure 8 reports endline math scores by baseline-ability quintile $\times$ compliance tercile. Within every quintile, top-tercile compliers (above 72% of prescribed activities) score 0.3–0.5 SD above bottom-tercile compliers (below 56%). The within-quintile gap is largest at Q1 (+0.5 SD)—the lowest-baseline students, those who arrive at PAL with the largest foundational skill gaps. The gap remains positive across Q2–Q5 (+0.3 to +0.5 SD), so engagement with PAL's primarily grade-level content predicts higher learning at every point in the baseline-ability distribution.

D. The Grade Gradient and Tablet Congestion

The declining treatment effect across grades ($p = 0.091$; Table 3) could reflect several mechanisms: differential exposure (grade 6 had one year vs. two for grades 7–9), differences in instructional quality by grade, or a binding supply constraint on devices. Figure 9 distinguishes among these explanations by splitting schools at the median of total enrollment.

All treatment schools received exactly 30 tablets, regardless of enrollment. Appendix Figure D1 shows the full distribution of section sizes relative to this cap: 44.7% of treatment sections exceed 30 students and 59.6% of treatment students are in a sharing-required section. In small schools, section enrollment (students per grade-section) stays at or below 30 across all grades—every student gets a device during PAL periods

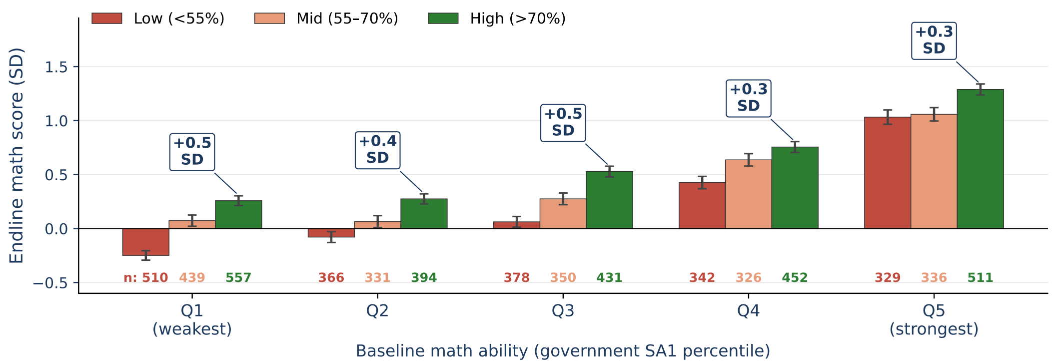

FIGURE 8. ENDLINE MATH SCORE BY BASELINE ABILITY QUINTILE × COMPLIANCE TERCILE

Notes: Mean endline math score (in SD units, standardized to control-group mean and SD) by baseline math ability quintile (x-axis; from government SA1 percentile rank within school × grade) and compliance tercile (color; equal-frequency cuts at compliance ≈ 0.56 and 0.72). Error bars are ±1 standard error. Cell counts shown below each bar (range 343–538 students). Boxed annotations report the bottom-to-top tercile gap within each quintile. Treatment students only ($N = 6{,}052$); compliance defined as in §C. The within-quintile gradient is a selection-on-observables comparison conditioning on baseline ability, not a separate randomization; unobserved motivation correlated with both engagement and learning would bias it upward, so the spread bounds rather than identifies the per-unit return.

(Panel A). In large schools, sections exceed the 30-tablet threshold and grow with grade (median 32 at grade 6, 40 at grade 9), requiring students to share devices or take turns.

This physical constraint translates directly into usage. In small schools, PAL usage hours are relatively flat across grades (Panel B). In large schools, usage declines at higher grades where congestion is most severe. The treatment effect pattern mirrors this: in small schools, effects are uniformly large across all grades (0.58–0.65 SD) and statistically indistinguishable ($p = 0.90$; Panel C). In large schools, effects decline from 0.42 SD at grade 7 to 0.22 SD at grade 9, and the gradient is marginally significant ($p = 0.07$).

The parallel pattern across all three panels—flat in small schools, declining in large—is consistent with tablet congestion driving the grade gradient. When every student has a device, PAL produces large and uniform gains regardless of grade. The grade gradient in the pooled estimates reflects the fact that larger schools, where older students face the most congestion, pull down the average effect at higher grades. This mechanism has a direct policy implication: providing additional tablets to high-enrollment schools should eliminate the gradient and raise the average treatment effect toward the 0.58–0.65 SD observed in unconstrained settings.

E. Dose-Response

We formalise the congestion mechanism using an instrumental variables strategy. Because all treatment schools received 30 tablets regardless of enrollment, the interaction of treatment status and section enrollment generates exogenous variation in per-student PAL usage: students in larger sections mechanically receive

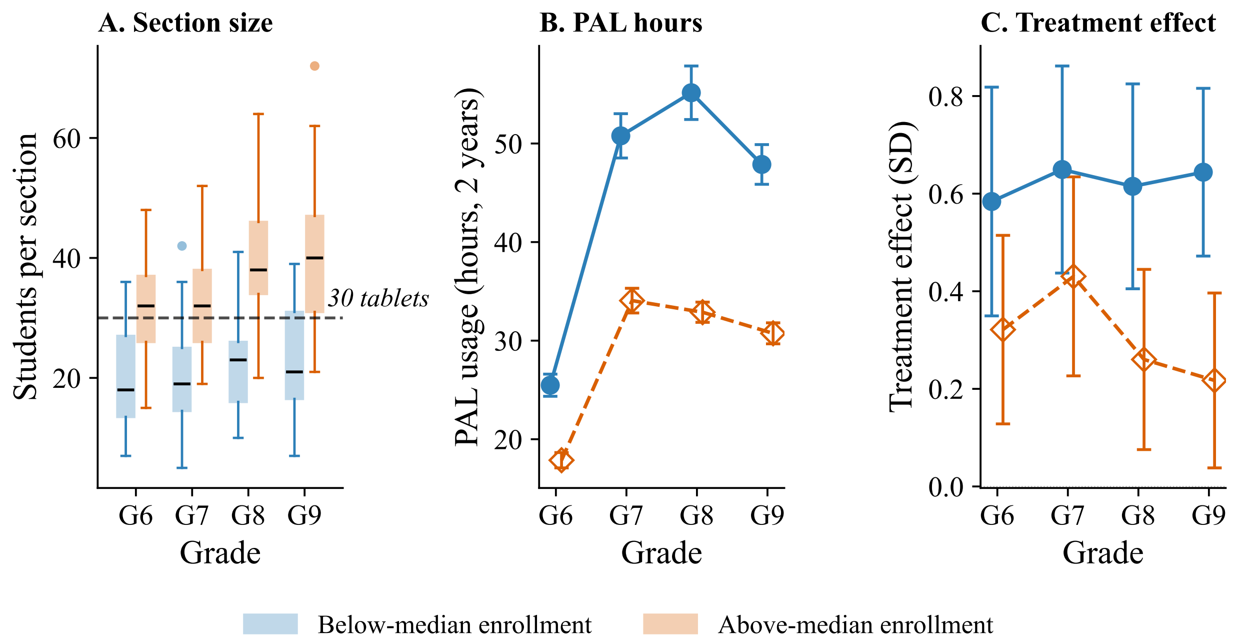

FIGURE 9. THE GRADE GRADIENT IN TREATMENT EFFECTS AND TABLET CONGESTION

Notes: Schools are split at the median of total enrollment in grades 6–9 (median = 138). *Panel A*: section enrollment by grade; the dashed line marks the 30-tablet supply. *Panel B*: mean PAL usage hours (2 years) by grade with 95% CIs. *Panel C*: ITT estimates on endline math scores. In small schools, effects are uniformly large across grades ($p = 0.90$); in large schools, effects decline with grade ($p = 0.07$). All regressions include controls, strata FE, and school-clustered SEs.

less tablet time. We instrument for usage hours using this interaction (equations 2–3). The first-stage $F = 11.2$ exceeds the conventional Staiger-Stock (1997) rule-of-thumb of 10 and the Stock-Yogo (2005) 15%-maximal-bias threshold of 8.96, while falling below their 10%-bias threshold of 16.38; the reduced form is consistent with the first stage (Appendix Figure F1); section enrollment has no effect on control student scores ($\beta = 0.001$, $p = 0.75$; Appendix Figure G2), supporting the exclusion restriction.

Table 5 reports the formal estimates. The 2SLS estimate of the causal return to PAL usage is 0.011 SD per hour ($p < 0.001$), close to the within-treatment OLS estimate of 0.009 SD per hour. The 2SLS specification anchors the linear projection at zero usage by attributing the full T–C gap to usage hours: the slope is the per-hour return that translates control-mean baseline into the observed treatment-arm endline. Replicating the 2SLS separately by grade and by gender yields broadly consistent per-hour estimates (Appendix Table F3).

Figure 10 characterizes the within-school shape of the dose-response. The relationship is S-shaped: approximately flat below 10 hours, steepest between 10 and 30 hours (slopes of 0.019 and 0.013 SD per hour), and flattening to 0.005 SD per hour above 30 hours. The IV slope of 0.011 SD per hour is the average per-hour return implied by the within-treatment regression; the spline shows this average masks substantial

TABLE 5—DOSE-RESPONSE: IV ESTIMATES USING TABLET CONGESTION

	First Stage Usage (hrs) (1)	Reduced Form Math score (2)	OLS Math score (3)	2SLS Math score (4)
Treatment × Section enrollment	-0.370***	-0.0103***		
	(0.110)	(0.0036)		
PAL usage (hours)			0.0093***	0.0110***
			(0.0018)	(0.0014)
Treatment	48.3***	0.728***		
	(4.2)	(0.126)		
Controls	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes
Grade FE	Yes	Yes	Yes	Yes
Control mean		0.001		0.001
First-stage F	11.2			11.2
Observations	14,123	14,123	6,665	14,123

Notes: Dependent variable in columns (2)–(4): IRT math score standardized to control mean and SD. Column (1): first stage regresses PAL usage hours on treatment × section enrollment and controls. Column (2): reduced form regresses math score on the same instrument. Column (3): OLS of math score on usage hours (treatment group only; biased by selection into usage). Column (4): 2SLS using treatment × section enrollment as instrument for usage hours, with the linear projection anchored at the control mean at zero usage (structural equation drops the treatment dummy so the slope attributes the full T–C gap to usage hours). The instrument exploits the fact that all treatment schools received 30 tablets regardless of enrollment; larger sections share more students per tablet, reducing per-student PAL time. Section enrollment does not affect control student scores ($\beta = 0.001$, $p = 0.75$; see Figure 9). All regressions include controls, 21 strata FE, and grade FE. Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

variation across exposure regions. Scaling 0.011 linearly to the 36 two-year usage hours observed in grade 9 predicts approximately 0.40 SD, close to the 0.33 SD we estimate—and more generally, grade mean hours (the diamonds in Panel A) bracket the 10–40 hour range that spans both the steep learning zone and the flattening one. The pooled grade gradient therefore reflects more than differences in exposure alone; tablet congestion (Section D) is one contributor, and grade-specific variation in content difficulty and measurement are plausible others that we do not attempt to separate here. Appendix F reports the shape comparison across specifications.

F. Effects on Government Test Scores

Table 6 examines whether treatment effects appear on government-administered math assessments, which are set and graded by individual schools. Effects are positive and significant for several grade-year combinations: grade 6 in Year 2 (0.34 and 0.27 SD on SA1 and SA2), grade 8 in Year 1 SA1 (0.24 SD), and grade 8 in Year 2 SA1 (0.36 SD). Other cells are positive but not statistically significant. Effects tend to be

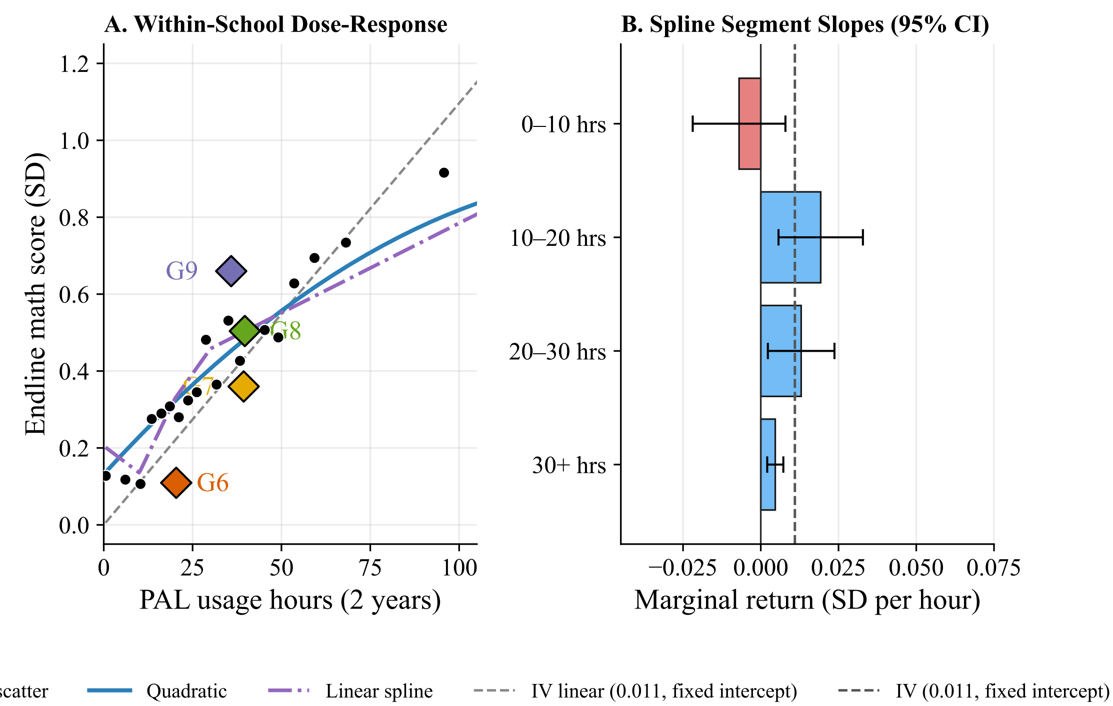

FIGURE 10. DOSE-RESPONSE SHAPE AND THE IV SLOPE

Notes: Treatment students only, within-school (school fixed effects), school-clustered standard errors. *Panel A*: binscatter of endline math scores by PAL usage hours (20 equal-frequency bins; $N = 6{,}742$) with quadratic ($\beta_2 = -0.00003$, $p = 0.002$) and linear-spline (knots at 10, 20, 30 hours; joint $p = 0.012$) fits, conditional on covariates. The dashed gray line is the IV linear prediction (slope $+0.011$ SD per hour, matching the 2SLS estimate in Table 5, anchored at the control mean at zero usage). Diamonds mark the raw mean score and mean hours for each grade at endline (G6, G7, G8, G9). *Panel B*: slope of the linear spline by hours range, with 95% CIs (school-clustered). Slopes are 0–10 hrs -0.007 $[-0.022, +0.008]$; 10–20 hrs $+0.019$ $[+0.006, +0.033]$; 20–30 hrs $+0.013$ $[+0.002, +0.024]$; 30+ hrs $+0.005$ $[+0.002, +0.007]$. The dashed gray reference marks the IV slope of $+0.011$ SD per hour. The shape is concave: average per-hour returns mask substantial variation across exposure regions, with the steepest slopes in the 10–20 hour range and flattening above 30 hours.

stronger on SA1 (mid-year) than SA2 (end-of-year) assessments, potentially reflecting teachers' awareness of PAL content alignment with mid-year topics.

The grade 10 board exam is the one government outcome graded externally to the schools students attended (unlike the FA/SA assessments, which are school-graded and carry substantial grading heterogeneity across schools; Appendix H). For the cohort with one year of PAL exposure in grade 9, the math board exam shows a negative point estimate (-0.14 SD, $p = 0.04$), significant only with strata fixed effects; no student-level controls are available for this cohort (Appendix Table H1). An investigation of whether the effect reflects PAL displacing exam preparation finds the opposite pattern: it is concentrated among low-usage students, while high-usage students are indistinguishable from control (Appendix Table H2). We report the estimate because the external-grading structure makes it the cleanest government cross-school measure, while the specification sensitivity cautions against treating the negative direction as a settled finding.

TABLE 6—EFFECTS ON GOVERNMENT MATH ASSESSMENT SCORES

	Year 1 (AY 2023–24)		Year 2 (AY 2024–25)	
	SA1	SA2	SA1	SA2
	(1)	(2)	(3)	(4)
Grade 6			0.341***	0.268*
			(0.113)	(0.138)
Grade 7	0.138	0.148	0.181*	-0.013
	(0.100)	(0.119)	(0.098)	(0.130)
Grade 8	0.238***		0.364***	0.092
	(0.082)		(0.115)	(0.153)
Grade 9	0.025	0.131	0.118	0.018
	(0.087)	(0.104)	(0.098)	(0.101)
Grade 10†		-0.108		-0.136**
		(0.095)		(0.067)
Controls	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes

Notes: Dependent variable: government math assessment score, standardized within grade to control mean and SD. SA1 and SA2 are mid-year and end-of-year summative assessments administered and graded by schools. Grades 7–9 at endline had PAL for both years; grade 6 had PAL in Year 2 only (new cohort entering AY 2024–25); grade 10 had PAL in Year 1 only (as grade 9 in AY 2023–24, then aged out). †Grade 10 Year 2 column shows the state board exam (externally graded), not SA2. Grade 10 regressions include strata FE only (no student-level controls available). Cells left blank where treatment or control coverage is below 500 students. All other regressions include demographic controls, strata FE, and school-clustered standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Government school assessments have well-documented limitations as measures of absolute learning. The convergent validity between Year 2 SA2 scores and our independently administered endline is modest at the student level ($r = 0.35$) and negative at the school level ($r = -0.14$; Appendix Figure H1), consistent with school-specific grading standards introducing noise. The positive effects we observe on these assessments should therefore be interpreted as broadly consistent with the main IRT results, rather than as precise estimates of treatment effect magnitude.

VI. Cost Effectiveness

The PAL program is funded through state and central (Samagra Shiksha) government education budgets and implemented by the Andhra Pradesh state government's regular school administration. We calculate the government's cost of delivering the program using administrative expenditure data from the 524-school combined sample where state-deployed PAL is in use (444 program schools, 60 RCT treatment schools, and 20 remediation schools). In the first year (AY 2023–24), the per-student cost was \$33 (INR 2,861), which includes one-time lab setup costs: 30 tablets, charging infrastructure, earphones, and classroom preparation. In the second year (AY 2024–25), the per-student cost fell to \$19 (INR 1,616), reflecting recurring costs only: software licensing, Field Management Staff, government monitoring, and equipment replacements. The total cost of delivering PAL over the two-year study period was therefore \$52 per student (\$33 + \$19), or \$26 per student-year.[5] Costs are annualized assuming a 3-year tablet depreciation schedule with annual replacement of headphones and SD cards; software licensing and vendor service rates were set at program inception in 2019 and have remained nominally constant in state contracts since.

Hardware is the single largest cost component, accounting for approximately 36% of annualized expenditure. This cost is effectively recurring: a January 2026 inventory across the 524 schools where state-deployed tablets are in use found that only 57% of tablets remained functional, with 29% of schools reporting zero working tablets. Tablet replacement is driven by both physical degradation (a four-year deployment window for the program rollout cohort, with a two-year COVID-related shutdown from 2020 to 2022) and evolving software requirements—the program has already transitioned from 8-inch to 10-inch devices at higher unit cost. Software licensing accounts for only 5% of costs, though current pricing reflects a volume discount across 524 schools that may not persist. Monitoring and implementation support—Field Management Staff visits, state dashboards, and district follow-up—account for 11%.

To place these costs in context, we express learning gains in two complementary metrics. First, using our own data: in control schools, the cross-grade learning gradient is 0.22 SD per grade, meaning the treatment effect of 0.39 SD is equivalent to approximately 1.8 additional years of learning in the Andhra Pradesh government school system. Second, to enable comparison with the broader literature, we convert to Learning-Adjusted Years of Schooling (LAYS), which expresses gains relative to a high-performance global benchmark of 0.8 SD per year (Angrist et al., 2020; Filmer et al., 2018):

$$\text{LAYS} = \frac{\hat{\beta}}{\delta_h} = \frac{0.39}{0.80} = 0.49$$

The PAL intervention produced learning equivalent to just under half a year of frontier-quality schooling.

[5] All costs in 2024–25 prices, converted at \$1 = INR 86 (Indian Rupees). Per-student figures assume 204 students per school, the average across the 80 research schools.

Given a total program cost of \$52 per student over two years, this translates to:

$$\text{LAYS per \$100} = \frac{0.49}{0.52} \times 1 = 0.94$$

PAL's cost-effectiveness of 0.94 LAYS per \$100 falls within the range reported for adaptive and computer-assisted learning programs (0.24–2.66 LAYS per \$100) (Angrist et al., 2020). The Global Education Evidence Advisory Panel classified adaptive learning software as "Promising but Limited Evidence," noting that no study had demonstrated cost-effectiveness at scale in the public sector (Global Education Evidence Advisory Panel, 2023). Our estimates suggest that government-delivered adaptive learning can be cost-effective, though it is not among the cheapest interventions per unit of learning gained—the cost is driven by hardware, a general infrastructure constraint rather than a feature specific to the intervention design.

VII. External Validity

This section asks whether the experimental per-hour return and the dose-response shape generalize to the program at scale and past the experimental window. Sections V–VI establish PAL's causal effects within our evaluation schools: an intent-to-treat from random assignment, and a per-hour return of $+0.011$ SD identified by the binding tablet supply constraint within the treatment arm. The within-treatment dose-response curve gives that return a shape—flat below ten hours of two-year usage, steep between ten and thirty, and flattening above. These are internal-validity claims: they hold for the 60 schools we randomized. But those 60 schools were added to a 444-school government program the state had been running since 2019, a rollout that has since scaled to roughly 1,200 government schools statewide. The policy question is what the per-hour return and the dose-response shape imply for the program at scale, and for its continuation past the experimental window. This section turns from internal to external validity by extrapolating the curve over the program's usage distribution to project the at-scale effect, validating that projection against three Year-3 endline samples that span the dose-response, and characterizing the delivery infrastructure under which the program operates.

Three samples were assessed at Year-3 endline in February–March 2026 on a parallel math assessment, with Year-3 ability placed on the Year-2 IRT scale through Mean–Sigma linking on 45 common anchor items: a random 80-school subset of the parent RCT (40 treatment, 40 control); a 40-school subsample of the 444-school program rollout; and 40 of the roughly 700 Kasturba Gandhi Balika Vidyalaya (KGBV) residential schools the state added to the rollout in January 2025. The first sample re-tests the original

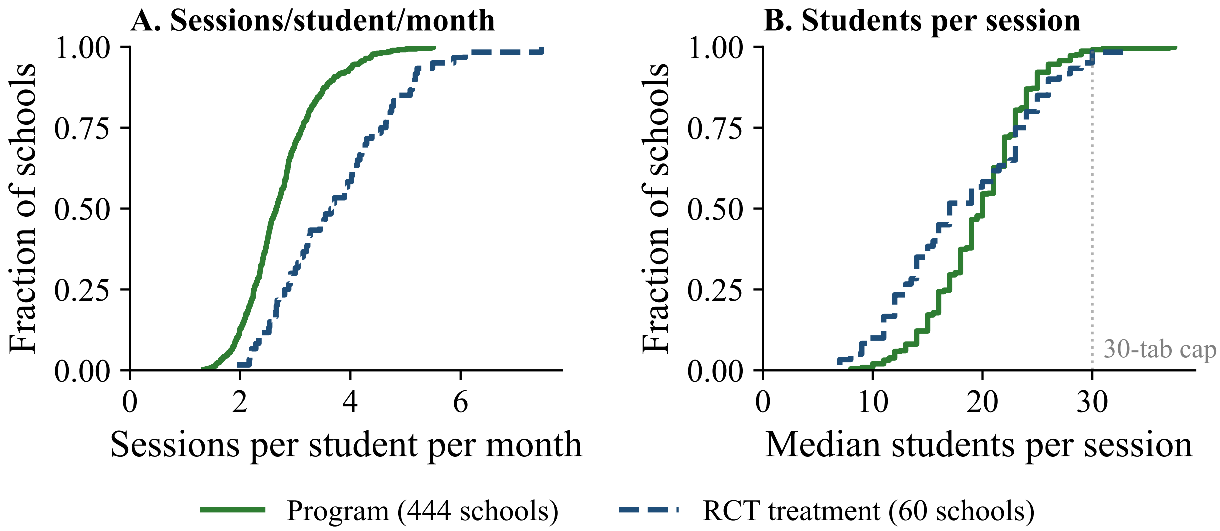

FIGURE 11. SESSION-LEVEL COMPARISON: PROGRAM VS. RCT TREATMENT SCHOOLS

Notes: ECDFs derived from day-level ConveGenius usage data, both arms in AY 2024–25. A session is defined as a (school × grade × day) combination with at least 5 active math students. *Panel A*: ECDFs of sessions attended per student per month, computed as school-level means across 444 program schools (green solid) and 60 RCT treatment schools (blue dashed). Program median = 2.7, RCT median = 3.7. *Panel B*: ECDFs of median students per session at the school level. Program median = 20, RCT median = 17; session sizes are essentially equivalent across arms. Together the two panels show the dose gap is driven by session frequency rather than by batch size.

experiment one year past the experimental window; the second and third reach out-of-experiment populations the state operates under different delivery configurations.

Program students receive roughly one-fifth the per-student math-PAL dose of RCT students—6.9 versus 35.3 hours of cumulative math usage over AY 2023–25—and the gap is structural rather than operational (Table 7). Program schools have more than twice the grade-6–9 enrollment of RCT schools (326 vs. 155) under the same 30-tablet cap, producing a per-grade-cohort student-to-tablet ratio of 4.4:1 in program schools versus 1.2:1 in the RCT. Program PAL infrastructure splits weekly tablet-time across three subjects: math is allocated two of the four weekly PAL periods, with one each for Telugu and English. The RCT intervention was math-only with two weekly periods, so each tablet supplies roughly half the math time per program-school student that it does in the RCT. Session-level evidence shows the resulting dose gap is driven by session frequency rather than by batch size: program-school median session size is 20 students versus 17 in RCT treatment schools, while program students attend 2.7 PAL sessions per active month versus 3.7 in the RCT (Figure 11). The bottleneck is in tablet-time allocation, not operational capacity to deliver PAL.

The structural dose gap is compounded by hardware attrition. Three inventory snapshots across the 444 program schools document a nine-month attrition rate of approximately 19% in mean working tablets per

TABLE 7—PAL USAGE BY COHORT: YEARS 1–3

	RCT Treatment (60 schools)	Program (all)	Program (>0 tabs)	KGBV (40 schools)
Schools	60	444	375	40
Students (active on PAL)	6,532	134,090	110,458	—
Mean enrollment (grades 6–9)	155	326	319	159
Working tablets (reported, mean)	27.1	19.0	22.5	—
Peak monthly active students per grade (mean)	26.5	66.5	65.1	—
Student-tablet ratio per grade-cohort (mean)	1.2	4.4	4.4	—
Math usage Y1+Y2, hours (mean)	35.3	6.9	7.1	—
Math usage Y1+Y2, hours (median)	31.1	5.5	5.7	—
Months active Y1+Y2 (mean)	6.9	5.2	5.3	—
Math usage Y3, hours (mean)	5.4	0.64	—	13.6
% math-active in Y3	60%	17%	—	97%
Y3 θ_z (mean, linked)	+0.32	+0.21	—	+0.54

Notes: RCT treatment schools used PAL for math only. Program schools used PAL for math, Telugu, and English (3 subjects). KGBV residential schools joined the AP rollout in January 2025; Years 1 and 2 rows are not applicable. Years 1 and 2 usage statistics are for math only in both groups, summed across AY 2023–24 and AY 2024–25. Year-3 rows are computed on Year 3 endline-panel students (n: RCT treatment 2,967; Program 5,957; KGBV 5,452); Year 3 θ_z is standardized to the RCT-control mean and standard deviation. "Active" students are those with any recorded PAL usage across the relevant period (Years 1 and 2 for the upper block, Year 3 for the lower block). "Working tablets (reported)" is from government/CG infrastructure records. "Peak monthly active students per grade" is computed per school-grade pair: the maximum, across months of AY 2024–25, of the count of students in that grade with any usage that month; averaged across school-grade pairs. Grade-level disaggregation avoids pooling four cohorts who use the same tablets at different periods. "Student-tablet ratio per grade-cohort" is the mean of grade-cohort enrollment / school working tablets, computed across (school $\times$ grade) cells. A PAL period serves one grade-cohort sharing the school's tablet allocation; the ratio exceeds 1 when the cohort is larger than the working tablet count. For program schools we use student-grade counts as the cohort proxy (section-level enrollment data are not available in program records). Program schools were part of the government's PAL rollout (17 of 26 AP districts) beginning in 2019. RCT schools began implementation in AY 2023–24. All schools—both program and RCT treatment—have a 30-tablet cap regardless of enrollment.

school between April 2025 and January 2026; by the January 2026 snapshot 34% of program schools have zero working tablets, and 83 schools have moved from a working inventory to zero working tablets over the nine-month window (Figure 12). Because the original 2019–2022 deployment has not been systematically replaced, the dose at scale reflects accumulating hardware attrition layered on top of the structural three-subject, four-grade allocation across a 30-tablet cap. The structural dose gap on its own would predict roughly half the per-student math hours of the math-only RCT configuration even at full hardware availability; the hardware attrition then compresses delivery further, into the near-zero region observed at Year-3.

Applying the within-treatment dose-response curve over the program's usage distribution gives two complementary projections of the at-scale effect. Linearly scaling the IV slope of +0.011 SD per hour (Figure 10, Table 5) to the program two-year mean of 6.9 hours (Table 7) implies a linear-IV upper bound of approximately +0.076 SD. Integrating the within-treatment piecewise-linear shape (Figure 10) over

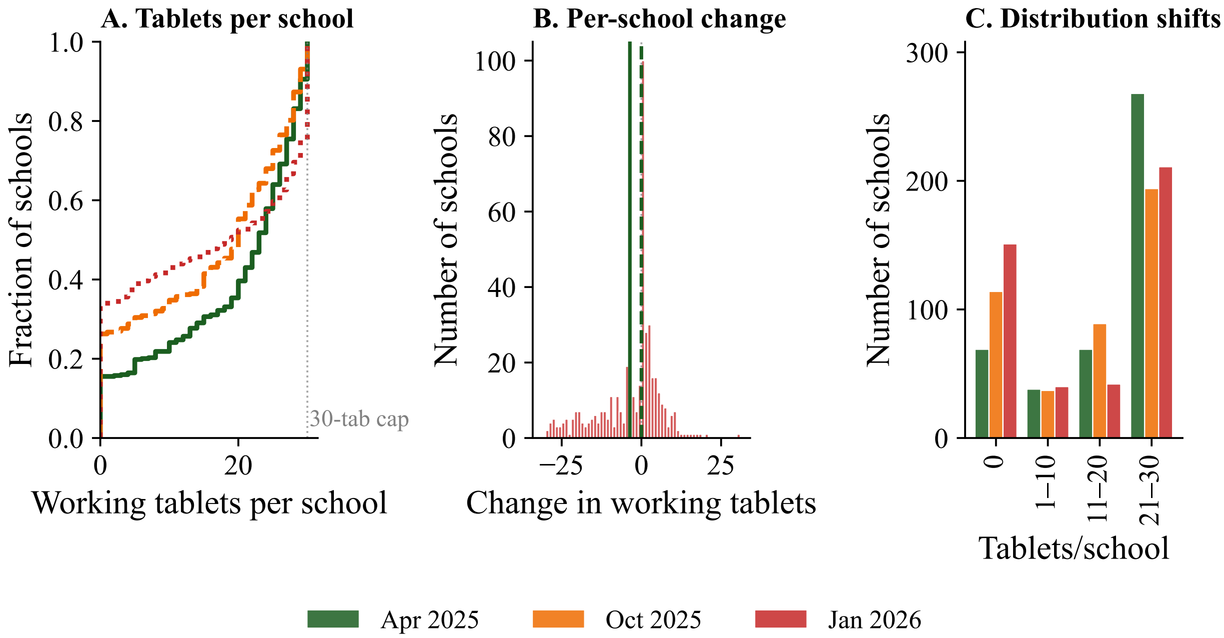

FIGURE 12. WORKING TABLET INVENTORY ACROSS 444 PROGRAM SCHOOLS, 2025–2026

Notes: Three inventory snapshots: April 2025 (end of AY 2024–25; $n = 444$, median 23), October 2025 (start of AY 2025–26; $n = 434$, median 20), and January 2026 (mid AY 2025–26; $n = 444$, median 18). *Panel A*: empirical CDF of working tablets per school at each snapshot. *Panel B*: per-school change between April 2025 and January 2026 (mean -3.6, median 0); 174 of 434 schools (40%) lost tablets, 142 (33%) were stable, 118 (27%) gained. *Panel C*: distribution of schools across 0/1–10/11–20/21–30 working-tablet buckets. Mean attrition over the nine-month window is -3.6 tablets per school (-19%).

the program student-level usage distribution gives a substantially smaller projected ATE of approximately -0.02 SD (95% Monte Carlo CI $[-0.09, +0.05]$), reflecting that the curve is approximately flat below ten hours of usage where the bulk of the program-school distribution sits. The two projections together place the at-scale effect well below the experimental ITT, with the within-treatment shape suggesting near-zero returns in the 0–10 hour region.

The Year 3 cross-sample evidence places the three samples at three distinct levels of cumulative PAL exposure (Figure 14): the 444-school program rollout at near-zero exposure, the 40 KGBV residential schools at mid-range exposure, and the original RCT treatment arm at the high end during the experimental window with declining post-experimental delivery.

The figure shows weekly math PAL minutes per enrolled student across three academic years for the three samples. The 60 RCT treatment schools reach their three-year peak in Year 1 at roughly 70 minutes per enrolled student per week, sustain 30–50 minutes per week through Year 2, and collapse to under 10 minutes per week in Year 3. The 444-school program rollout runs at roughly one-third the RCT level through Year 1 and Year 2 (peak $\sim$25 minutes per week in late Year 2) and similarly collapses in Year 3. The 40 KGBV residential schools are absent until the January 2025 launch, ramp to ∼50 minutes per week within months, and sustain 20–40 minutes per week through Year 3 even as RCT and program decline.

Cross-sample comparisons confirm this projection (Table 7). Program-rollout students average 0.64 math PAL hours over AY 2025–26 with 83% recording zero hours, and post a pooled Year-3 θ_z statistically indistinguishable from the RCT-control mean—consistent with the within-treatment dose-response's near-zero return at near-zero exposure. KGBV residential students at 13.6 hours per student (97% math-active) post a pooled Year-3 θ_z of +0.54 SD. At grade 7, where the samples overlap most cleanly, KGBV's Year 3 mean of +0.42 SD approaches the Year-2 RCT treatment grade-7 ITT of +0.51 SD.

Within each governance mode, regressing Year-3 linked θ on cumulative PAL math hours with school fixed effects recovers concave shape across all three samples (Figure 13; per-hour coefficients in Appendix Table F2). The linear within-school slope orders by mean exposure: +0.025 SD per hour in program schools (3-year cumul mean 6.6 hrs, near the activation region), +0.015 in KGBV (mean 13.6 hrs), and +0.010 in the RCT-T 3-year panel (mean 24.3 hrs, well into the flattening region). The RCT-T 3-year slope matches the IV slope of +0.011 identified on the RCT 2-year sample (Figure 10). The slope ordering across the three samples is consistent with a single concave dose-response whose averaged slope depends on where each sample's mean sits along the curve. The cross-sample comparisons remain descriptive—the program and KGBV panels are not random—but the per-hour return reproduces across distinct delivery contexts.

The figure also overlays state-coordinator review meeting dates ($n = 23$, red verticals) and WhatsApp DNO usage nudges ($n = 64$, orange verticals) on the cohort trajectories. Some review meetings precede a visible rise in weekly usage—notably the late-2023 cluster where the RCT line trends upward across the November– December 2023 meeting dates, and the post-launch KGBV ramp through February–March 2025. Other meetings sit on flat or declining stretches. The overlay is descriptive only: it does not isolate meeting-driven uplift from coincident calendar variation (AY rhythm, exam weeks, holidays) or from the selection that schedules meetings during periods of perceived under-usage.

Within the original RCT treatment schools, the Year-3 delivery infrastructure contracted across multiple layers and delivery collapsed in the chronic under-implementer tail. Year-1 and Year-2 saw approximately weekly state-DNO communications via the State PAL Coordinator's WhatsApp channel (about 46 messages each year), with field-staff visits cadencing at 1–4 per RCT treatment school per month over AY 2024–25. Across the partial Year 3 window through 27 October 2025, the WhatsApp channel produced 4 messages across 4 months—roughly one-quarter the Years 1–2 rate—as the State PAL Coordinator's bandwidth shifted

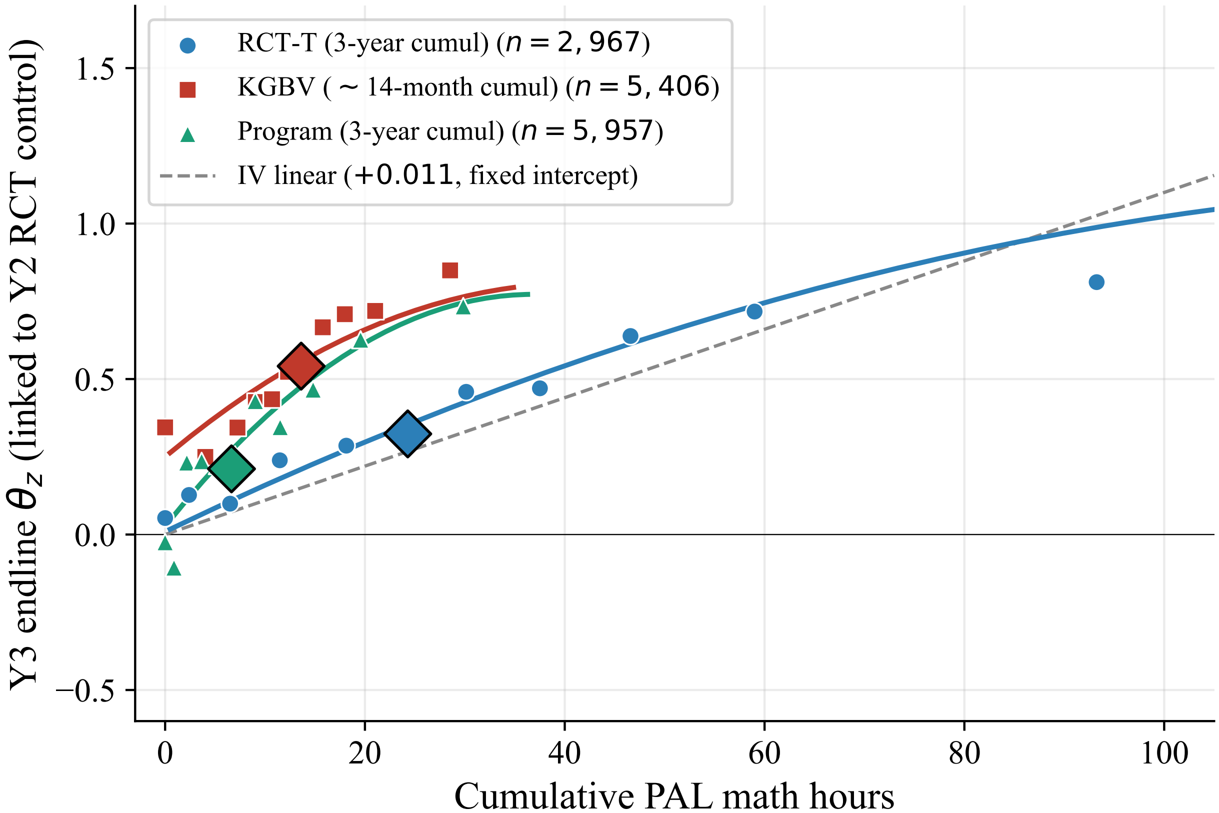

FIGURE 13. CROSS-SAMPLE WITHIN-SCHOOL DOSE-RESPONSE ON CUMULATIVE PAL HOURS

Notes: Within-school binscatter and quadratic fit (zero-bin plus 10 active deciles per sample) of Y3 endline IRT θ (linked to the Y2 RCT-control scale via Mean–Sigma on 45 anchor items) on cumulative PAL math hours, for three governance modes: RCT-T treatment arm (blue circles, 3-year cumul AY 2023–26, $n = 2{,}967$), KGBV residential girls (red squares, ∼14-month cumul since the January 2025 launch, $n = 5{,}406$), and program-rollout subsample (green triangles, 3-year cumul AY 2023–26 with Y1 student-level hours recovered via SHA-256 hash linkage to the cleartext Y3 student_code, $n = 5{,}957$). Sample-mean diamonds mark each sample's mean hours and mean Y3 θ. Quadratic fits clipped to each sample's observed support (approximately 0–100 hrs RCT-T, 0–35 hrs KGBV, 0–12 hrs program). Gray dashed reference is the fixed-intercept IV slope of $+0.011$ identified causally on the RCT 2-year sample (Figure 10). Each sample's per-hour slope orders by where its mean sits on the curve — consistent with a single concave dose-response whose averaged slope depends on the sample's exposure region. Per-hour coefficients with significance tests are reported in Appendix Table F2.

to the KGBV residential expansion that launched in January 2025 (Figure 5). At the school level, this contraction landed unevenly: six of the 60 RCT treatment schools delivered effectively zero PAL in Year 3 to their Grades 6–8 cohort, and an additional 28 delivered fewer than five hours per student. The zero-bin schools were chronic under-implementers from the start—their Year-1 and Year-2 cohort doses were already roughly half those of the high-implementing arm before any Year-3 collapse—so the Year 3 pattern is a continuation of variation present throughout the experiment, with the chronic tail dropping to zero in Year 3.

The consequence for veteran panel students is attenuation of the year-on-year treatment effect. The Year-2 treatment premium of $+0.33$ SD narrows to $+0.15$ SD at Year 3 (Table 8, Panel B); the covariate-adjusted

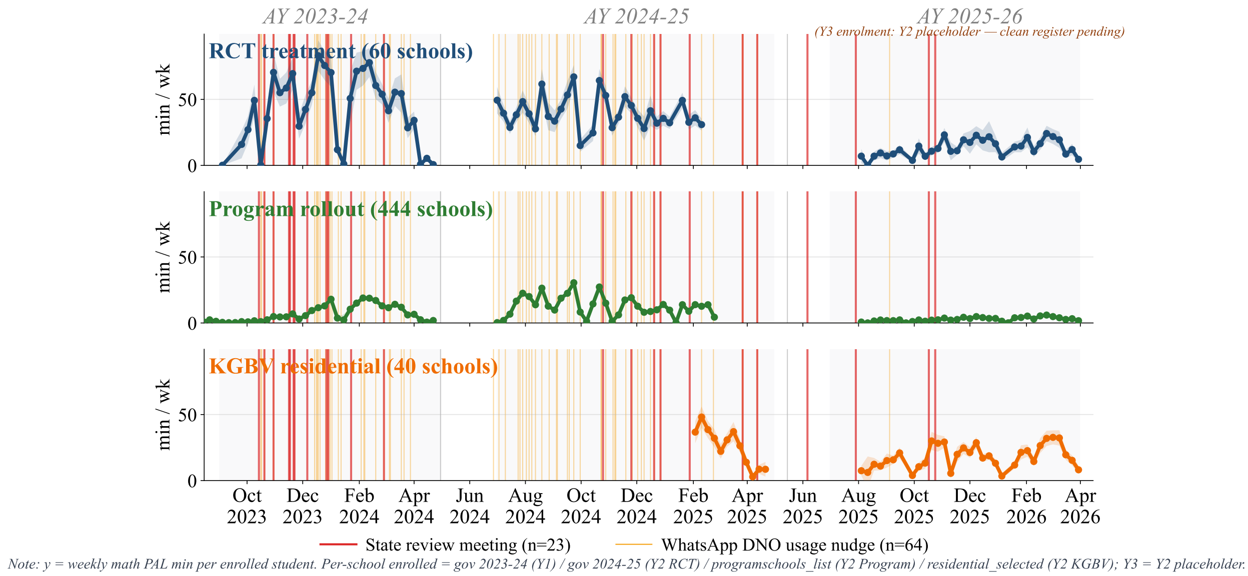

FIGURE 14. THREE-YEAR MATH PAL USAGE: RCT TREATMENT, PROGRAM ROLLOUT, AND KGBV RESIDENTIAL

Notes: Weekly mean math PAL minutes per enrolled student across three academic years, by governance mode. RCT treatment (60 schools, blue) operated AY 2023–24 and 2024–25 as an intensive math-only intervention; AY 2025–26 reflects post-experimental delivery. Program rollout (444 schools, green; 391 in Y1 due to a 2023–24 ConveGenius export gap) shares tablet-time across math, Telugu, and English. KGBV residential schools (40 schools, orange) joined the AP rollout in January 2025; pre-2025 traces do not exist. Per-school enrollment is taken from each year's official register; AY 2025–26 uses the AY 2024–25 register as a placeholder while the Y3 register is being finalized. Vertical markers: red for state-coordinator review meetings ($n = 23$); orange for WhatsApp DNO usage nudges ($n = 64$). End-of-AY exam season and summer breaks shaded gray.

change in the $T - C$ gap is -0.21 SD ($p < 0.05$, controls + strata FE). Within-arm Year-2-to-Year-3 changes are not displayed: 46 of the 70 Year-3 test items are physically the same questions panel students answered at the Year-2 endline, and familiarity with these specific items inflates both arms' raw Year-3 scores above one year of pure schooling growth. The shared inflation cancels in the $T - C$ comparison but makes the within-arm changes difficult to interpret as ability gain. The attenuation is consistent with two channels developed above: saturation among veterans whose Year-1 and Year-2 dose was already in the flat segment of the dose-response curve, and delivery collapse in the chronic-implementer schools.

In levels, however, the cumulative effect persists. Treatment students at Year 3 ($\theta_z = +0.534$) remain 0.539 SD above where control students were at Year 2 (-0.005; Table 8, Panel A); on raw percent-correct for the 45 anchor items, treatment students at Year 3 (51.6%) sit above controls at Year 2 (41.4%). The cumulative two-year experimental advantage is retained even as the year-on-year $T - C$ gap narrows.

These delivery patterns sit within a wider Year-3 contraction across the layers of the delivery infrastructure that produced Year-1 and Year-2 dose. The same small State PAL coordination unit that ran the program in 444 schools across 17 districts during the trial—a State Project Director, an Additional SPD, a State PAL

TABLE 8—YEAR-3 ATTENUATION IN THE RCT PANEL

	Y2 (θ_z)	Y3 (θ_z^{linked})
Panel A. Endline ability levels by wave		
Control	-0.005	$+0.381$
Treatment	$+0.329$	$+0.534$
Panel B. Treatment effect ($T-C$)		
Unadjusted means	$+0.334$	$+0.153$
Controls + strata FE	$+0.297^{***}$	-0.110
	(0.072)	(0.068)
N (panel students)	3,092	3,092

Notes: Panel sample: 3,092 students measured at both Year-2 and Year-3 endlines (G7+G9 at Y3, drawn from the G6 and G8 Year-2 cohorts). Y2 θ_z is the Year-2 endline IRT score; Y3 θ_z^{linked} is the Year-3 IRT score linked to the Year-2 scale (Mean–Sigma, 45 anchor items). *Panel A* reports ability levels by wave; *Panel B* reports the treatment-control difference at each wave (*unadjusted* is the simple difference of means; *controls + strata FE* adds student covariates—gender, age, SES, parent education, within-school $\times$ grade Y0 baseline percentile—and randomization strata fixed effects, with standard errors clustered at school). Within-arm Year-2-to-Year-3 changes are not reported: 46 of the 70 Year-3 test items are physically the same questions panel students answered at the Y2 endline, so familiarity inflates both arms' raw Y3 scores; the inflation is shared and cancels in the $T-C$ comparison reported in Panel B but makes within-arm changes difficult to interpret as one year of ability growth. The covariate-adjusted change in the $T-C$ gap from Y2 to Y3 is -0.21 SD ($p < 0.05$) and is the formal test of year-on-year attenuation. The unadjusted Y2 T–C on the panel (0.334) is below the headline pooled ITT (0.39 SD) because the panel restricts to two of the four Year-2 cohorts: it drops the highest-effect G7-at-Y2 cohort (+0.51 SD; Section V), reassigned to the tablet-vs-paper measurement-equivalence sub-study at Y3, and the G9-at-Y2 cohort, which aged out of the eligible Y3 sample. ** $p < 0.05$; *** $p < 0.01$.

Coordinator, and one DNO per district—by AY 2025–26 was responsible for approximately 524 regular middle schools spanning all 26 districts and a roughly 700-school KGBV residential expansion that the state launched in January 2025, a near-threefold increase in the school count under central oversight against an unchanged team and the same per-school 30-tablet hardware base now visibly aging. The contracted FMS workforce experienced a contract gap from the close of AY 2024–25 through late July 2025, with hiring resuming in batches and remaining roughly a quarter below plan through December 2025; effective field presence was below the Year-2 baseline for most of AY 2025–26. Device replacements in the original 524 schools, including the RCT treatment arm, did not begin until December 2025, layered on the 444-school nine-month attrition documented above. Tablet attrition therefore extends past the program rollout into the RCT treatment arm: in the Year-3 HM survey, 52% of RCT treatment headmasters report a tablet shortage, against 82% in residential schools and 85% in program schools. With monitoring contracting, FMS field presence below the Year-2 baseline, and hardware degrading, teachers exercised discretion. Across 164 PAL-school teachers surveyed at Year-3, 29% report having reduced (18%) or canceled (11%) PAL sessions over the year; among those, 75% cite competing school activities (parent meetings, events), 40% syllabus-completion pressure, 36% technical issues with tablets, electricity, or internet, and 23% extra revision delivered through the regular math period. Teacher discretion is uneven across arms: in schools where the HM still reports PAL is scheduled at the full two periods per week, 45% of RCT treatment teachers nonetheless report delivering less, against 24% in program schools and 14% in residential schools.

Students report the same contraction. At the Year-2 endline, treatment students reported a mean of 2.55 PAL labs per week against the scheduled target of two, with 91% reporting two or more labs and only 0.6% reporting none. By the Year-3 endline one year later in the same RCT treatment arm, 53% of students report PAL ran every week, 29% some weeks, and 19% either before exams only (9%) or not at all (10%). The drop is sharper in the program rollout sample, where only 33% of students report PAL ran every week and 31% report it ran only before exams or not at all. KGBV residential students sit at the high-delivery end (69% every week; 12% before exams only or not at all). When students who reported less-than-every-week were asked why, the dominant cited reasons were hardware: tablets were not working (31% RCT treatment, 32% program, 24% residential) and not enough tablets for all students (34%, 36%, 45% respectively); 21% of RCT treatment students additionally report that "no PAL period happened" that week—the lab simply did not run. The student-reported decline tracks the teacher-reported reductions and the working-tablet attrition.

The within-school dose-response shape generalizes beyond the experimental sample: in KGBV residential schools operating under a different governance configuration, the curve is concave (Figure 13) with the same shape as the trial, and the local slope at low cumulative hours matches the trial 2SLS slope of +0.011 SD per hour. The dose at which that return is delivered, however, does not generalize. The 444-school program rollout sits below the activation region of the curve by structural design—a 4.4:1 student-to-tablet ratio with math allocated two of four weekly PAL periods—and recent hardware attrition has compressed delivery further still. Within the original RCT treatment arm, schools whose Year-1 and Year-2 implementation was already low collapsed to zero in a Year-3 setting where the delivery infrastructure contracted across multiple layers, narrowing the year-on-year $T-C$ gap from +0.33 SD at Year 2 to +0.15 SD at Year 3—though the cumulative two-year experimental advantage persists in levels. Translating the per-hour return into learning at program scale therefore depends on the integrity of the delivery infrastructure—state monitoring, contracted field workforce, working-tablet stock, and tablet-time allocation across subjects—rather than on changes to the intervention design itself.

VIII. Discussion

This paper contributes three causal estimates to the literature on adaptive learning at government scale. The first, identified by school-level randomization, is a pooled intent-to-treat effect of +0.39 SD on math achievement after two years of state-delivered exposure. The second, identified by the interaction of treatment status with class-section enrollment under a binding 30-tablet-per-school supply constraint, is a per-hour causal return of 0.011 SD. The third, descriptive within the treatment arm, has two parts: a concave dose-response in cumulative platform hours (flat below 10, steep 10–30, flattening above; Figure 10) and a within-quintile compliance gradient in which endline scores rise with engagement at every baseline-ability level, largest at the bottom quintile (Figure 8).

Our +0.39 SD treatment effect is in the range of recent adaptive-learning evaluations in low- and middle-income countries. The Rajasthan in-school adaptation of Mindspark produced +0.22 SD in math over 18 months (Muralidharan and Singh, 2025); the Khan Academy field experiment in 83 Uttar Pradesh boarding schools produced +0.44 SD over 31 weeks (Oreopoulos et al., 2026). Our 2SLS per-hour return of 0.011 SD—obtained by anchoring the dose-response at zero usage and identifying off the interaction of treatment status with section enrollment under the 30-tablet cap—sits within the range these studies span and is the cross-study-comparable per-hour figure. A complementary translation maps the effect onto the foundational-skill classification used by the Annual Status of Education Report (ASER; ASER Centre, 2024), the standard reference for arithmetic mastery in rural India. ASER reports for rural Andhra Pradesh Grade 8 that 48% of students can perform basic division (down from 52% in 2022), 30.9% can subtract but not divide, and 14.2% remain at the "recognise 11–99" floor. Replicating this classification on our endline (Appendix Table A1) places the Grade 7–8 control share with both subtraction and Grade-3 division at 54.8%, rising to 61.5% under treatment—a 6.7-pp gain that more than reverses the 4-pp decline ASER documents for rural AP Grade 8 between 2022 and 2024. Our intervention was delivered by AP government teachers and schools through the regular state administration, supported by approximately one external field coordinator per twelve schools.

The heterogeneity in our gains points to engagement, not precision routing, as the operative channel. The platform's per-chapter diagnostic gate routes lower-baseline students to slightly more below-grade content—24.2% of modules at Q1 against 22.1% at Q5—a small differential that cannot on its own account for the +0.48 SD effect at Q1 against +0.31 at Q5. The within-quintile compliance gradient does more work: top-tercile compliers outperform bottom-tercile by $+0.3$ to $+0.5$ SD across every quintile, with the largest gap at Q1. Item-level effects are larger on harder items and at every content grade level, with the largest differential on Grade 4–6 content—consistent with the program helping students reach material they would otherwise leave incorrect rather than reinforcing material they already had right. We test directly whether the platform's two-grade-below-enrolled remediation cap binds. A pre-registered within-school experiment embedded in 20 high-usage treatment schools across two academic years randomly assigned students in grades 6–9 either to the standard platform configuration or to a variant that allowed remediation up to five grade levels below enrolled grade (Cupito et al., 2026, in preparation). The dose contrast is meaningful: students in the deeper-cap arm logged roughly two additional hours per year on content the standard configuration does not render. The aggregate learning effect on the endline math assessment is small ($+0.05$ SD, $p = 0.09$), concentrated in the Grade 6 cohort and the lowest pre-PAL math tercile, and the Grade 6 gain registers specifically on items one to two grades below enrolled grade rather than at the deepest rendered content level. Deeper remediation has bottom-loaded returns where prerequisite gaps are deepest but does not detectably help students whose foundations are already adequate to access at-grade content with platform-supplied scaffolding. Grade 6 students gained from deeper remediation after one academic year of the experimental contrast, while the Grades 7–9 cohorts show no differential gain after two academic years. The standard configuration's diagnostic gate and completion-based remediation taper, given two academic years of exposure, deliver enough below-grade content to close the foundational-gap channel for the modal middle-school student. The Grade 6 concentration is over-determined: this cohort has the shortest cumulative experimental exposure and the largest deeper-remediation dose under the experimental contrast, and our data are consistent with each as a contributing channel without fully adjudicating among them.

Three threads of evidence place depth of targeting as one component, not the engine, of these programs' effects. Even without PAL, foundational skills accrue through ordinary grade-level instruction: among 711 Grade-7 control panel students re-tested as Grade-8s one year later, the share answering Grade-3 division anchors correctly rises from 70.9% to 82.3%. Our cap-bind experiment shows that allowing remediation up to five grades below produces gains only at Grade 6 and the lowest pre-PAL tercile, not for the modal student. The recent literature points the same way: experimental isolation of personalization within Mindspark gives a null average effect (de Barros and Ganimian, 2024); cross-study variation in TaRL evaluations tracks who delivers, time allocated, and monitoring intensity rather than depth of level-matching (Ardington et al., 2026); the TaRL approach decomposes into nine components of which level-alignment is one (Kaffenberger et al., 2026). For state adoption this matters: the 2-grade-back design is operationally simpler than the 4–5 grade configurations of the comparison studies and produces gains in the same range.

The Year-3 evidence in Section VII qualifies the persistence of these effects. In panel students whose Year-1 and Year-2 implementation continued, the Year-2 treatment premium of $+0.33$ SD narrows to $+0.15$ SD at Year 3; in the KGBV residential sample where Year-3 delivery was sustained, the within-school dose-response shape reproduces. The attenuation is not evidence that the program, the schools, or the teachers have failed—it tracks a Year-3 contraction across multiple layers of the delivery infrastructure that had produced Year-1 and Year-2 dose. The system has not yet built the redundancy to deliver dose autonomously when its central layers contract in tandem.

The per-student two-year delivery cost of $52 places this program within the published range of adaptive learning interventions on a cost-effectiveness basis. At program scale, however, students average approximately 3.5 hours of math PAL per year (Table 7) rather than the $\sim$ 18 hours per year accumulated in our evaluation schools, and the cross-sample evidence reported in Section VII shows that the per-hour return is preserved across delivery contexts that successfully move students into the activation region of the dose-response. The marginal policy investment that would produce additional learning is therefore reallocation toward more engaged hours per student rather than further refinement of the platform's adaptive design: smaller per-tablet ratios, longer cumulative exposure within the activation-to-saturation band of the dose-response, and more reliable hardware delivery in schools where the program currently produces little usage. The Kasturba Gandhi Balika Vidyalaya residential schools reported in Section VII demonstrate that high-dose PAL delivery within the AP government system is operationally feasible. The harder margin is sustaining that dose in regular government schools, where shared tablet stock across subjects, structurally larger cohorts, and accumulating hardware attrition together compress per-student exposure to a level below the activation region of the dose-response curve.

Year-3 reveals an asymmetry in how the state allocated investment. While the FMS contract was lapsing and tablet replacements in the original 524 schools were deferred until December 2025, the state launched a 700-school KGBV residential expansion in January 2025 and added 278 PMShri schools in December 2025—an instance of ribbon-cutting bias playing out in education delivery (Glaeser and Poterba, 2021). The state's coordination bandwidth was itself a casualty: the same small State PAL coordination unit that ran the program at 444 schools at trial start was responsible for roughly 1,200 schools by Year 3—premature load-bearing in the sense of Andrews et al. (2017). The Year-3 dose distribution tracks the cost: trial-arm dose fell, program-rollout schools sit below the activation region of the dose-response, and only the newly launched KGBV residential schools—where the state's expansion energy was directed—operate at a dose comparable to what the trial demonstrated. Translating the per-hour return into delivered learning at scale therefore depends on whether the maintenance side of the investment ledger scales alongside the adoption side.

At the school level, the asymmetry surfaced as teacher-level discretion. Across 164 PAL-school teachers surveyed at Year-3, 29% report having reduced or canceled PAL sessions over the year, citing competing school activities (75%), syllabus-completion pressure (40%), and hardware or electricity issues (36%)—the resource constraints and accountability slack that the literature on street-level discretion documents (Lipsky, 2010). Discretion is uneven across arms: in schools where the headmaster reports PAL is scheduled at the full two periods per week, 45% of RCT treatment teachers nonetheless report delivering less, against 24% in program schools and 14% in residential schools—tracking inversely with the layers of monitoring still operating around each arm. The trial-arm chronic tail's dose collapse is the proximate consequence of these decisions, taken in classrooms where multiple upstream supports were degrading at once.

IX. Conclusion

Adaptive learning programs have produced large measured math gains in NGO-run efficacy trials. Whether such programs can deliver gains of similar magnitude when funded, staffed, and operated by a state government using its existing schools and teachers has been the central open question for the field. Our results suggest that they can, when the state sustains the delivery infrastructure that produces engaged hours per student.

Two years of PAL exposure raised endline math ability by $+0.39$ SD, with broad gains across grade, gender, and baseline ability, and a per-hour causal return of $+0.011$ SD identified by treatment$\times$section-enrollment variation under the 30-tablet supply constraint. Cost-effectiveness, at \$52 per student over two years, is in the range reported for adaptive-learning programs in other settings.

These gains were generated under a delivery infrastructure that operated in tandem during the experimental window: sustained state-level monitoring of school-level implementation (weekly usage reporting through the State PAL Coordinator's WhatsApp channel, scheduled review meetings with district nodal officers and headmasters, real-time dashboards), a field-staff workforce visiting schools at a ratio of approximately one staff member per twelve schools, and a tablet stock that was new at the start of the trial. The Year-3 follow-up reported in Section VII shows that effects attenuate where these layers contracted in tandem, while the per-hour return reproduces wherever the program continues to be delivered.

At program scale, where 30 tablets are shared across three subjects and four grades, students currently average approximately 3.5 hours of math PAL per year (Table 7) against approximately 18 hours per year in our evaluation schools—below the activation region of the within-school dose-response. The marginal policy investment that would translate the per-hour return identified here into program-scale learning is therefore reallocation of scarce tablet-time toward more engaged hours per student, not more sophisticated personalization.

What this experiment establishes is that the operational machinery of a state government can deliver large adaptive-learning gains in its own classrooms. What it does not yet establish is whether the state will maintain the delivery infrastructure—technical-support workforce, hardware, and central coordination bandwidth—at the same pace as it expands to new schools. The open question for state-led adaptive learning at scale is that asymmetry between adoption and maintenance.

References

Andrabi, Tahir, Jishnu Das, and Asim Ijaz Khwaja, "Report Cards: The Impact of Providing School and Child Test Scores on Educational Markets," *American Economic Review*, 2017, *107* (6), 1535–1563.

Andrews, Matt, Lant Pritchett, and Michael Woolcock, *Building State Capability: Evidence, Analysis, Action*, Oxford University Press, 2017.

Angrist, Noam, David K. Evans, Deon Filmer, Rachel Glennerster, F. Halsey Rogers, and Shwetlena Sabarwal, "How to Improve Education Outcomes Most Efficiently? A Comparison of 150 Interventions Using the New Learning-Adjusted Years of Schooling (LAYS) Metric," Technical Report Policy Research Working Paper 9450, World Bank, Washington, DC 2020.

Ardington, Cally, Alicia Menendez, and Jack Thunde, "The Limits of a 'Great Buy': What Rigorous Evidence Reveals about Teaching at the Right Level," Technical Report, African Foundational Learning Data Hub (AFLEARN), DataFirst, University of Cape Town January 2026.

ASER Centre, "Annual Status of Education Report (Rural) 2024," Available online from ASER Centre (Pratham) 2024. Nationwide household survey of rural learning outcomes and enrollment in India.

Banerjee, Abhijit, Rukmini Banerji, James Berry, Esther Duflo, Harini Kannan, Shobhini Mukerji, Marc Shotland, and Michael Walton, "From Proof of Concept to Scalable Policies: Challenges and Solutions, with an Application," *Journal of Economic Perspectives*, 2017, *31* (4), 73–102.

Banerjee, Abhijit V., Shawn Cole, Esther Duflo, and Leigh Linden, "Remedying Education: Evidence from Two Randomized Experiments in India," *The Quarterly Journal of Economics*, 2007, *122* (3), 1235–1264.

Barnett, Inka, Proochista Ariana, Stavros Petrou, Mary E Penny, Le Thuc Duc, S Galab, Tassew Woldehanna, Javier A Escobal, Emma Plugge, and Jo Boyden, "Cohort profile: the Young Lives study," *International journal of epidemiology*, 2013, *42* (3), 701–708.

Beg, Sabrin, "Digitization and development: Property rights security, and land and labor markets," *Journal of the European Economic Association*, 2022, *20* (1), 395–429.

Bold, Tessa, Mwangi Kimenyi, Germano Mwabu, A Ng'ang'a, and J Sandefur, "Experimental evidence on scaling up education reforms in Kenya," *Journal of Public Economics*, 2016, p. 168.

Cupito, Emily, Alex Eble, Guthrie Gray-Lobe, Saloni Gupta, Michael Kremer, Sabareesh Ramachandran, and Wendy Wong, "Testing Deeper Remediation: A Within-School RCT of Personalized Adaptive Learning in India," 2026. Working paper, in preparation.

de Barros, Andreas and Alejandro J. Ganimian, "Which Students Benefit from Computer-Based Individualized Instruction? Experimental Evidence from Public Schools in India," *Journal of Research on Educational Effectiveness*, 2024, *17* (2), 318–343.

Duflo, Esther, Rema Hanna, and Stephen P. Ryan, "Incentives Work: Getting Teachers to Come to School," *American Economic Review*, 2012, *102* (4), 1241–1278.

Filmer, Deon P., F. Halsey Rogers, Noam Angrist, and Shwetlena Sabarwal, "Learning-Adjusted Years of Schooling (LAYS): Defining a New Macro Measure of Education," Technical Report Policy Research Working Paper 8591, World Bank, Washington, DC 2018.

Glaeser, Edward L. and James M. Poterba, "Economic Analysis and Infrastructure Investment," in Edward L. Glaeser and James M. Poterba, eds., *Economic Analysis and Infrastructure Investment*, University of Chicago Press, 2021.

Global Education Evidence Advisory Panel, "Cost-Effective Approaches to Improve Global Learning: Smart Buys 2023," Technical Report, World Bank and Foreign, Commonwealth & Development Office, Washington, DC 2023.

Gray-Lobe, Guthrie, Anthony Keats, Michael Kremer, Isaac Mbiti, and Owen Ozier, "Can Education be Standardized? Evidence from Kenya," BFI Working Paper 2022-68, Becker Friedman Institute for Economics 2022.

Kaffenberger, Michelle, Noam Angrist, Yue-Yi Hwa, Heather Leigh Kayton, Matthew Jukes, and Jonathan Stern, "Core components of teaching at the right level: Unpacking the black box of proven programmes into a set of 'core components' by systematically combining multiple sources of rigorous evidence with implementer insights," Working Paper 2026/001, What Works Hub for Global Education 2026.

Lipsky, Michael, *Street-level Bureaucracy: Dilemmas of the Individual in Public Services*, 30th anniversary expanded ed., Russell Sage Foundation, 2010.

Muralidharan, Karthik and Abhijeet Singh, "Adapting for Scale: Experimental Evidence on Technology-Aided Instruction in India," NBER Working Paper 34205, National Bureau of Economic Research September, revised November 2025.

_ **and Alejandro J. Ganimian**, "Disrupting education? Experimental evidence on technology-aided instruction in India," *American Economic Review*, 2019, *109* (4), 1426–1460.

_ **and Venkatesh Sundararaman**, "Teacher Performance Pay: Experimental Evidence from India," *Journal of Political Economy*, 2011, *119* (1), 39–77.

_ , Paul Niehaus, and Sandip Sukhtankar, "Building state capacity: Evidence from biometric smartcards in India," *American Economic Review*, 2016, *106* (10), 2895–2929.

_ , _ , _ , and Jeffrey Weaver, "Improving last-mile service delivery using phone-based monitoring," *American Economic Journal: Applied Economics*, 2021, *13* (2), 52–82.

Okunogbe, Oyebola and Victor Pouliquen, "Technology, taxation, and corruption: evidence from the introduction of electronic tax filing," *American Economic Journal: Economic Policy*, 2022, *14* (1), 341–372.

Oreopoulos, Philip, Oliver Keyes-Krysakowski, and Deepak Agarwal, "How In-School Supervised Ed-Tech Support Produces Massive Learning Gains: A Khan Academy Field Experiment in India," NBER Working Paper 34683, National Bureau of Economic Research January 2026.

Singh, Abhijeet, "Improving administrative data at scale: Experimental evidence on digital testing in Indian schools," *The Economic Journal*, 2024, *134* (661), 2207–2223.

TABLE A1—ASER-EQUIVALENT CLASSIFICATION, CONTROL VS TREATMENT ENDLINE

	Subtract, not divide	Cannot subtract	Both ($\geq$ G3 arithmetic)	n
Y2 grade 7–8 control endline	16.4%	28.8%	54.8%	3,609
Y2 grade 7–8 treatment endline (2yr PAL)	13.0%	25.5%	61.5%	3,368
Treatment − Control	*−3.5 pp*	*−3.3 pp*	*+6.7 pp*	—
ASER 2024 AP grade 8 (ASER Centre, 2024)	30.9%	14.2%	48%	—

Notes: Subtraction anchor: `g2q3` (12−4 word problem). Division anchor: any of `g3q3` (250 ÷ 10) or `g3q5` (65 ÷ 5). ASER "11–99" floor refers to students who can recognize 11–99 but not subtract.

Appendices

A. Translating the ITT into the ASER 2024 Classification

This appendix maps the items in our Math Master Challenge to the foundational-skill classification used by the ASER survey, replicates the classification on our control endline, and re-expresses the two-year ITT in the same categorical language used by ASER and by national policy audiences.

ASER Centre (2024) reports for rural Andhra Pradesh grade 8 that 48% of students can perform basic division (down from 52% in 2022), 30.9% can subtract but cannot divide, and 14.2% are stuck at the "recognize 11–99" floor. We replicate the classification on our control endline samples to anchor our control arm's foundational-skill profile against this published reference, and translate the two-year ITT into the same categorical language.

Anchor items. We identify items in our test by *operation content* rather than item-grade label, since calibrated IRT difficulty in our instrument is not monotonic in the question's grade tag. The subtraction anchor (`g2q3`, "Lata has 12 chocolates and gives 4 to her friend") is a word problem without borrowing. The division anchors (`g3q3`, 250 ÷ 10; `g3q5`, 65 ÷ 5) are clean two-digit-by-one-digit divisions without remainders, materially easier than ASER's three-digit-by-one-digit-with-remainder protocol.

Headline. Two years of PAL raise the share of treated grade 7–8 students with full grade-3 arithmetic mastery from 54.8% in control schools to 61.5% in treatment schools, a 6.7-percentage-point gain that more than reverses the 4-percentage-point decline in basic-division mastery ASER documents for rural AP grade 8 between 2022 and 2024 (52% → 48%). The shift breaks down into roughly equal parts: −3.3 pp move off the "cannot subtract" floor, and −3.5 pp clear the subtraction-to-division cliff—both flowing into the "both-operations" bucket.

Why control levels differ from ASER. Our control "both-operations" rate (54.8%) is 6.8 pp higher than ASER's "can divide" rate (48%), consistent with our division anchor being easier than ASER's. Our higher "cannot subtract" rate (28.8% vs 14.2% at ASER's 11–99 floor) is consistent with two contrasts: our 100%-government-school sample lacks the higher-performing rural private-school enrollees ASER's household-based survey includes, and our subtraction item is a word problem requiring reading comprehension while ASER's is bare arithmetic.

TABLE B1—DISTRIBUTION OF ASSESSMENT ITEMS BY CONTENT GRADE AND TOPIC

Content grade	Algebra	Geometry	Measurement	Number Systems	Statistics	Total
Grade 2	1	2	1	5	1	10
Grade 3	2	2	2	3	1	10
Grade 4	2	2	2	3	1	10
Grade 5	2	2	2	3	1	10
Grade 6	0	2	2	4	2	10
Grade 7	1	2	2	3	2	10
Grade 8	2	1	1	1	1	6
Grade 9	2	1	1	1	1	6
Total	12	14	13	23	10	72

Notes: Topic classification follows the Indian NCERT curriculum convention, where Number Systems covers arithmetic on specific number types (whole numbers, fractions, decimals, integers) and Algebra begins with variables and unknowns. Students in grade k answer all items from grade 2 through grade k, yielding 50 items (grade 6) to 72 items (grade 9).

B. Development of the Test Instrument

A. Item Development and Piloting

We developed the Math Master Challenge assessment to measure mathematics achievement across the wide range of ability levels in our sample. Items were drawn from three sources: (i) all questions from every chapter of the Andhra Pradesh state math textbooks for grades 2 through 9; (ii) grade-specific math items from the Young Lives Study (Barnett et al., 2013); and (iii) items from a previous evaluation of adaptive learning technology in India (Muralidharan and Ganimian, 2019). From this bank, we selected an initial set of items targeting 10 questions per grade for grades 2–7 and 6 questions per grade for grades 8–9, with two items per topic per grade across five broad mathematical domains: algebra, geometry, measurement, statistics, and number systems.

We piloted the instrument in five schools over the course of one week. Each day, we downloaded the pilot data and estimated a two-parameter logistic (2PL) IRT model, dropping items that exhibited poor discrimination (low slope parameters) or extreme difficulty, with the goal of constructing an instrument whose test information function was sufficiently wide to capture the range of learning levels in our sample.

B. Final Instrument

The final assessment comprises 72 unique multiple-choice items spanning the grade 2 through grade 9 curriculum. Each student answered items from grade 2 up to the content level of their enrolled grade, yielding between 50 items (grade 6 students) and 72 items (grade 9 students). The assessment was administered on tablets via SurveyCTO in both English and Telugu, with three randomized question orderings (set assignment uniform at $P = 0.33$ per student) to mitigate order and copying effects. Students completed the assessment in approximately 40–50 minutes on average.

Table B1 shows the distribution of items across content grades and topics. Items from grades 2–7 have 10 questions each, while grades 8–9 have 6 questions each. Number systems has the most items (23), reflecting its central role in the Indian mathematics curriculum from basic arithmetic through rational numbers.

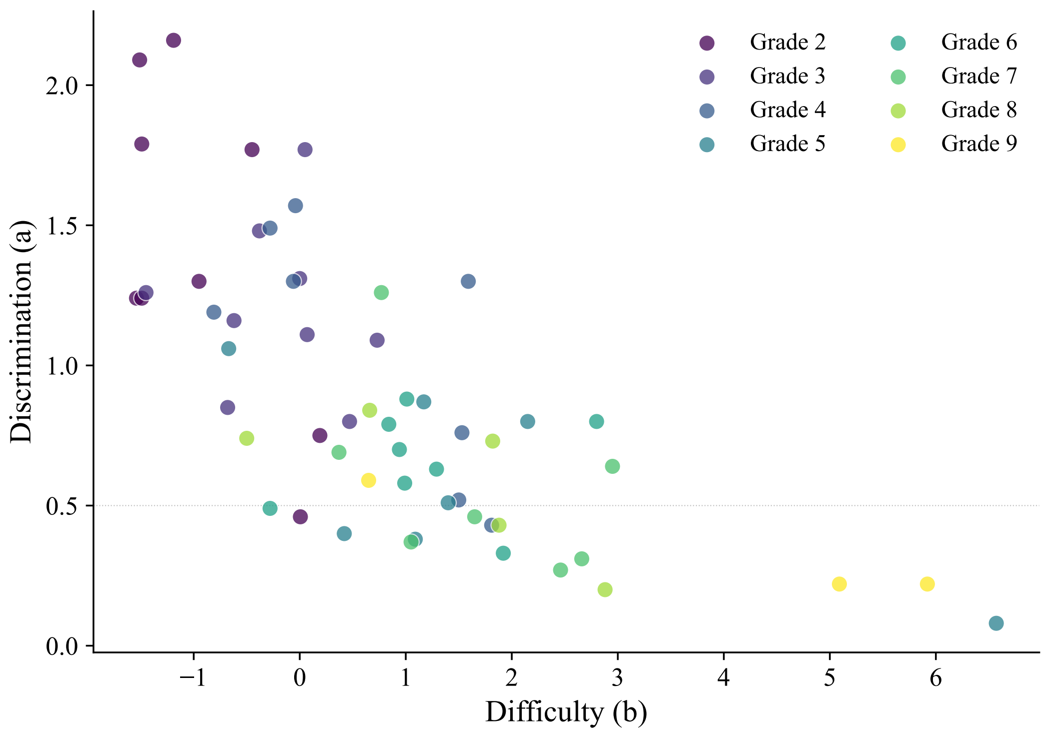

FIGURE B1. ITEM DISCRIMINATION AND DIFFICULTY PARAMETERS

Notes: Each point represents one of the 72 items on the Math Master Challenge, colored by the curriculum grade from which the item was drawn. Discrimination (a) reflects how well the item differentiates between students of different ability levels; difficulty (b) indicates the ability level at which a student has a 50% probability of answering correctly. Parameters estimated via a 2PL IRT model. The dashed line at $a = 0.5$ marks a common threshold for acceptable discrimination.

C. *Psychometric Properties*

We estimate a 2PL IRT model, which allows both discrimination (a_j) and difficulty (b_j) parameters to vary across items. Under this model, the probability that student i answers item j correctly is:

$$P(X_{ij} = 1 \mid \theta_i) = \frac{1}{1 + \exp\left(-a_j(\theta_i - b_j)\right)} \tag{4}$$

where θ_i is the latent ability of student i. We standardize the estimated ability parameters to the control group mean and standard deviation, following Muralidharan and Ganimian (2019) and Muralidharan and Singh (2025).

Across all 72 items, discrimination parameters range from 0.08 to 2.16 (mean = 0.88), and difficulty parameters range from -1.54 to 6.57 (mean = 0.84). Lower-grade items are easier and more discriminating (grade 2: mean $a = 1.42$, mean $b = -0.94$), while higher-grade items are harder with lower discrimination (grade 9: mean $a = 0.34$, mean $b = 3.89$), reflecting the intended difficulty progression across the curriculum. Figure B1 displays the joint distribution of discrimination and difficulty parameters, colored by content grade.

Internal consistency is high across all grade levels: Cronbach's α ranges from 0.90 (grade 6, 50 items) to 0.92 (grade 9, 72 items).

Figure B2 presents the test information function by enrolled grade. The instrument provides the most precise measurement in the ability range $\theta \in [-2, 1]$, with peak information near $\theta \approx -0.7$. The standard error of measurement remains below 0.35 across most of the ability distribution. Higher-grade students receive additional items that extend measurement precision into the upper ability range.

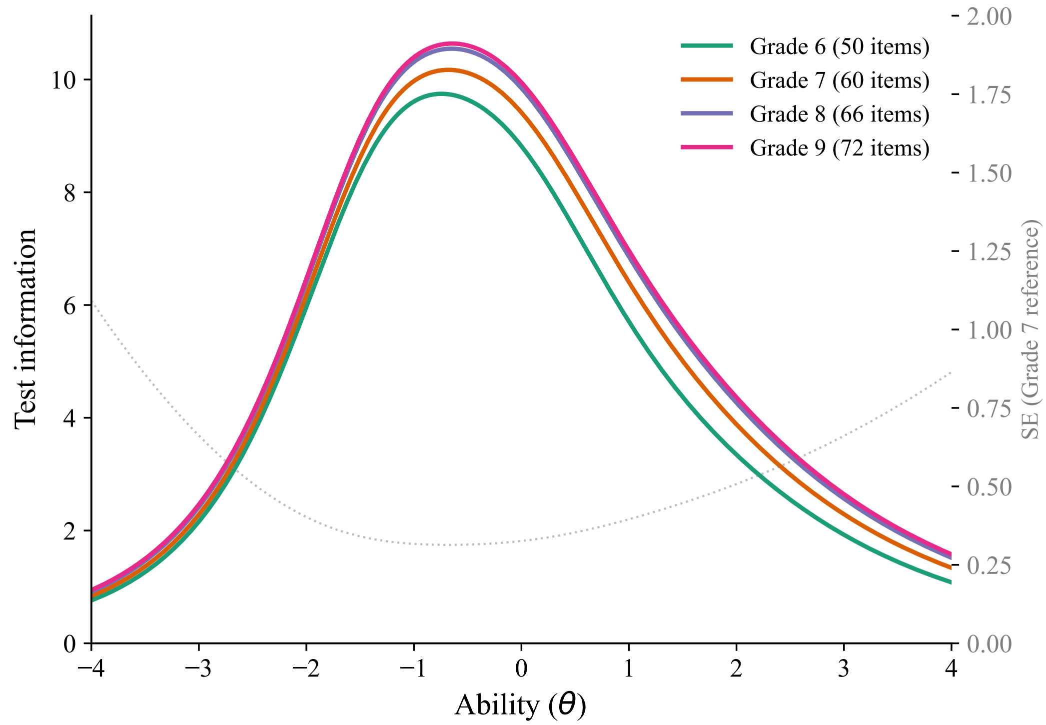

FIGURE B2. TEST INFORMATION FUNCTION BY GRADE

Notes: Test information $I(\theta) = \sum_j a_j^2 P_j(\theta)(1 - P_j(\theta))$ for each enrolled grade's item set. Higher information indicates more precise measurement at that ability level. The dashed gray line shows the corresponding standard error $\text{SE}(\theta) = 1/\sqrt{I(\theta)}$ for grade 7 as a reference. Students in higher grades answer additional items from higher curriculum levels, extending the instrument's measurement range.

C. Tablet vs. Paper Assessment-Mode Equivalence

A natural concern with our tablet-administered endline is that PAL exposure may inflate measured ability through tablet familiarity rather than learning, biasing the headline ITT upward. We test this directly using a sub-study embedded in the Year-3 endline (AY 2025–26) that re-tested the *same* Grade 8 RCT students on *the same items* using both modes. Each of 1,953 Grade 8 students in our 80 RCT schools sat both a tablet (SurveyCTO) and a paper (Optical Mark Recognition, OMR) Math Master Challenge in randomized order, with a counter-balanced crossover assigning opposite question sets to the two modes (Tab Set 1 students received OMR Set B and vice versa). Twenty-six "common" items appear on both modes for every student, providing within-student same-item measurement; 40 set-specific items contribute cross-form information through the IRT.

Table C1 reports treatment-effect specifications analogous to the main paper's Table 3, with mode (tablet vs. paper) and its interaction with treatment as the key terms. Panel A uses 2PL IRT θ from a pooled fit across all items administered in the dual-mode design; Panel B restricts the IRT to the items every student answered on both modes (a fully balanced design that does not rely on cross-form linking). Across all six (panel $\times$ specification) cells, the mode $\times$ treatment interaction lies between -0.09 and $+0.003$ SD and is not statistically significant in any specification. The mode main effect is at most -0.09 SD, with $p \geq 0.13$ throughout.

A model-free verification on the common items reaches the same conclusion: mean percent correct is 50.4% on tablet and 50.7% on paper, with a within-student Pearson correlation of 0.83 on per-student common-items scores—indicating coherent measurement of the same latent ability across modes. The

TABLE C1—G8 MODE COMPARISON: TABLET VS PAPER ASSESSMENT (Y3 RCT STUDENTS)

	(1) Model 1 (main)	(2) Model 1 (robust)	(3) Model 2 (1st exam)
Panel A. Primary specification: 2PL IRT on 67 items			
1.mode	0.006	0.006	-0.063
	(0.031)	(0.031)	(0.062)
1.mode × 1.treatment	-0.092	-0.094	-0.019
	(0.060)	(0.060)	(0.094)
R^2	0.914	0.913	0.095
Panel B. Robustness: 2PL IRT on 26 common items only			
1.mode	-0.038	-0.038	-0.094
	(0.030)	(0.030)	(0.062)
1.mode × 1.treatment	-0.070	-0.071	0.003
	(0.045)	(0.045)	(0.090)
R^2	0.922	0.922	0.087
N	3,906	3,886	1,953
Fixed effects	Student	Student	Mandal cluster

Notes: Outcome is θ standardized to the control × tablet reference cell (mean 0, SD 1). Panel A uses 2PL IRT θ from a pooled fit across all items administered in the dual-mode design; Panel B uses 2PL IRT θ from a fit restricted to the items every Grade 8 RCT student answered on both modes (a fully balanced spec). Models 1 use student fixed effects; Model 2 (first exam) uses mandal-cluster strata FE plus covariates (gender, age, SES, parental education with missing flags). Standard errors clustered at school. A negative interaction would indicate that treatment students gain a relative tablet advantage. * p<0.10, ** p<0.05, *** p<0.01.

DiD interaction on percent correct is -1.32 pp (SE 0.91, $p = 0.15$), consistent in sign and SD-equivalent magnitude with Panel A and Panel B.

We conclude that mode of administration does not differentially advantage treatment students on the tablet endline. The tablet-familiarity channel cannot account for the headline $+0.39$ SD ITT.

D. Implementation: Tablets, Effort, and the Chapter Pipeline

This appendix documents how PAL is actually delivered—the 30-tablet cap and within-section sharing, the split between active practice and passive viewing, student-reported implementation challenges, and the standardized six-stage module sequence that proceeds uniformly across ability levels.

Figure D1 shows the distribution of class-section sizes across the 60 treatment schools. Because every treatment school received 30 tablets regardless of section enrollment, sections above 30 students faced a mechanical sharing constraint: 44.7% of the 300 treatment sections exceed this count, with the median oversized section holding 37 students (a 1.23:1 student-to-tablet ratio). 59.6% of treatment students were in a sharing-required section.

Students spend slightly more time watching videos (54%) than answering questions (46%; Table D1). Eighty percent of treatment students report enjoying PAL "a lot," though 37% report language comprehension and video loading as implementation challenges (Table D2).

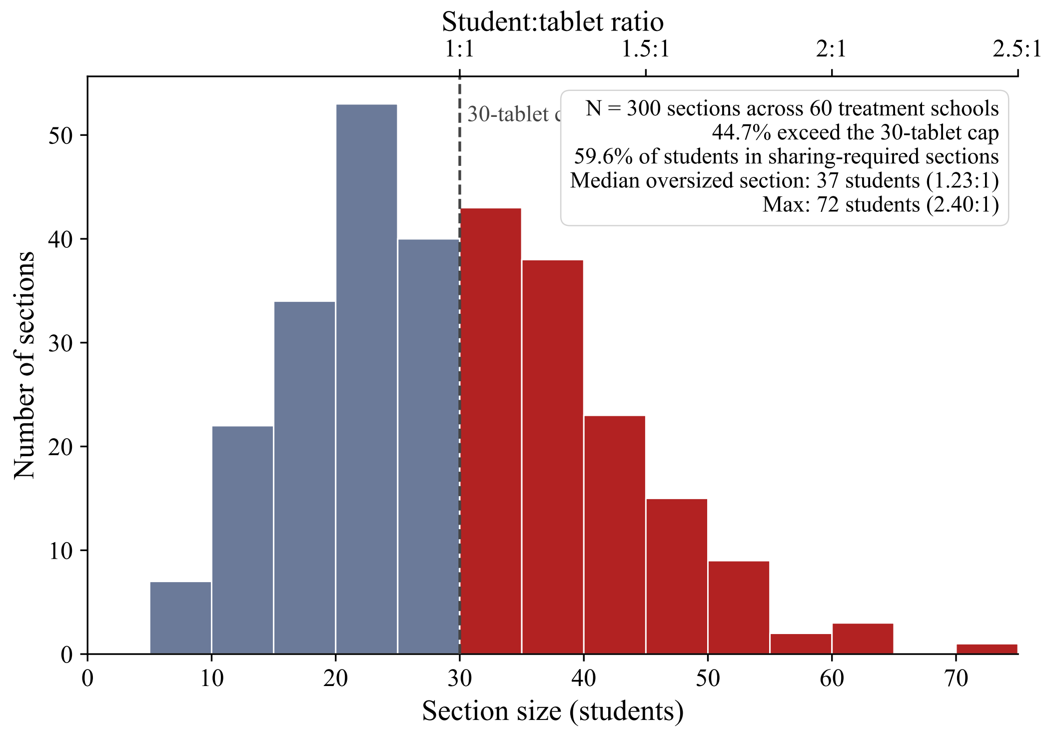

FIGURE D1. CLASS SECTION SIZES RELATIVE TO THE 30-TABLET CAP

Notes: Treatment schools only; $N = 300$ class sections across 60 schools, defined as unique school-grade-section combinations. Section enrollment was missing for 1.1% of treatment students; these were imputed using the mean section enrollment within each student's school-grade cell. Sections ≤ 30 students (blue-gray, 166 of 300) could accommodate every student on a dedicated tablet; sections > 30 (red, 134 of 300) required sharing. The top axis shows the implied student-to-tablet ratio. 59.6% of treatment students were in a sharing-required section.

TABLE D1—TIME DIVISION BY MODULE TYPE (TREATMENT GROUP ONLY)

	Overall	Grade 6	Grade 7	Grade 8	Grade 9
Total hours	34.2	20.4	39.4	39.7	35.9
Video hours	18.9	11.2	21.5	21.6	20.6
Question hours	15.3	9.2	17.9	18.1	15.3
% Video	53.6%	51.7%	52.6%	52.8%	56.9%
N	6,742	1,540	1,599	1,769	1,834

Notes: Treatment group only. Total hours is cumulative PAL usage across 2 academic years. Video hours = passive content viewing. Question hours = active practice (answering questions). Grade 6 had 1 year of PAL; grades 7–9 had 2 years.

A. *PAL Instructional Pipeline*

Using module-level platform data (8.78 million learning events across 12,019 students in the 60 treatment schools), we reconstruct the instructional pipeline that students experience within each chapter. Figure D2 shows the standardized six-stage sequence: a short diagnostic on prerequisite content, below-grade remediation (videos and practice), a baseline assessment at grade level, at-grade instruction with formative knowledge checks, and a chapter endline. The chapter sequence within each grade is fixed and identical for all students.

TABLE D2—PAL EXPERIENCE (TREATMENT GROUP ONLY)

Panel A: Student experience (% reporting)	
Enjoy PAL lab "a lot"	80.3%
PAL helped understand math "a lot"	73.2%
Like watching PAL videos	87.5%
Prefer additional PAL period over math	66.3%
PAL less difficult than regular math	57.5%
Prefer English in PAL lab	52.1%
Always log in with own credentials	74.3%
Always get teacher help to log in	34.4%
PAL classes per week, mean (SD)	2.6 (1.4)
Panel B: Implementation quality (% reporting no problem)	
Video quality is good	88.4%
Knows how to use PAL	87.1%
Enough time allotted	84.1%
Enough tablets available	78.5%
Screen easy to see	77.3%
Lab not too noisy	74.5%
Headphones work	73.8%
No login problems	70.7%
Videos load fine	63.4%
Language understandable	62.8%

Notes: Treatment group only ($N \approx 6{,}740$). Panel A: percentage of students giving the top/positive response to each PAL experience question from the endline student survey (tablet-based, administered by independent enumerators). Panel B: percentage reporting no problem for each implementation dimension (binary: 1=good/no problem, 0=has problem). The most common implementation challenges are language comprehension (37% report problems) and video loading speed (37%).

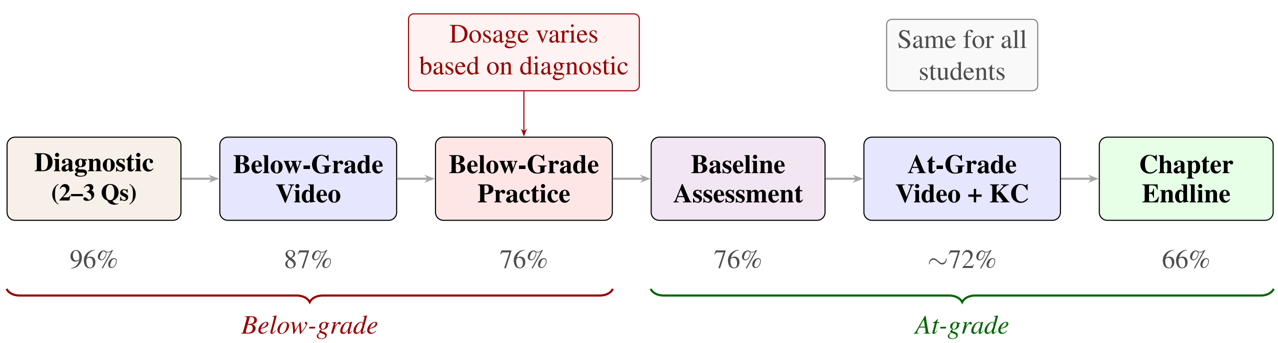

FIGURE D2. PAL CHAPTER PIPELINE

Notes: Standardized chapter pipeline based on 246,829 chapter journeys across 12,019 students. Percentages indicate the fraction of students reaching each stage. The below-grade practice stage (highlighted) is the primary locus of dosage variation; the at-grade stages are uniform across students.

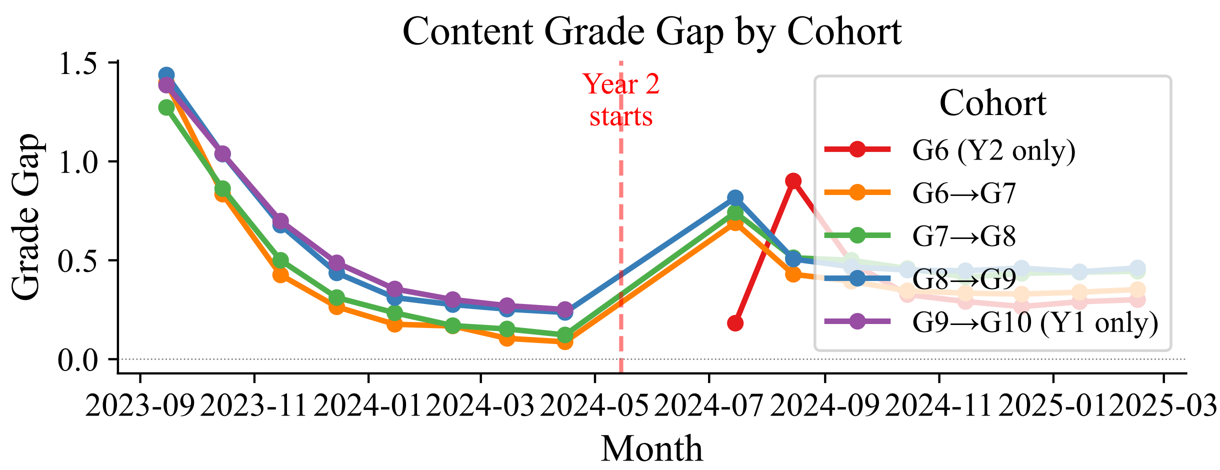

FIGURE D3. CONTENT GRADE GAP BY COHORT

Notes: Grade gap = student's enrolled grade − average content grade of modules used that month. Each line tracks a cohort across its years of PAL exposure: three cohorts (G6→G7, G7→G8, G8→G9) span both academic years; the G9→G10 cohort aged out after Year 1; the Grade 6 cohort entered in Year 2 only. $N = 10{,}920$ students across 60 treatment schools (excluding repeaters and mid-year dropouts).

TABLE D3—PAL USAGE BY BASELINE ABILITY

Baseline quintile	Total usage (hours)	Video (hours)	Practice (hours)	Video share (%)	Below-grade content (%)
Q1	33.9	19.0	14.9	57	33
Q2	33.9	19.1	14.8	56	32
Q3	32.1	18.0	14.1	56	32
Q4	35.3	19.4	15.9	55	30
Q5	35.6	19.0	16.6	54	29
All	34.2	18.9	15.3	55	31
Q5 − Q1	+1.8			-3	-4

Notes: Treatment students with non-missing baseline scores and PAL usage data ($N = 6{,}742$). Total usage is the sum of video and practice hours over two academic years (2023–24 and 2024–25). Below-grade content share is the average fraction of modules within each chapter journey that cover content below the student's enrolled grade.

Figure D3 shows the content grade progression over time, tracking five cohorts defined by their enrolled grades across the two study years. All cohorts begin approximately 1.4 grade levels below their enrolled grade and converge toward grade-level material over roughly four months. At the start of Year 2, the system resets continuing cohorts back to below-grade content regardless of Year 1 progress. The Grade 9→10 cohort appears only in Year 1 (students aged out after grade 9); the new Grade 6 cohort appears only in Year 2.

Table D3 shows that the student experience is largely uniform across the baseline ability distribution. Total usage hours, the video–practice split, and the share of below-grade content are all flat across baseline ability quintiles. The Q5–Q1 gap in video share is 3 percentage points; in below-grade content share, 4 percentage points. Figure D4 provides the most direct evidence: the content grade progression for the bottom and top terciles of baseline ability overlaps almost completely in both years. The system moves all students through the same content at the same speed.

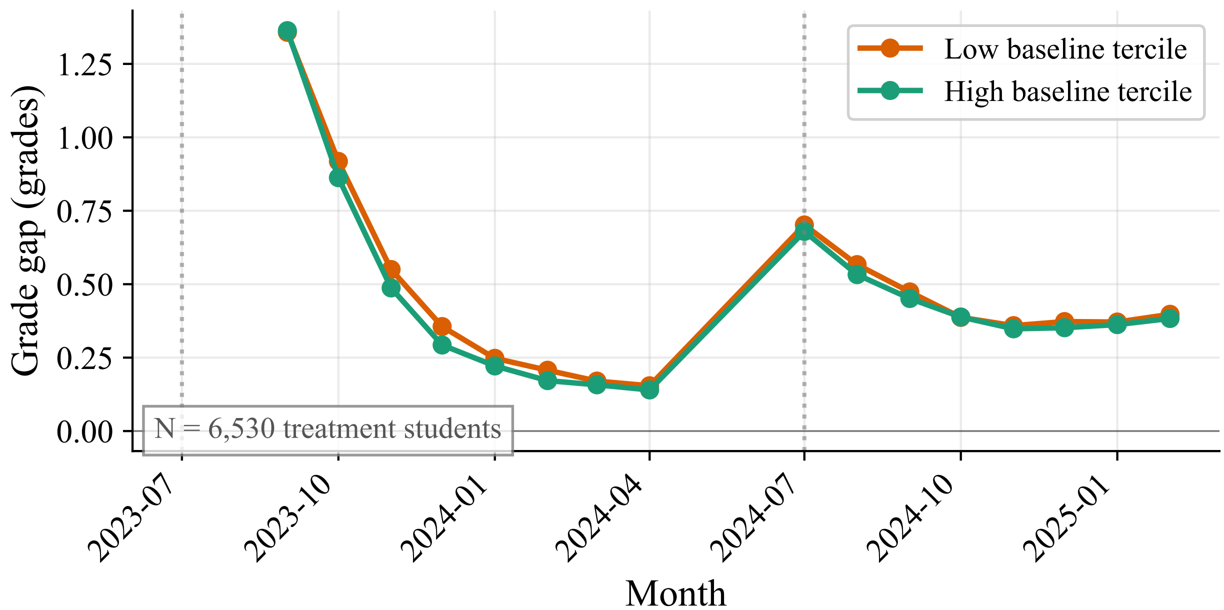

FIGURE D4. CONTENT GRADE PROGRESSION BY BASELINE ABILITY

Notes: Grade gap = student's enrolled grade − average content grade used that month. Bottom and top terciles defined by within-school × grade percentile rank of government math score. Lines overlap almost completely in both years, indicating identical progression trajectories regardless of actual ability. $N = 12{,}019$ students across 60 treatment schools.

E. Heterogeneity: Gender, Baseline Ability, and Data Coverage

This appendix resolves why girls gain more than boys (usage quantity, not per-hour productivity), documents the across-quintile gradient in baseline ability, and rules out that the gradient is a data-coverage artefact of asymmetric baseline extracts.

We examine treatment effect heterogeneity along several dimensions. Girls gain more than boys (+0.46 vs +0.33 SD), and this advantage persists across all grades (Figure I1). Effects are uniform across SES terciles ($p = 0.62$; Table E2) but decline across baseline ability quintiles, with the largest gains for the lowest-baseline students (Figure E1, Panel C). Girls accumulate more PAL usage hours and report higher engagement than boys (Figure E2); Table E1 provides the full regression results for the gender gap diagnosis.

Baseline ability in our regressions is the student's within-school × grade percentile rank on the pre-PAL government math SA1 assessment, with a cohort-specific source: grade 7–9 at endline use AY 2022–23 SA1, while grade 6 at endline (new cohort, PAL exposure only in 2024–25) use AY 2023–24 SA1—their grade-5 pre-PAL year. Using 2022–23 for grade 6 would measure them as grade 4 students two years pre-endline.

The large treatment effect for the "Missing baseline" subgroup in Table 4 (0.50 SD) reflects overlapping data-coverage issues, not a true subgroup effect, and these issues differ by cohort. For the 2-year cohort (grades 7–9 at endline), missingness reflects a *data-extraction asymmetry*: the AP government delivered AY 2022–23 data for the 524 program + treatment schools in one extract and the 60 control schools in another, with no overlap. In grade 7, this gives 87–89% coverage for treatment vs. 26–29% for control (Table E3, Panel A); in grades 8–9, coverage is balanced at 78–93% in both arms. For the 1-year cohort (grade 6 at endline), missingness is *structural and not closed by choice of baseline year*: grade 6 students are brand-new entrants to the study schools in 2024–25, so their pre-PAL records live in whatever government school they attended as grade 5. Treatment grade 6 students came from the 444 program schools (captured by the 524-school extract, ∼97% coverage), while control grade 6 students came from schools outside the

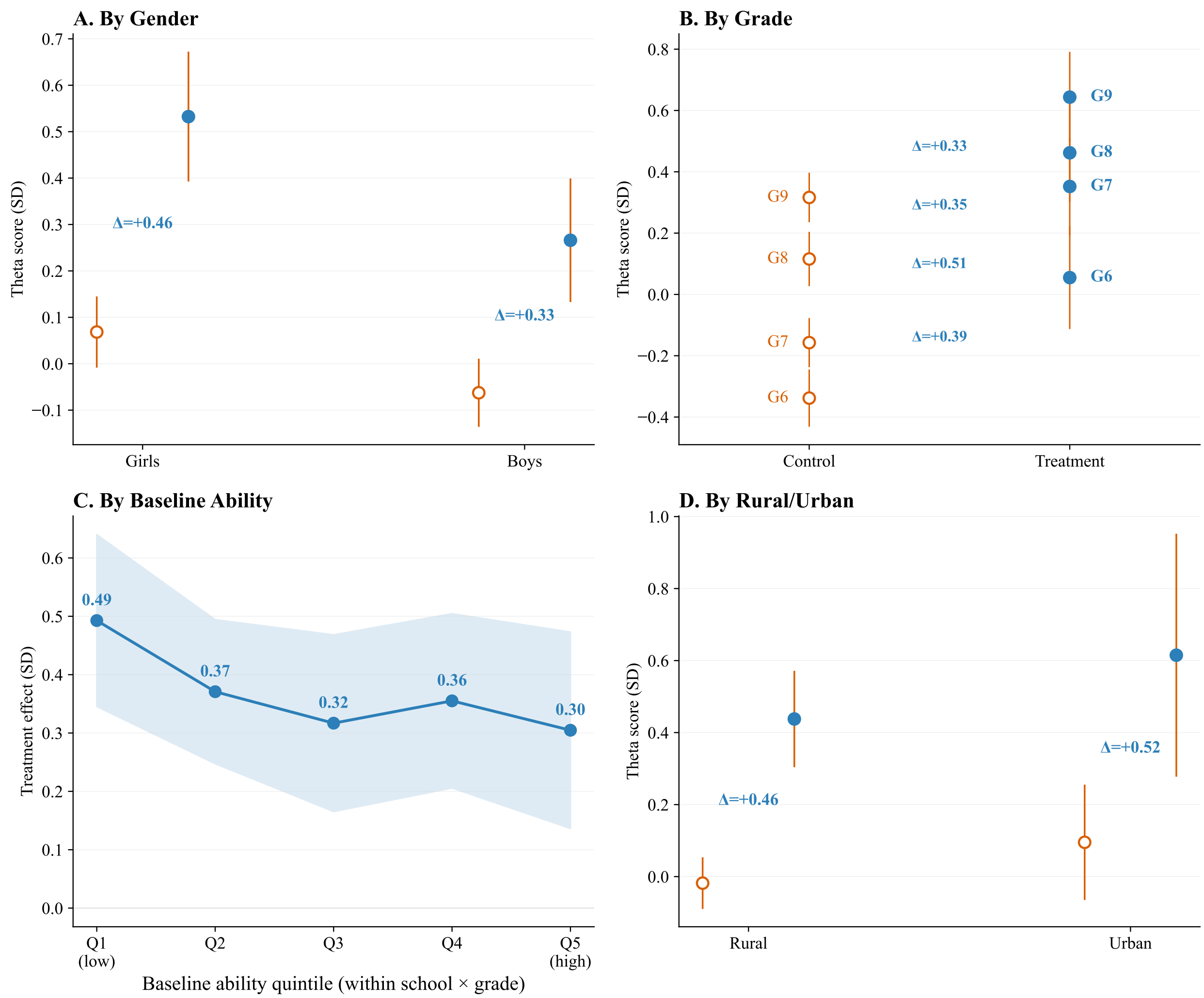

FIGURE E1. HETEROGENEOUS TREATMENT EFFECTS BY SUBGROUP

Notes: Open circles show control group means; filled circles show adjusted treatment group means (control mean + regression-adjusted treatment effect); whiskers show 95% school-clustered CIs. Δ annotations show the ITT estimate from regression with controls and strata FE. *Panel A*: Girls start slightly higher and gain more (+0.46 vs +0.33 SD). *Panel B*: Control scores rise with grade (older students know more) while treatment effects decline—consistent with the congestion mechanism (Figure 9). *Panel C*: Treatment effects (line with 95% CI band) decline across baseline ability quintiles (within-school × grade rank), with the largest gains for the lowest-baseline students. *Panel D*: Rural and urban schools show similar effects. Panels A, C, D include grade FE; Panel B estimates are within-grade.

60-school control extract, yielding only ∼29% coverage. Switching the grade 6 baseline source from 2022–23 to 2023–24 slightly improves total coverage (55% → 61%) but does not close the treatment–control gap, because both years draw from the same asymmetric extracts.

A linear probability model confirms that missingness is predicted entirely by the treatment × grades 6–7 interaction and school category, not by student characteristics (Panel B). Restricting to grades 8–9—where coverage is balanced—the treatment effect among missing-baseline students is 0.46 SD; in grades 6–7, where coverage is asymmetric or structural, it is inflated to 0.56 SD (Table E4). The inflated pooled estimate reflects compositional imbalance in the missing-baseline subgroup, not a real effect on students without baseline.

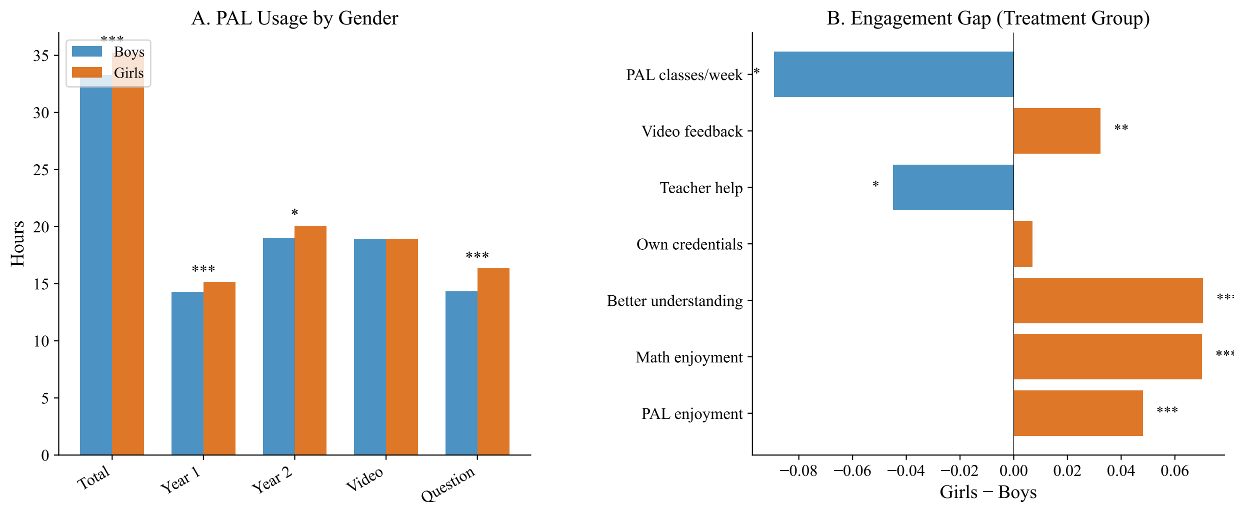

FIGURE E2. GENDER DIFFERENCES IN PAL USAGE AND ENGAGEMENT

Notes: Panel A: Mean PAL usage hours by gender (treatment group only). Girls accumulate more total hours than boys, driven primarily by more time on questions (active practice) rather than videos. *Panel B*: Gender gap in engagement measures (girls − boys, treatment group). Girls report higher PAL enjoyment, math enjoyment, and perceived understanding. Significance stars from regressions with strata FE and school-clustered SE.

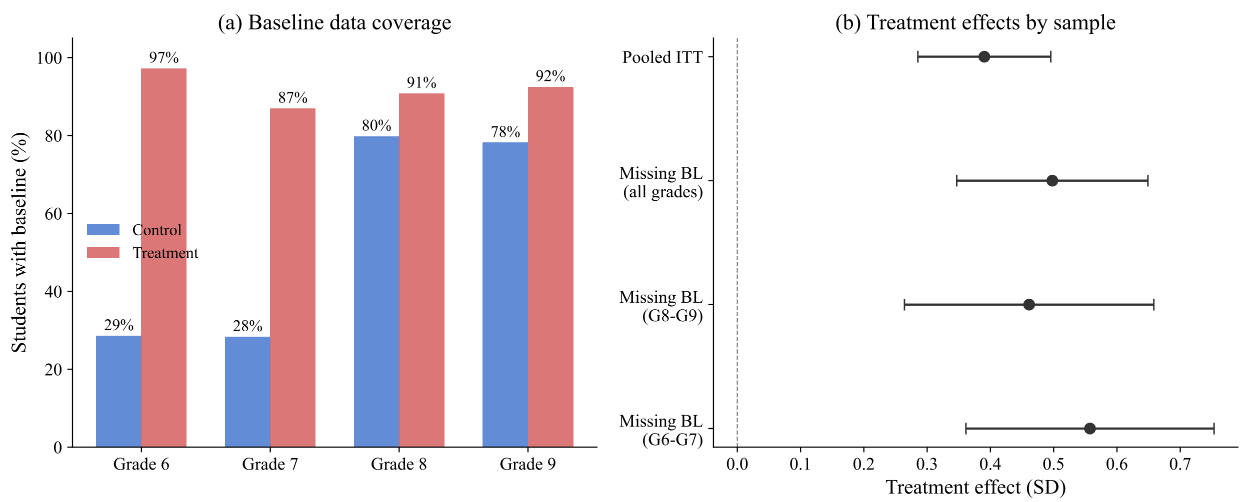

FIGURE E3. BASELINE DATA COVERAGE AND TREATMENT EFFECT DECOMPOSITION

Notes: Panel (a) shows the percentage of students with a valid pre-PAL government baseline math score (AY 2022–23 SA1 for grades 7–9 at endline; AY 2023–24 SA1 for grade 6) by treatment status and endline grade. The grade 7 imbalance is a data-extraction artifact (treatment and control data delivered in separate extracts without overlap); the grade 6 imbalance is structural (new-cohort students' pre-study schools are not in either extract). Panel (b) shows treatment effects for the pooled sample and for missing-baseline students overall, in grades 8–9 (balanced coverage), and in grades 6–7 (asymmetric coverage). Whiskers show 95% confidence intervals.

F. Dose-Response: IV Diagnostics, Shape Robustness, and Subgroup IV

This appendix supports the IV identification behind Table 5 and Figure 10: the first-stage and reduced-form scatters, the shape robustness of the dose-response curve under alternative fixed effects, and the subgroup IV replications by grade and gender.

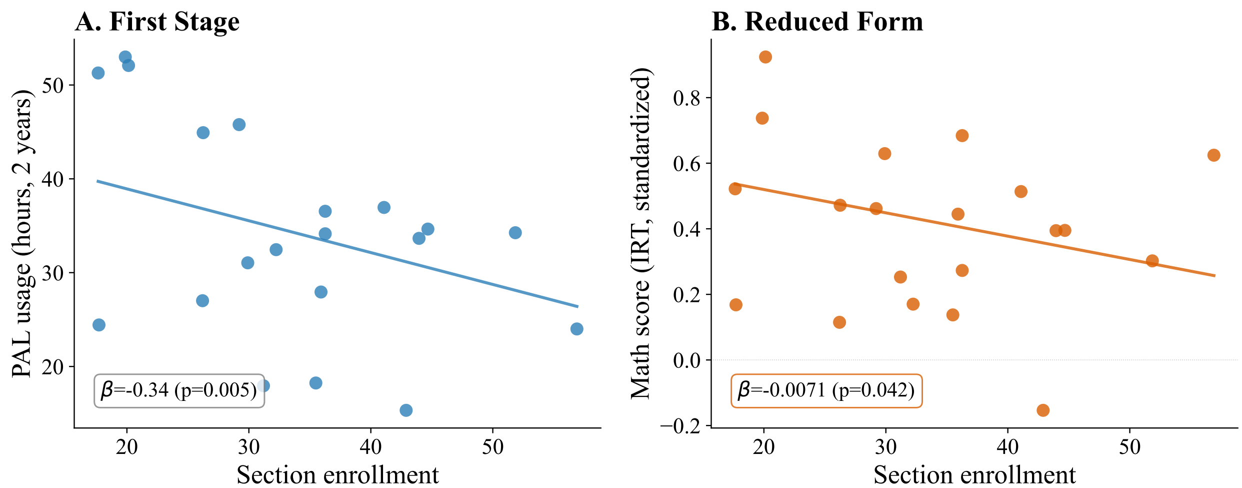

FIGURE F1. IV DIAGNOSTICS: FIRST STAGE AND REDUCED FORM

Notes: Treatment students only. *Panel A*: bin means of PAL usage hours by section enrollment, with a fitted regression line conditional on controls, grade FE, and strata FE (school-clustered SE). Each additional student in a section reduces PAL usage by 0.34 hours ($p = 0.005$). *Panel B*: bin means of endline math scores by section enrollment, same covariate structure. Each additional student reduces scores by 0.007 SD ($p = 0.042$). Bin means are formed within grade.

First Stage and Reduced Form

Figure F1 visualises the first-stage and reduced-form relationships that identify the IV in Table 5. Panel A shows that each additional student in a section reduces PAL usage by 0.34 hours over two years ($p = 0.005$). Panel B shows that each additional student reduces endline math scores by 0.007 SD ($p = 0.04$). The ratio of these slopes delivers the 2SLS estimate reported in Table 5.

Shape of the Dose-Response

Figure 10 in the main text characterizes the within-school shape of the dose-response. Table F1 supports that figure by reporting the pooled quadratic and linear-spline specifications under two alternative sets of fixed effects. The quadratic coefficient on hours^2 is negative across all six specifications; under 21 mandal-cluster strata FE the coefficient is significant at the 5% level ($p = 0.024$), and under 60 school FE (within-school identification only) precision tightens to $p = 0.002$. The linear spline's joint Wald test rejects linearity under school FE ($p = 0.012$). The shape finding—steep learning between 10 and 30 hours, flat above—is robust to including the one-year Grade 6 cohort and to retaining students with 100+ hours of usage.

Cross-Sample Per-Hour Returns

Figure 13 in §VII overlays within-school dose-response binscatter and quadratic fits across three governance modes (RCT-T 3-year cumul, KGBV $\sim$14-month cumul, program 3-year cumul). Table F2 reports the underlying linear and quadratic per-hour coefficients with school-clustered standard errors. The cross-sample slope ordering 0.025 (program) > 0.016 (KGBV) > 0.011 (RCT-T) tracks each sample's mean position on the dose-response curve — programs are concentrated near the steep activation region, KGBV in the steepening mid-range, and RCT-T in the flattening high-dose region. The RCT-T 3-year linear slope of $+0.0105$ matches the IV slope of $+0.011$ identified on the RCT 2-year sample (Figure 10). All three samples show concave shape (negative hours-squared, $p \leq 0.02$).

Subgroup Replications

Table F3 replicates the IV analysis from Table 5 separately by grade and by gender. Per-hour estimates are broadly consistent across subgroups (0.008–0.018 SD per hour), though statistical power is limited within any single subgroup. Girls accumulate approximately 2 more hours of PAL usage than boys over two years; per-hour returns are similar.

TABLE E1—GENDER DIAGNOSIS: WHY DO GIRLS GAIN MORE FROM PAL?

	Boys	Girls	Difference	SE
Panel A: Ruling out alternative explanations				
Baseline ability (observed, within school×grade pctile)				
Control group	0.454	0.532		
Treatment group	0.456	0.508		
Gender TE gap	T×F interaction			
Without baseline control	$+0.170^{***}$			(0.049)
With baseline control	$+0.162^{***}$			(0.047)
Attendance (days present, last 2 weeks)				
Control group	10.2	10.4		
Treatment group	10.5	10.6		
Treatment × Female			-0.180^{**}	(0.080)
School composition (T×F×Moderator)				
Enrollment (G6–9)	$+0.079^{*}$			(0.046)
Urban	+0.015			(0.136)
% female in school	+0.009			(0.063)
Tablets/student	+0.043			(0.047)
	Boys	Girls	Difference	SE
Panel B: Usage and engagement (treatment group)				
PAL usage (hours)				
Total hours (2 years)	33.3	35.2	$+2.0^{***}$	(0.7)
Year 1 hours	14.3	15.1	$+1.2^{***}$	(0.5)
Year 2 hours	19.0	20.1	$+0.7^{*}$	(0.4)
Video hours	18.9	18.9	-0.1	(0.4)
Question hours	14.3	16.3	$+2.0^{***}$	(0.4)
Engagement (higher = better)				
PAL enjoyment	3.74	3.79	$+0.05^{***}$	(0.02)
Math enjoyment	3.65	3.71	$+0.07^{***}$	(0.02)
Better understanding	3.66	3.72	$+0.07^{***}$	(0.02)
Own credentials	3.63	3.63	+0.01	(0.02)
Teacher help	3.04	2.97	-0.04^{*}	(0.03)
Video feedback	2.85	2.88	$+0.03^{**}$	(0.01)
PAL classes/week	2.59	2.51	-0.09^{*}	(0.05)
Challenges (total)	7.32	7.91	$+0.58^{***}$	(0.09)
	Boys TE	Girls TE	Interaction	SE
Panel C: Topic-specific treatment effects (standardized)				
Algebra	0.291	0.425	$+0.160^{***}$	(0.047)
Number systems	0.309	0.434	$+0.159^{***}$	(0.047)
Geometry	0.314	0.385	$+0.101^{**}$	(0.046)
Measurement	0.341	0.438	$+0.125^{**}$	(0.049)
Statistics	0.263	0.375	$+0.145^{***}$	(0.050)

Notes: Exploratory analysis investigating mechanisms behind the gender gap in treatment effects (Table 4, Panel A). Panel A tests whether baseline ability, attendance, or school characteristics explain the gap. Panel B compares PAL usage and engagement by gender within the treatment group. Panel C tests whether the gender advantage is concentrated in specific topics. All regressions include strata FE; standard errors clustered at school level. No multiple testing correction. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE E2—HETEROGENEOUS TREATMENT EFFECTS BY SES TERCILE

	Low SES (1)	Middle SES (2)	High SES (3)
Treatment	0.350***	0.422***	0.381***
	(0.059)	(0.066)	(0.060)
Observations	4,746	5,182	4,287
p (equality)		0.619	

Notes: Each column estimates the ITT within a SES tercile. SES index is PCA PC1 of 9 asset indicators. All regressions include controls, strata FE, grade FE, and school-clustered SE. p (equality) tests that treatment effects are equal across terciles. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Baseline coverage by treatment status and grade

TABLE E3—BASELINE DATA AVAILABILITY: DIFFERENTIAL COVERAGE BY TREATMENT STATUS

	Control		Treatment	
	Has baseline	%	Has baseline	%
Grade 6	506/1,773	28.5%	1,497/1,540	97.2%
Grade 7	497/1,755	28.3%	1,390/1,599	86.9%
Grade 8	1,479/1,854	79.8%	1,606/1,769	90.8%
Grade 9	1,634/2,091	78.1%	1,696/1,834	92.5%

Panel B: Linear probability model—predictors of missing baseline

	Coefficient	(SE)
Treatment	-0.1243***	(0.0264)
Grades 6–7	0.5462***	(0.0428)
Treatment × Grades 6–7	-0.5054***	(0.0470)
Has primary section (Cat. 6)	-0.1290***	(0.0243)
Female	-0.0095	(0.0100)
SES index	0.0049	(0.0046)
Age	0.0204***	(0.0034)
Observations	14,215	
R^2	0.376	
Strata FE	Yes	

Notes: Panel A shows the number and percentage of students with 2022–23 government baseline math scores (SA1) by treatment status and endline grade. The differential coverage for grades 6–7 reflects a data extraction artifact: the government provided broader baseline data for treatment-eligible schools than for control schools (see text). Panel B reports coefficients from a linear probability model with missing baseline as the dependent variable. Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE E4—TREATMENT EFFECTS BY BASELINE DATA AVAILABILITY AND GRADE

	(1) Full sample	(2) Has baseline	(3) G8–G9 only	(4) G8–G9 has baseline	(5) Missing BL G8–G9	(6) Missing BL G6–G7
Treatment	0.390***	0.361***	0.336***	0.313***	0.461***	0.557***
	(0.054)	(0.059)	(0.059)	(0.062)	(0.101)	(0.100)
Control mean	0.000	0.126	0.222	0.242	0.146	-0.254
N	14,215	10,305	7,548	6,415	1,133	2,777

Notes: Each column reports the ITT treatment effect from OLS of standardized math scores on treatment assignment with controls (SES, parental education, age, gender, missing flags), strata fixed effects, and grade fixed effects appropriate to the sample. Columns (1)–(2) include all grades with full and baseline-restricted samples. Columns (3)–(4) restrict to grades 8–9, where baseline coverage is balanced across arms (∼80–93% in both treatment and control). Columns (5)–(6) isolate students missing baseline data: column (5) in grades 8–9 (balanced missingness), column (6) in grades 6–7 (where control students are disproportionately missing due to the data extraction artifact). Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE F1—SHAPE OF THE POOLED DOSE-RESPONSE

	Strata FE			School FE		
	Linear (1)	Quadratic (2)	Spline (3)	Linear (4)	Quadratic (5)	Spline (6)
Usage hours	0.0091***	0.0128***	-0.0023	0.0068***	0.0101***	-0.0070
	(0.0018)	(0.0029)	(0.0078)	(0.0012)	(0.0018)	(0.0076)
Usage hours^2 ($\times 10^3$)		-0.036**			-0.033***	
		(0.016)			(0.010)	
Slope change at 10 hrs			+0.0235**			+0.0263**
			(0.0115)			(0.0118)
Slope change at 20 hrs			-0.0061			-0.0063
			(0.0111)			(0.0109)
Slope change at 30 hrs			-0.0082			-0.0083
			(0.0074)			(0.0058)
Spline joint Wald *p*			0.086			0.012
Strata FE (21)	Y	Y	Y			
School FE (60)				Y	Y	Y
Controls + grade FE	Y	Y	Y	Y	Y	Y
Observations	6,742	6,742	6,742	6,742	6,742	6,742

Notes: Treatment-group students only. Dependent variable is endline IRT math score standardised to the control mean and SD. Columns (1)–(3) use the 21 mandal-cluster strata FE; columns (4)–(6) replace strata FE with 60 school FE, absorbing between-school composition. The quadratic coefficient on hours^2 is negative across all six specifications, indicating concavity in the pooled within-treatment dose-response. Precision tightens substantially under school FE. Spline knots at 10, 20, and 30 hours; rows report the change in slope at each knot. Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE F2—CROSS-SAMPLE WITHIN-SCHOOL DOSE-RESPONSE ON CUMULATIVE PAL HOURS

	RCT-T (3-year)		KGBV (∼14-month)		Program (3-year)	
	Linear (1)	Quadratic (2)	Linear (3)	Quadratic (4)	Linear (5)	Quadratic (6)
PAL hours	+0.0105***	+0.0155***	+0.0155***	+0.0270***	+0.0249***	+0.0405***
	(0.0012)	(0.0020)	(0.0032)	(0.0055)	(0.0030)	(0.0058)
PAL hours^2		-0.00005***		-0.00033**		-0.00055***
		(0.00001)		(0.00014)		(0.00016)
Students	2,967	2,967	5,406	5,406	5,957	5,957
Schools	40	40	40	40	40	40
Mean hours	24.3	24.3	13.6	13.6	6.6	6.6
Median hours	16.5	16.5	12.8	12.8	3.2	3.2
School FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Within-school OLS of Y3 endline IRT θ (linked to the Y2 RCT-control scale via Mean–Sigma on 45 anchor items) on cumulative PAL math hours. Each sample's exposure window matches its actual PAL exposure: RCT-T is 3-year cumul (AY 2023–24 + 2024–25 + 2025–26); KGBV is cumul since the residential rollout's January 2025 launch (∼14 months through endline); program is 3-year cumul (AY 2023–26), with Y1 student-level hours recovered via SHA-256 hash linkage to the cleartext Y3 student_code. Per-hour slopes are descriptive within-school and not IV-identified; the RCT-T 3yr linear slope of $+0.0105$ is close to the fixed-intercept IV slope of $+0.011$ reported in Figure 10, identified causally on the RCT 2-year sample. Standard errors (in parentheses) clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE F3—IV ESTIMATES BY GRADE AND GENDER

	By Grade					By Gender	
	Grade 6	Grade 7	Grade 8	Grade 9	Pooled	Boys	Girls
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: First Stage (Dep. var.: PAL usage hours)							
Treatment × Section enrollment	-0.359***	-0.518**	-0.581***	-0.405***	-0.370***	-0.411***	-0.315***
	(0.106)	(0.247)	(0.158)	(0.135)	(0.110)	(0.133)	(0.101)
First-stage *F*	11.5	4.4	13.5	9.0	11.2	9.5	9.7
Panel B: Reduced Form (Dep. var.: Math score)							
Treatment × Section enrollment	-0.008	-0.006	-0.011**	-0.009**	-0.010***	-0.012***	-0.008*
	(0.006)	(0.007)	(0.006)	(0.005)	(0.004)	(0.004)	(0.004)
Panel C: IV Estimates							
IV (2SLS) estimate	0.0180	0.0125	0.0084	0.0088	0.0110	0.0096	0.0124
OLS estimate	0.016***	0.010***	0.006***	0.011***	0.009***	0.009***	0.009***
	(0.006)	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)
Panel D: Exclusion Restriction (Control group only)							
Section enrollment	-0.000	0.006**	-0.002	-0.000	0.001	0.005*	-0.003
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grade FE	—	—	—	—	Yes	Yes	Yes
Mean usage, T group (hrs)					34.4	33.6	35.4
Observations	3,282	3,328	3,607	3,906	14,123	7,232	6,842

Notes: Each column replicates the IV analysis from Table 5 within a single grade (columns 1–4), pooled (column 5), or by gender (columns 6–7). The instrument is treatment × section enrollment: treatment schools received 30 tablets regardless of enrollment, so larger sections share more students per tablet. Panel A: first stage (section enrollment reduces PAL usage hours). Panel B: reduced form (section enrollment reduces math scores in treatment schools). Panel C: 2SLS and OLS estimates of the return per PAL hour. The 2SLS estimate is stable across grades (0.008–0.018 SD/hour) and similar for boys and girls, indicating the grade gradient in treatment effects (Table 3) reflects differences in usage quantity, not in the per-hour productivity of PAL. The gender gap likewise reflects quantity: girls use PAL ≈2 hours more than boys, but the per-hour return is similar. Panel D: exclusion restriction — section enrollment should not affect control student scores. This holds in all grades except grade 7 ($p = 0.01$). Grade 7 also has a weak first stage ($F = 4.4$); the IV estimate for grade 7 should be interpreted with caution. Grade 6 students had 1 year of PAL; grades 7–9 had 2 years. All regressions include controls and strata FE; pooled and gender columns also include grade FE. Standard errors clustered at the school level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE G1—ENDLINE ATTRITION ANALYSIS AND LEE BOUNDS

	Pooled	Grade 6	Grade 7	Grade 8	Grade 9
Panel A: Attrition Rates					
Treatment	0.237	0.221	0.210	0.234	0.275
Control	0.198	0.186	0.201	0.185	0.216
Difference	0.033**	0.035	0.009	0.049	0.059
	(0.013)				
N (enrolled)	18,039	4,117	4,186	4,567	5,169
Panel B: Differential Attrition Predictors					
	Covariate effect	Treatment × Covariate	p (interaction)	N	
Female	-0.0405***	-0.0279**	0.042	18,039	
Age	0.0233***	0.0120*	0.073	18,039	
Urban	-0.0081	0.0902**	0.015	18,039	
Baseline math (SA1)	-0.0030***	0.0008	0.186	10,638	
Panel C: Lee Bounds on Pooled ITT					
Original ITT			0.390*** (0.054)		
Lower bound			0.270*** (0.047)		
Upper bound			0.481*** (0.051)		
Trim proportion			0.049 (332 students)		

Notes: Panel A: attrition rates from the enrollment frame (18,039 students in 120 schools) to endline assessment completion. Difference from OLS of attrition indicator on treatment with strata FE and school-clustered SE. Panel B: each row regresses the attrition indicator on treatment, the baseline covariate, and their interaction (plus strata FE). The interaction tests whether attrition is differentially predicted by the covariate across treatment and control. Panel C: Lee (2009) bounds trim the treatment group outcome distribution to equalize attrition rates. The lower bound trims the top of the treatment distribution; the upper bound trims the bottom. Both bounds being positive rules out attrition bias as an explanation for the main treatment effect. All regressions include strata FE. Panel C includes demographic controls and grade FE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G. Robustness of the Main ITT

This appendix confirms that the 0.39 SD headline estimate survives differential attrition, alternative control sets, multiple-testing corrections, non-parametric inference, and the congestion IV's exclusion restriction.

Table G1 presents attrition analysis and Lee (2009) bounds. Differential attrition is modest (3.3 percentage points, treatment higher) but statistically significant. Lee bounds remain large and positive (0.27–0.48 SD), ruling out attrition bias as an explanation for the main treatment effect.

We assess the sensitivity of the main ITT estimate to alternative specifications and inference procedures. The treatment effect is stable across three control specifications (0.38–0.39 SD; Table G2). The gender interaction survives Bonferroni, Holm, and Benjamini-Hochberg corrections (Table G3). Randomization inference, which does not rely on asymptotic assumptions, confirms the main result (RI $p < 0.001$; Figure G1).

Figure G2 visualises the exclusion-restriction placebo for the congestion IV. We plot raw within-grade binned means of endline math scores against section enrollment, separately for treatment and control. Conditioning on grade, controls, and strata FE, section enrollment has a negative effect on T scores ($\beta = -0.007$, $p = 0.07$) but is flat for C ($\beta = +0.001$, $p = 0.75$). The flat control line is direct visual evidence that section size does not affect learning in the absence of PAL, supporting the exclusion restriction.

TABLE G2—SENSITIVITY OF ITT TO CONTROL SPECIFICATION

	No controls (1)	Demographics only (2)	Demographics + baseline (3)
Treatment	0.379***	0.390***	0.393***
	(0.054)	(0.054)	(0.055)
R^2	0.106	0.137	0.238
Observations	14,215	14,215	14,215

Notes: Dependent variable: IRT math score standardized to control. Column (1): treatment + grade FE + strata FE only. Column (2): adds demographic controls (SES, parental education, age, gender). Column (3): adds within-school×grade baseline ability percentile (28.8% missing, imputed at median with missing flag). The treatment effect is stable across specifications. All regressions include strata FE and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE G3—MULTIPLE HYPOTHESIS TESTING CORRECTIONS

Hypothesis	Unadjusted p	Bonferroni	Holm	BH (FDR)
Gender	0.0005***	0.0049	0.0049	0.0049
Grade 7 interaction	0.3680	1.0000	0.9208	0.3680
Grade 8 interaction	0.2733	1.0000	0.9208	0.3514
Grade 9 interaction	0.1236	1.0000	0.6181	0.2225
Baseline Q2 interaction	0.0064***	0.0580	0.0516	0.0290
Baseline Q3 interaction	0.2302	1.0000	0.9208	0.3453
Baseline Q4 interaction	0.0607*	0.5460	0.4247	0.1681
Baseline Q5 interaction	0.3494	1.0000	0.9208	0.3680
Urban	0.0747*	0.6723	0.4482	0.1681

Notes: Unadjusted p-values from treatment × subgroup interaction tests. Bonferroni and Holm control family-wise error rate. BH controls false discovery rate. All from regressions with controls, strata FE, grade FE, and school-clustered SE.

H. Government School Assessments: Validity and the Grade 10 Board

This appendix explains why govt FA/SA scores are useful only as within-school rank controls (not cross-school level measures), reports the negative Grade 10 board exam estimates on the externally graded cohort, and runs a displacement test that rules out PAL-driven exam-prep crowd-out as the mechanism.

Government-administered school assessments (FA and SA exams) are set and graded by individual schools, with teachers proctoring their own students. These assessments have well-documented limitations as measures of absolute learning: teacher-proctored school exams inflate scores by approximately 16–20 percentage points relative to independently monitored tests (Singh, 2024), and grading standards vary across schools, making scores non-comparable.

Figure H1 decomposes the rank and level structure of Year 2 SA2 scores against independently administered endline IRT scores, in the control group only. Two patterns emerge. First, within any given school, govt scores preserve student rank: the median within-school correlation is $+0.47$ and is positive in 98% of 60 control schools (Panel A, gray lines). Teachers rank their own students accurately. Second, across schools

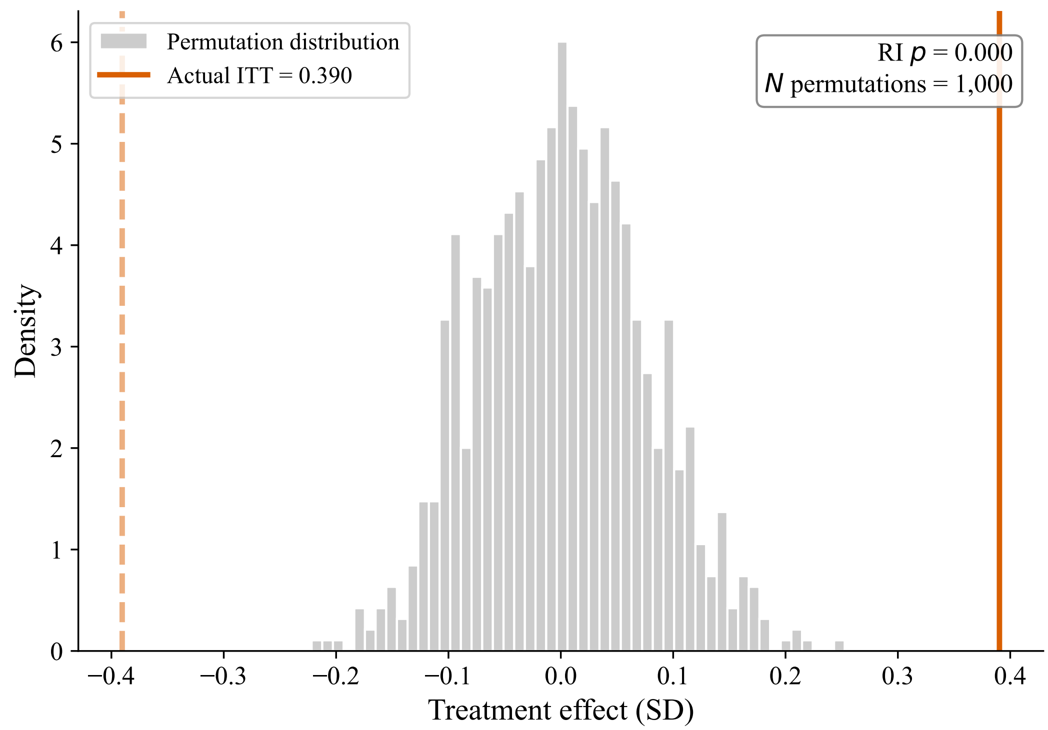

FIGURE G1. RANDOMIZATION INFERENCE: PERMUTATION DISTRIBUTION OF TREATMENT EFFECTS

Notes: Distribution of treatment effects from 1,000 random permutations of treatment assignment within strata (at the school level). The solid vertical line shows the actual ITT estimate. The RI p-value is the fraction of permuted effects at least as extreme (in absolute value) as the actual effect. This non-parametric test does not rely on asymptotic assumptions about the error distribution.

the signal disappears: the school-level correlation is -0.14 (Panel B), and the intra-class correlation of govt scores (0.21) is roughly three times that of the endline IRT (0.068). This excess between-school variance is visible in Panel A's marginal kdensities—school-mean govt scores (top, dashed) span nearly as widely as individual students, while school-mean endline scores (right, dashed) collapse to a tight band near zero. Most cross-school variation in govt scores reflects grading heterogeneity rather than ability. These patterns justify the use of govt scores only as within-school ranked baseline controls, and motivate the externally graded Grade 10 board exam as the cleaner cross-school measure.

Table H1 reports Grade 10 board exam effects for the cohort that had one year of PAL exposure in grade 9. Unlike the school-graded FA/SA assessments discussed in the preceding sub-appendix, the board exam is administered and graded externally, removing the grading heterogeneity that dominates cross-school variation in govt scores (Figure H1). Point estimates on the six subjects are all negative (-0.08 to -0.15 SD), but results are sensitive to the inclusion of strata fixed effects: no subject is significant at 5% without strata FE. The uniform direction across non-math subjects is puzzling given PAL is a math-only intervention; one possible explanation is differential exam preparation displacement during grade 9. We report these estimates because the external-grading structure makes them the cleanest cross-school government measure, while the specification sensitivity cautions against treating the negative direction as a settled finding.

To investigate whether the negative board exam effects reflect displacement of exam preparation by PAL usage, Table H2 decomposes the treatment effect by Year 1 PAL usage intensity. If PAL displaced exam preparation, the negative effect should be largest among high-usage students who spent the most time on tablets instead of studying. The evidence points in the opposite direction: the negative effect is concentrated entirely among low-usage students (-0.37 SD composite, $p < 0.001$), while high-usage students

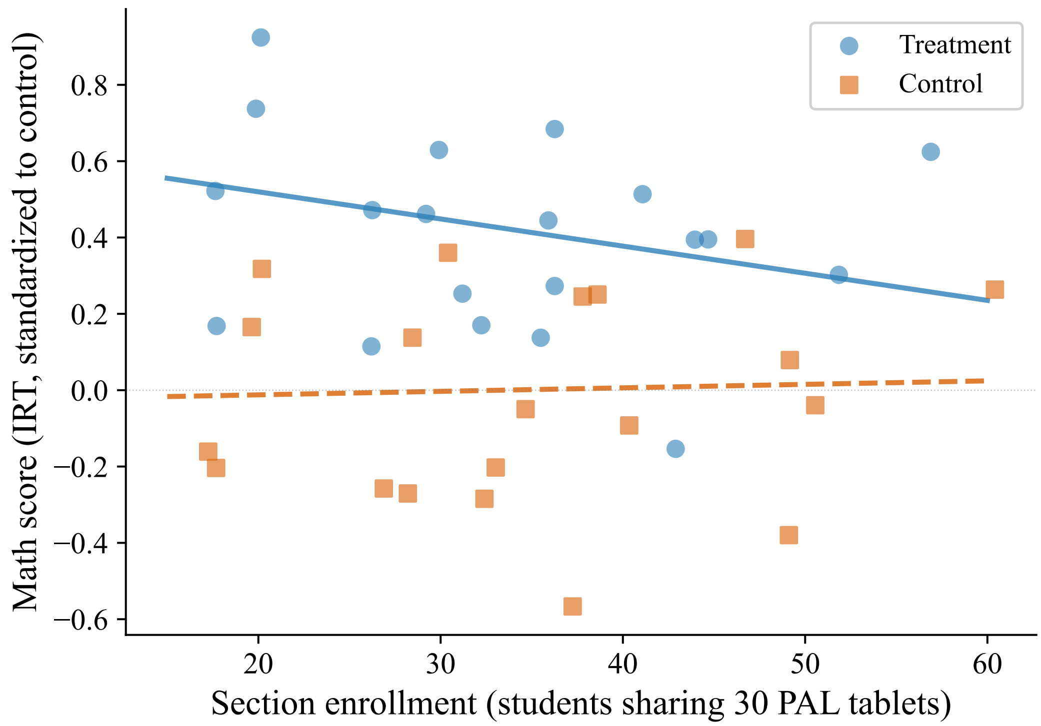

FIGURE G2. EXCLUSION-RESTRICTION PLACEBO: SECTION ENROLLMENT AND MATH SCORES, TREATMENT VS. CONTROL

Notes: Dots: raw within-grade bin means of endline IRT math scores (5 bins per grade, pooled across grades 6–9). Lines: regression-predicted slopes from full specifications with controls, strata FE, grade FE, and SE clustered at the school level. Treatment slope: $\beta = -0.007$, $p = 0.07$ (larger sections reduce T scores, consistent with congestion). Control slope: $\beta = +0.001$, $p = 0.75$ (flat, consistent with section size not affecting learning in the absence of PAL). The flat control line provides visual evidence for the exclusion-restriction placebo on the congestion IV (§E).

are statistically indistinguishable from control (+0.01 SD, $p = 0.93$). Within the treatment group, each additional hour of PAL predicts higher board scores across all six subjects (+0.015 SD/hour, $p < 0.001$), though this gradient likely reflects selection—higher-ability students engage more with the platform—rather than a causal benefit of usage. Splitting at median school enrollment, the negative effect appears in small schools (−0.14 SD) but not large schools (−0.00 SD), the opposite of the grade 6–9 endline pattern where small schools show the largest positive effects. Taken together, these results indicate that the negative board exam effect is not attributable to PAL usage displacing exam preparation time. The most parsimonious interpretation is that the effect reflects specification sensitivity—it is significant only with strata fixed effects and no student-level controls are available for this cohort—combined with possible school-level confounders that are absorbed differently across strata.

I. Supplementary Results

A. Additional Heterogeneity

Tables I2 and I3 examine treatment effect heterogeneity by school urbanicity and infrastructure resources, respectively; effects are similar across rural and urban schools (Table I2) and across schools with above- versus below-median infrastructure resources (Table I3). Treatment-effect heterogeneity by caste cannot be estimated because caste records in government enrollment data are available for treatment students only; Table I1 therefore reports treatment-group mean math scores by caste category for descriptive context.

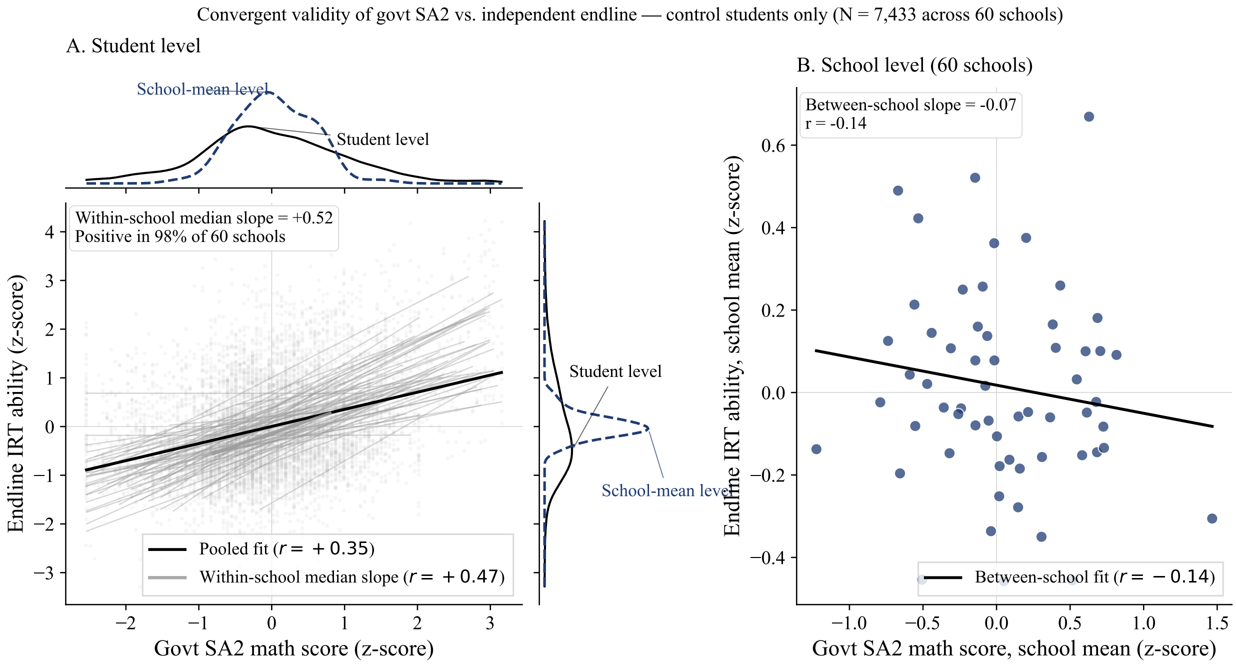

FIGURE H1. CONVERGENT VALIDITY OF GOVERNMENT SCHOOL ASSESSMENTS

Notes: Control students only ($N = 7{,}433$ across 60 schools). Outcome and regressor both z-scored to the control group's mean and SD. *Panel A*: student-level scatter of Year 2 govt SA2 math scores against independently administered endline IRT scores. Gray lines show OLS fits within each of the 60 control schools; the black line is the pooled student-level OLS fit. The gray fit at center of the panel marks the median within-school slope. Marginal kdensities on each axis compare the student-level distribution (solid black) with the school-mean distribution (dashed navy): schools differ moderately in mean govt score (top marginal) but barely in mean endline IRT (right marginal). *Panel B*: school-level means ($N = 60$) with OLS fit. Pooled student $r = +0.35$; within-school median $r = +0.47$ (positive in 98% of schools); between-school $r = -0.14$. The govt score's intra-class correlation (0.21) is $\approx 3\times$ that of the endline IRT (0.068); the excess between-school variance reflects grading heterogeneity rather than ability.

B. Gender Diagnosis

Figure I1 shows that girls gain more than boys in every grade, with the largest gap at grade 6. Figure I2 presents a comprehensive diagnostic of the gender gap in treatment effects, following a three-step narrative.

Step 1: Rule out. We test whether the gender gap can be explained by pre-existing differences. Baseline ability is balanced across gender within treatment status (Panel C). Attendance differences are small and work against girls, not in their favor (Panel B). School composition—enrollment, urbanicity, percent female, tablets per student—does not moderate the gender gap (Panel F). None of these channels account for the differential treatment effect.

Step 2: Identify. Girls spend approximately 2 more hours on PAL than boys, driven entirely by more time answering questions—active practice—rather than watching videos (Panel A). Girls also report higher enjoyment of both PAL and math more broadly (Panel D). The gender-stratified IV analysis (Table F3, columns 6–7) confirms that boys and girls have similar per-hour returns ($\approx$0.011 SD/hour), indicating the gap reflects differences in usage quantity rather than per-hour productivity.

Step 3: Assess breadth. The gender advantage is broad-based across all five math topics rather than concentrated in specific areas (Panel E), consistent with a general engagement mechanism rather than topic-specific content effects.

TABLE H1—GRADE 10 BOARD EXAM EFFECTS (1 YEAR PAL EXPOSURE)

	Math (1)	Science (2)	Social Sci. (3)	Lang 1 (4)	Lang 2 (5)	Lang 3 (6)
Panel A: With strata FE						
Treatment	-0.136**	-0.109*	-0.144**	-0.117*	-0.135*	-0.121*
	(0.067)	(0.065)	(0.067)	(0.070)	(0.071)	(0.070)
Panel B: Without strata FE						
Treatment	-0.138	-0.109	-0.152*	-0.139*	-0.121	-0.077
	(0.093)	(0.085)	(0.083)	(0.083)	(0.082)	(0.081)
Control mean (raw)	57.0	65.1	65.4	70.6	56.0	61.7
Observations	4,846	4,846	4,843	4,849	4,849	4,846

Notes: Grade 10 board exam scores for students who had PAL in grade 9 (AY 2023–24, 1 year exposure) and took the state board exam in 2024–25. Each column standardized to control mean and SD. Board exams are externally graded (state level). No student-level controls are available for this cohort (students not in endline sample). Standard errors clustered at the school level. Coverage: 98.4% treatment, 98.9% control. *Panel A* includes 21 strata fixed effects; *Panel B* shows results without strata FE. Point estimates are similar across panels (-0.08 to -0.15 SD), but standard errors are larger without strata FE (0.08–0.09 vs 0.07), and no subject is significant at 5% in Panel B. The negative coefficients across all six subjects suggest a possible effect, but interpretation requires caution: (i) this cohort had only 1 year of PAL exposure, (ii) the board exam was taken *after* students left treatment schools, (iii) the uniform direction across non-math subjects is puzzling given PAL is a math-only intervention, and (iv) results are sensitive to the inclusion of strata FE. One possible explanation is differential exam preparation: PAL may have displaced study time allocated to board exam review during grade 9. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

See Table E1 for the full regression results underlying each panel.

C. *Descriptive Usage Statistics*

Baseline ability and age are the strongest predictors of PAL usage, while SES is not ($p = 0.32$; Table I4). Algebra accounts for the largest share of usage across all grades, consistent with the assessment blueprint weighting (Figure I3).

D. *Compliance Descriptives*

Section C introduces compliance and reports the within-quintile dose-response gradient. This subsection adds one descriptive companion: the relationship between platform hours and compliance.

Hours on the platform and per-student compliance are correlated ($r = 0.57$, $N = 6{,}052$ treatment students with both measures) but not interchangeable (Figure I4). The decile-mean trace rises steeply over the first 30 hours of usage and then flattens, while individual-student variation in compliance at any given hours decile remains substantial throughout. The two metrics capture different margins of engagement: hours measure raw exposure during PAL periods, while compliance measures the fraction of the prescribed sequence the student walks. Two students with the same total hours can differ in compliance by 0.4 or more, depending on whether they complete most of what they open or click through and bail.

E. *Secondary Outcomes and Item-Level Texture*

PAL increases math enjoyment by 0.19 SD, with positive spillovers to other subjects (0.10–0.15 SD; Table I6, Panel A). Treatment students also report 0.22 fewer days absent over the last two weeks ($p = 0.003$;

TABLE H2—GRADE 10 BOARD EXAM: INVESTIGATING THE ROLE OF PAL USAGE

	Math	Science	Social Sci.	Lang 1	Lang 2	Lang 3	Composite
Panel A: Treatment effect by Year 1 PAL usage (vs. control)							
Low usage (4 hrs, N_T=819)	-0.381***	-0.335***	-0.360***	-0.373***	-0.391***	-0.327***	-0.369***
	(0.080)	(0.086)	(0.080)	(0.085)	(0.095)	(0.088)	(0.076)
Mid usage (14 hrs, N_T=818)	-0.077	-0.061	-0.092	-0.069	-0.052	-0.024	-0.061
	(0.080)	(0.070)	(0.073)	(0.087)	(0.084)	(0.082)	(0.068)
High usage (27 hrs, N_T=819)	0.021	0.026	-0.023	0.077	-0.004	-0.059	0.006
	(0.076)	(0.079)	(0.085)	(0.079)	(0.083)	(0.085)	(0.069)
Panel B: Within-treatment usage gradient (OLS)							
PAL hours (Year 1)	0.0139***	0.0145***	0.0156***	0.0180***	0.0190***	0.0098***	0.0154***
	(0.0039)	(0.0035)	(0.0037)	(0.0034)	(0.0036)	(0.0027)	(0.0030)
Panel C: Treatment effect by school size							
Small schools (N_T=1,209, N_C=1,206)	-0.152						-0.138
	(0.099)						(0.086)
Large schools (N_T=1,208, N_C=1,223)	0.018						-0.003
	(0.102)						(0.106)

Notes: This table investigates whether the negative Grade 10 board exam effects (Table H1) are driven by students who used PAL intensively. *Panel A*: treatment students are split into Year 1 usage terciles and each tercile is compared to the full control group (separate regressions). The negative effect is concentrated among low-usage students (-0.37 SD composite, $p < 0.001$); high-usage students are indistinguishable from control ($+0.01$ SD, $p = 0.93$). *Panel B*: within the treatment group, each additional hour of PAL predicts *higher* board scores ($+0.015$ SD/hour, $p < 0.001$), across all six subjects including non-math. This gradient likely reflects selection (higher-ability students engage more) rather than a causal benefit of usage. *Panel C*: splitting at median enrollment, the negative effect is concentrated in small schools, the opposite of the endline pattern (Figure 9). Taken together, these results indicate that the negative board exam effect is not attributable to PAL usage displacing exam preparation. All regressions include strata FE and school-clustered SE. No student-level controls are available for this cohort. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE I1—MATH SCORES BY CASTE CATEGORY (TREATMENT GROUP ONLY)

	SC	ST	BC-A/Other	BC-B/OBC
Mean IRT score (SD)	0.332	0.769	-0.047	0.430
	(1.066)	(1.211)	(1.059)	(1.150)
N	1,928	808	251	3,669
F-test (equality)	$F = 43.91, p = 0.000$			

Notes: Treatment group only. Caste information from government enrollment records, available for treatment students only. Scores are IRT math scores standardized to the overall control group mean and SD. Standard deviations in parentheses. SC = Scheduled Caste, ST = Scheduled Tribe, BC = Backward Class. F-test for equality of means across categories.

Panel B), though strong ceiling effects in enjoyment and the self-reported nature of attendance warrant caution in interpretation.

Treatment effects are remarkably uniform across all five math topics (0.31–0.38 SD; Table I7; Figure I5), indicating that PAL improves foundational math skills broadly rather than producing topic-specific gains. This uniformity is consistent with PAL's curriculum-aligned design, which covers all topics proportionally (Figure I3).

We examine treatment effects at the individual item level. Treatment effects are slightly larger on harder items (slope $= 0.10$; Figure I6), suggesting PAL helps students reach questions they would otherwise get

TABLE I2—HETEROGENEOUS TREATMENT EFFECTS BY URBAN/RURAL STATUS

	Rural (1)	Urban (2)
Treatment	0.456***	0.520***
	(0.057)	(0.151)
Observations	11,615	2,600
p (Rural = Urban)	0.075	

Notes: Each column estimates the ITT within rural or urban schools. p (Rural = Urban) tests equality of treatment effects from a pooled interaction model. All regressions include controls, strata FE, grade FE, and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE I3—TREATMENT EFFECT MODERATION BY SCHOOL RESOURCES

Moderator	Treatment × Moderator	SE	p
Enrollment (G6–9)	-0.134	(0.084)	0.109
Math teachers	-0.126	(0.078)	0.107
Total teachers	-0.143*	(0.077)	0.063
Classrooms	-0.043	(0.060)	0.474
Functional tablets	-0.066	(0.076)	0.380
Observations	14,215		

Notes: Each row shows the interaction of treatment with a standardized school-level moderator (mean 0, SD 1) from a separate regression. All regressions include treatment, the moderator, controls, strata FE, grade FE, and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

wrong rather than reinforcing already-known material. Treatment students outperform control students at every content grade level, with the largest gains on grade 4–6 content (7–10 pp; Figure I7). Both groups show the expected decline in accuracy as content difficulty increases.

F. Rank Invariance and Conditional Quantile Effects

Treatment students are more reshuffled across the ability distribution than control students (Figure I8), violating rank invariance and motivating the conditional quantile analysis. Conditioning on baseline quintile, treatment effects increase from the 10th to 75th percentile of the endline distribution within every quintile, then flatten at the 90th (Figure I9)—indicating that PAL creates within-group variance in addition to the across-quintile gradient documented in Table 4.

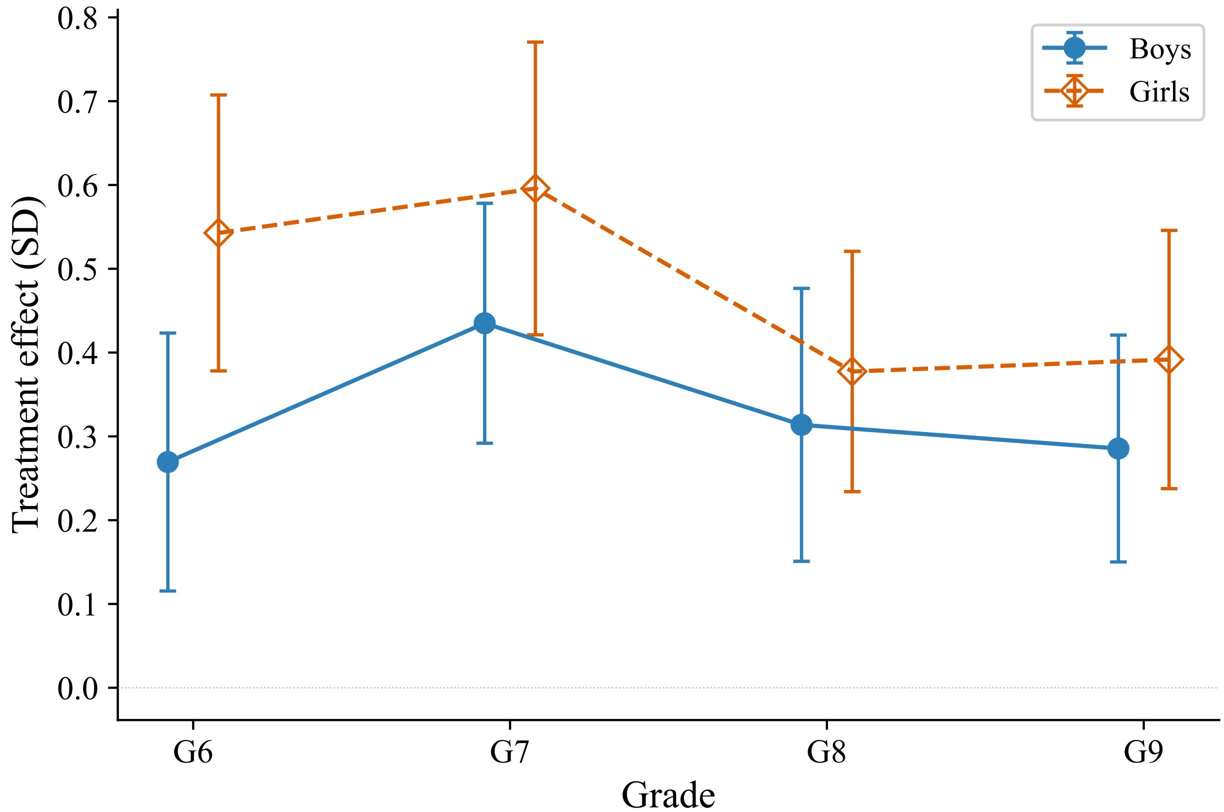

FIGURE I1. TREATMENT EFFECTS BY GENDER AND GRADE

Notes: ITT estimates of PAL treatment effects by gender and grade. Each point shows the treatment effect for a gender × grade subgroup, with 95% confidence intervals. Girls gain more than boys in every grade, with the largest gap at grade 6 (0.55 vs 0.27 SD). All regressions include demographic controls, strata FE, and school-clustered SE.

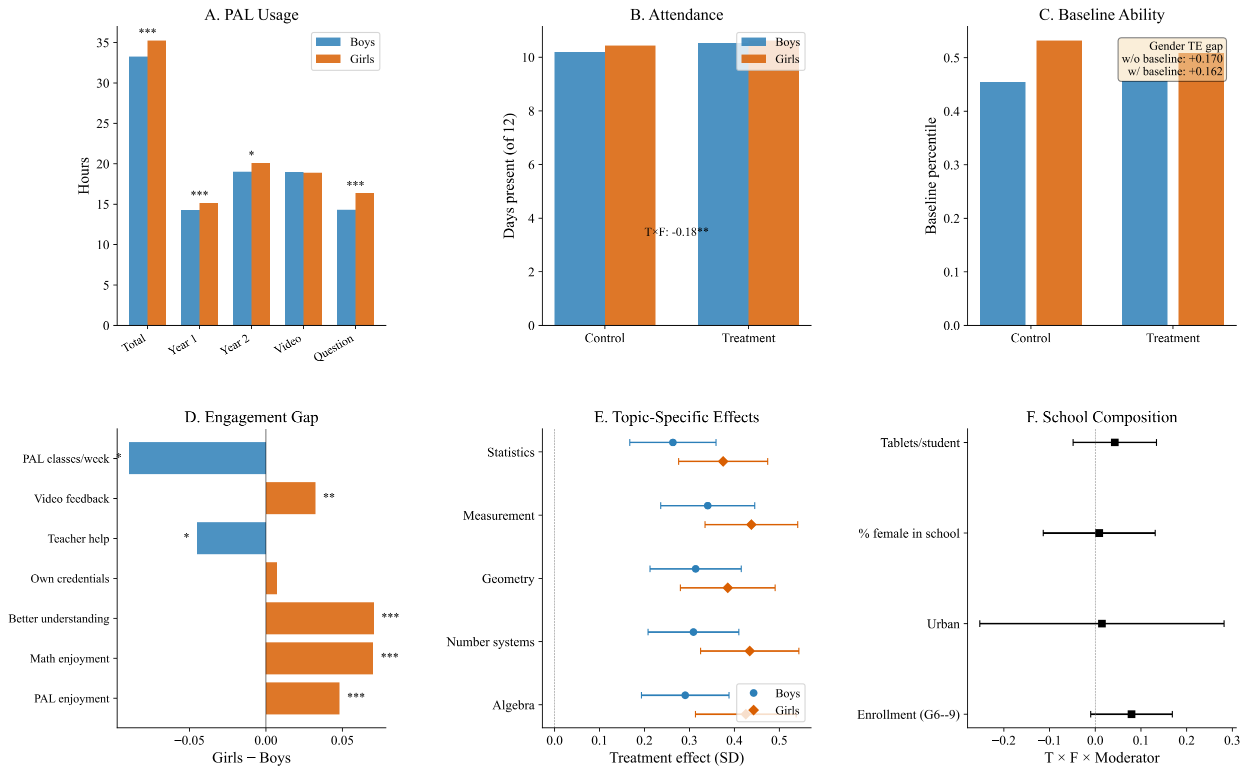

FIGURE I2. GENDER DIAGNOSIS: FULL MECHANISMS PANEL

Notes: Panel A: PAL usage by gender (treatment group). *Panel B*: Attendance by gender and treatment status. *Panel C*: Baseline ability by gender and treatment status. *Panel D*: Engagement gap (girls − boys, treatment group). *Panel E*: Topic-specific treatment effects by gender with 95% CIs. *Panel F*: School composition moderators (T×F×Moderator). All regressions include strata FE and school-clustered SE.

TABLE I4—PREDICTORS OF PAL USAGE (TREATMENT GROUP ONLY)

	Bivariate (1)	Joint (2)
Female	1.95***	1.27*
	(0.74)	(0.76)
Age	2.81***	3.03***
	(0.43)	(0.43)
SES index	-0.28	-0.30
	(0.28)	(0.28)
Baseline ability	1.70	1.35
	(1.42)	(1.39)
Enrollment (G6–9)	-0.06***	-0.07***
	(0.01)	(0.01)
Functional tablets	-0.08***	0.00
	(0.02)	(0.02)
R^2		0.273
Observations		6,742

Notes: Dependent variable: total PAL usage hours (2 years). Treatment group only. Column (1): bivariate regressions (one per row). Column (2): joint regression with all predictors. All regressions include strata FE and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

TABLE I5—PAL DOSAGE: TARGET VS. ACTUAL USAGE

	Year 1 (AY 2023–24)	Year 2 (AY 2024–25)	Total (2-year students)
School calendar			
Calendar weeks	31	33	64
Non-instructional weeks	9	12	21
PAL-eligible weeks	22	21	43
Target usage (hours)			
Scheduled (80 min/week)	29.3	28.0	57.3
Effective (60 min/week)	22.0	21.0	43.0
Actual usage (hours, treatment group mean)			
All students	14.7	19.5	34.2
2-year students (G7–9)	19.0	19.3	38.2
% of effective target			
2-year students (G7–9)	86%	92%	89%

Notes: Each school is allocated one 80-minute PAL period per week. The effective target of 60 minutes accounts for student transit to the computer lab and device setup time. Non-instructional weeks include holidays, formative assessment (FA) weeks, summative assessment (SA) weeks, and pre-exam revision weeks, during which PAL sessions are suspended. Year 1 ran late September 2023–April 2024; lab setup was staggered, with 49 schools starting in September and 11 in October. Year 2 ran July 2024–February 2025 (all 60 schools; capped at endline assessment start). Two-year students are grades 7–9 at endline, who had PAL in both academic years.

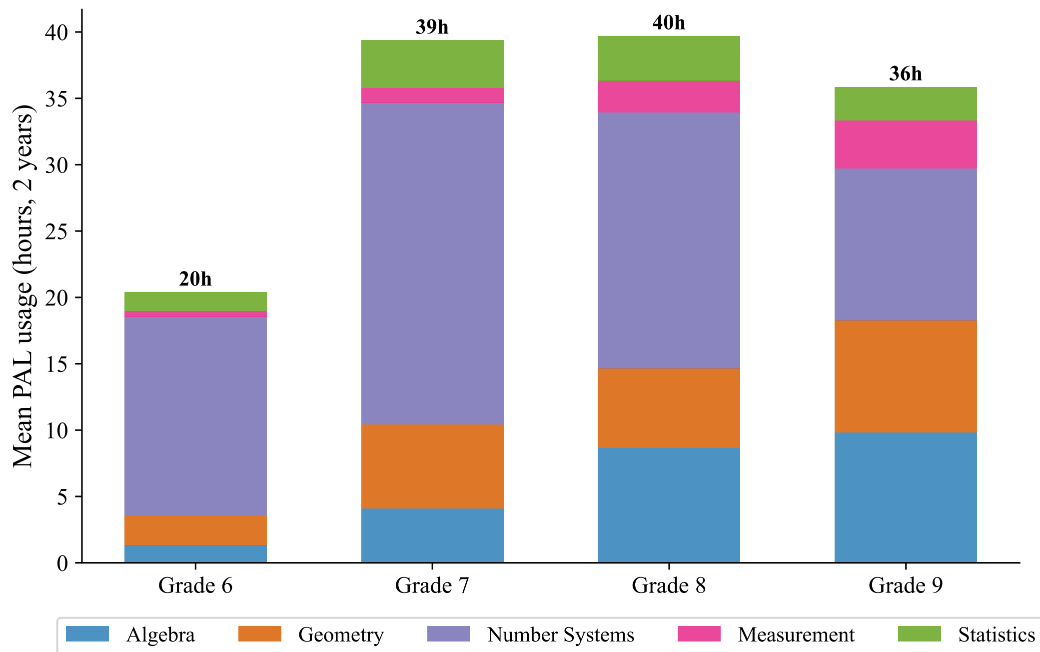

FIGURE I3. PAL USAGE BY MATH TOPIC AND GRADE

Notes: Mean PAL usage hours (cumulative over 2 academic years) by math topic and student grade. Treatment group only. Grade 6 had 1 year of PAL exposure; grades 7–9 had 2 years. Algebra accounts for the largest share of usage across all grades, consistent with the assessment blueprint weighting.

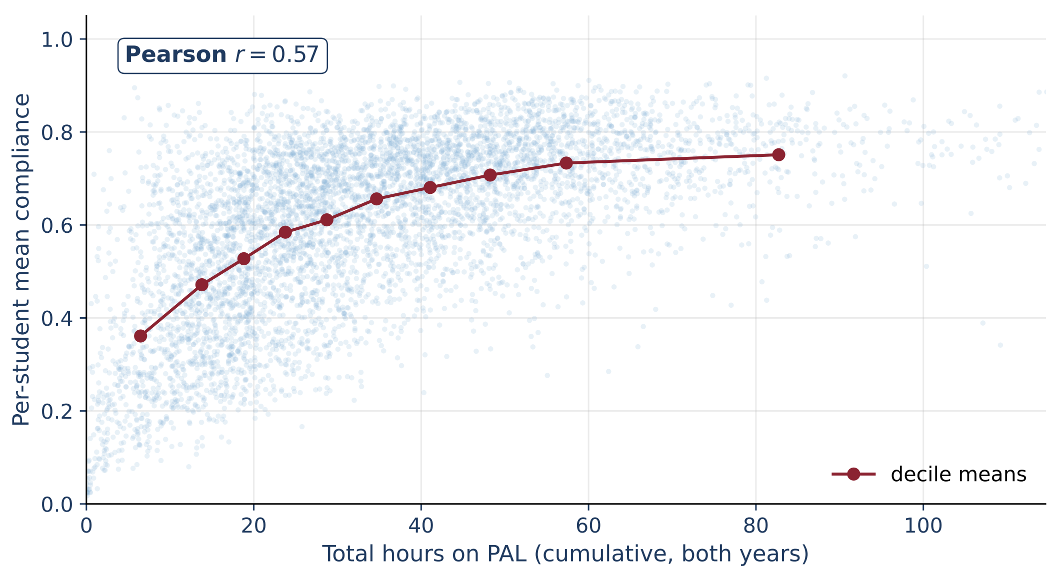

FIGURE I4. TOTAL PAL HOURS AND PER-STUDENT COMPLIANCE

Notes: Per-student total PAL usage (sum of 2023–24 and 2024–25 minutes, converted to hours) on the x-axis against per-student mean compliance on the y-axis, treatment students only ($N = 6{,}052$). Light points are individual students; dark red trace shows decile means of compliance plotted against decile means of hours. Pearson $r = 0.57$. Hours and compliance are correlated but not interchangeable: at any given hours decile, students vary substantially in the fraction of prescribed activities they complete, and the compliance–hours gradient flattens above ~ 30 hours, consistent with engagement having a separate margin from exposure.

TABLE I6—TREATMENT EFFECTS ON SECONDARY OUTCOMES

	Treatment effect	SE	Control mean
Panel A: Subject enjoyment (standardized)			
Math	0.190***	(0.025)	1.44
English	0.131***	(0.026)	1.40
Telugu	0.105***	(0.032)	1.34
Hindi	0.095	(0.065)	1.63
Science	0.144***	(0.029)	1.45
Social science	0.140***	(0.035)	1.53
Panel B: Attendance			
Days present (SD)	0.104***	(0.035)	10.31
Days absent	-0.22***	(0.07)	1.7
Observations		14,215	

Notes: Panel A: enjoyment rating for each subject, reverse-coded and standardized to control mean and SD (original scale: 1=A lot, 2=Somewhat, 3=Not at all, 4=Did not have; positive coefficients = treatment students enjoy the subject more). Strong ceiling effects—nearly all students report high enjoyment regardless of treatment. Panel B: days present out of 12 school days in last 2 weeks, self-reported by students during endline assessment ("In the last 2 weeks, how many days were you absent from school, excluding Sundays and holidays?"). Covers a 2-week window, not the full academic year. Days absent = 12 − days present (raw count). All regressions include controls, strata FE, grade FE, and school-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

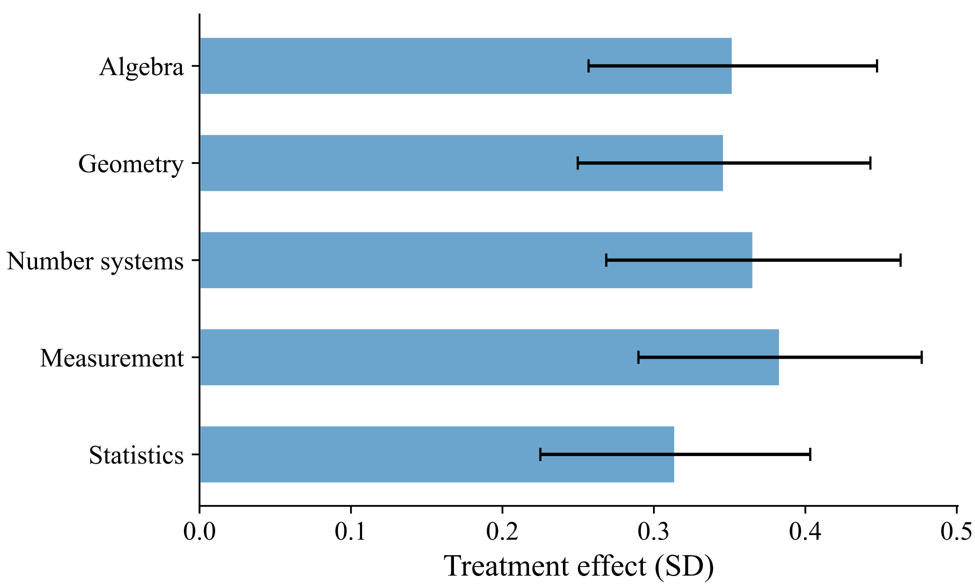

FIGURE I5. TREATMENT EFFECTS BY MATH TOPIC

Notes: Pooled ITT estimates for each of 5 math topics. Bars show point estimates; whiskers show 95% confidence intervals. Effects are remarkably uniform across topics (0.31–0.38 SD), indicating PAL improves foundational math skills broadly rather than topic-specifically. All regressions include controls, strata FE, grade FE, and school-clustered SE.

TABLE I7—TREATMENT EFFECTS BY MATH TOPIC

	Pooled (1)	Grade 6 (2)	Grade 7 (3)	Grade 8 (4)	Grade 9 (5)
Algebra	0.352***	0.345***	0.458***	0.327***	0.294***
	(0.049)	(0.067)	(0.070)	(0.059)	(0.054)
Geometry	0.346***	0.353***	0.441***	0.317***	0.286***
	(0.049)	(0.068)	(0.067)	(0.063)	(0.056)
Number systems	0.366***	0.397***	0.471***	0.320***	0.294***
	(0.050)	(0.069)	(0.065)	(0.061)	(0.058)
Measurement	0.383***	0.393***	0.493***	0.325***	0.322***
	(0.048)	(0.075)	(0.063)	(0.061)	(0.052)
Statistics	0.314***	0.417***	0.451***	0.245***	0.188***
	(0.045)	(0.070)	(0.066)	(0.056)	(0.053)
Controls	Yes	Yes	Yes	Yes	Yes
Strata FE	Yes	Yes	Yes	Yes	Yes
Grade FE	Yes				
Observations	14,215	3,313	3,354	3,623	3,925

Notes: Each row shows the ITT estimate for a different math topic. Topic scores are constructed from IRT subscores on items mapped to each topic, standardized to control mean and SD. Topics are not equally weighted in the overall score—algebra has 2–3× more items than statistics. All regressions include controls, strata FE (and grade FE for pooled). School-clustered SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

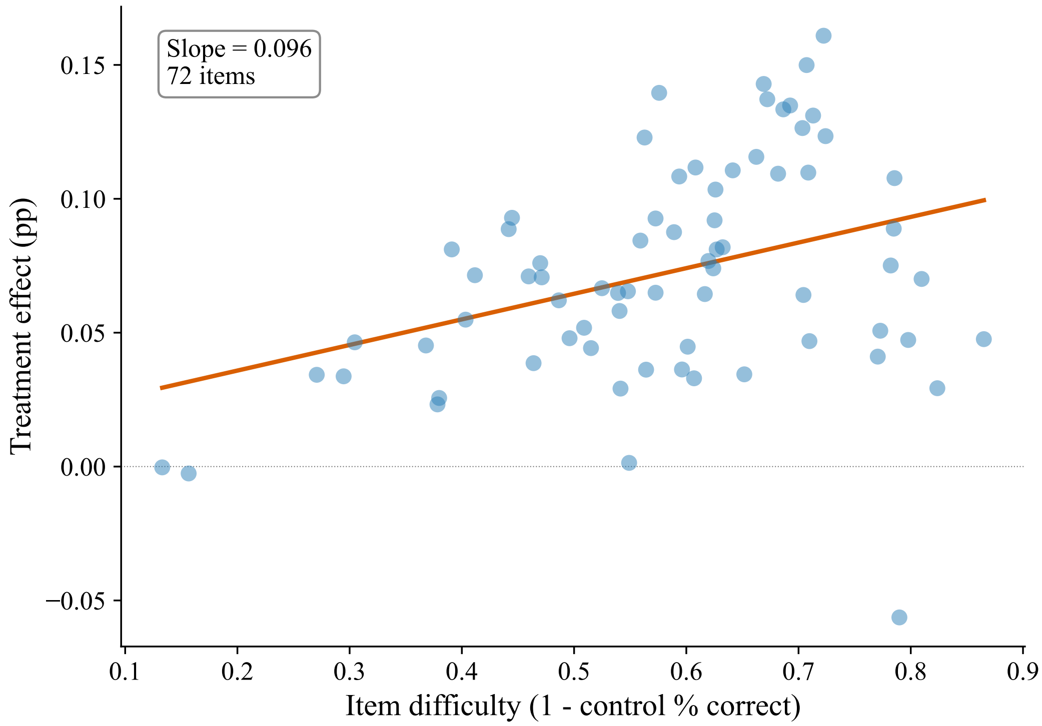

Figure I6. Item Difficulty and Treatment Effects

Notes: Each dot represents one of 72 endline assessment items. The *x*-axis shows item difficulty (1 minus control group percent correct); the *y*-axis shows the treatment effect on that item (percentage point difference, T−C). The fitted line shows the relationship between difficulty and treatment effect magnitude. All regressions include strata FE and school-clustered SE.

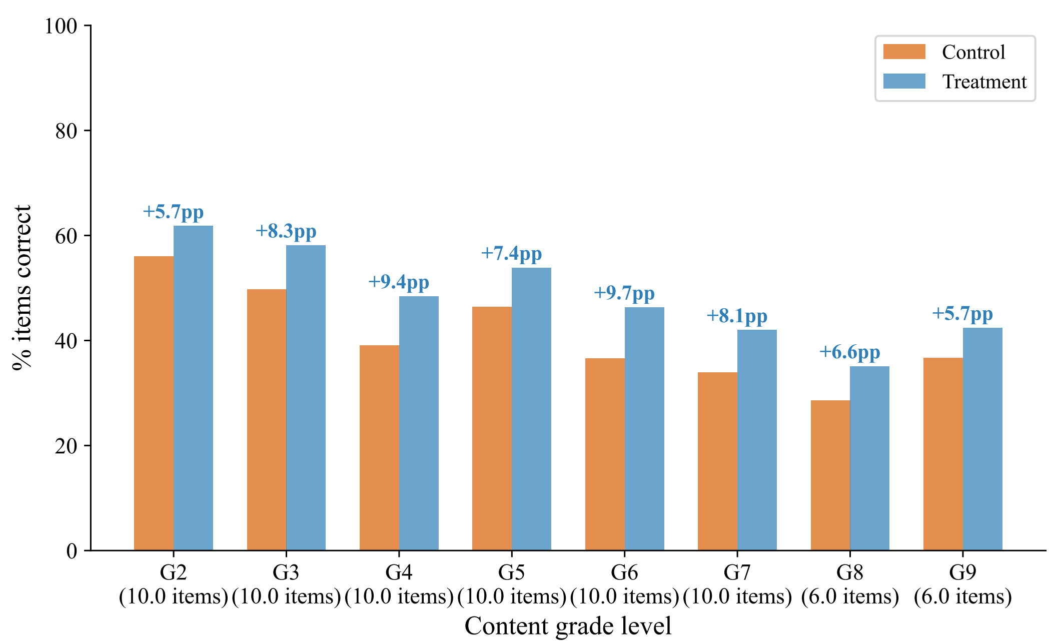

Figure I7. Content Mastery by Grade Level: Treatment vs. Control

Notes: Percent of items answered correctly by content grade level (grade 2 = easiest, grade 9 = hardest). Items are organized by the grade-level curriculum they test. Treatment students outperform control students at every content grade level, with the largest gaps on grade 4–6 content (7–10 pp). Both groups show the expected decline in accuracy as content difficulty increases.

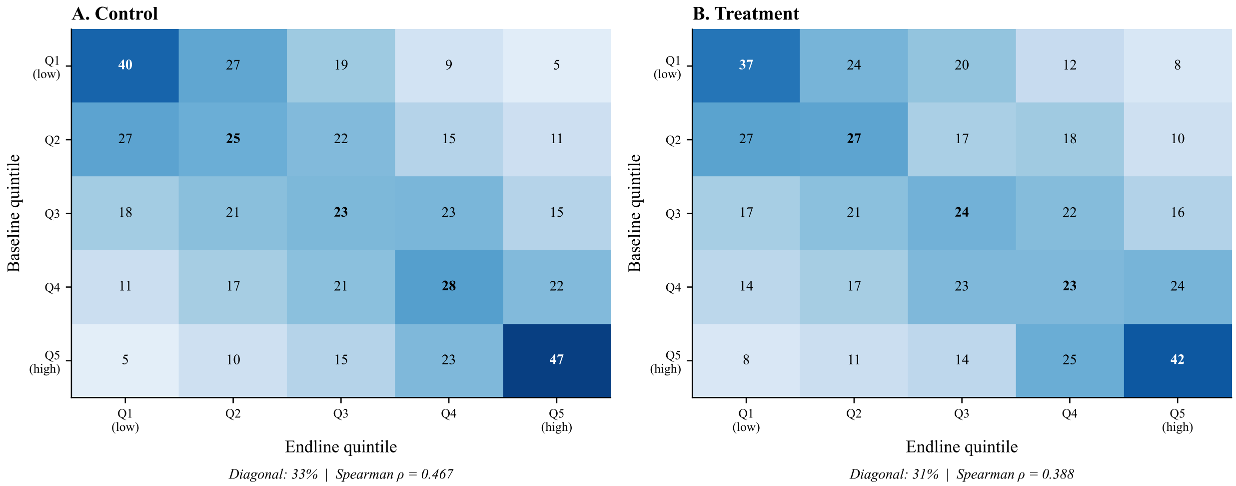

FIGURE I8. RANK INVARIANCE CHECK: BASELINE TO ENDLINE TRANSITION MATRICES

Notes: Each cell shows the percentage of students in a baseline quintile (row) who end up in a given endline quintile (column). Bold diagonal entries show the percentage staying in the same quintile. If rank invariance held perfectly, all mass would be on the diagonal. Treatment students are more reshuffled (diagonal average 31%) than control (33%), with a lower Spearman rank correlation (0.36 vs 0.47). This violation of rank invariance motivates the conditional quantile analysis in Figure I9. Sample restricted to students with non-missing baseline scores (72% of sample).

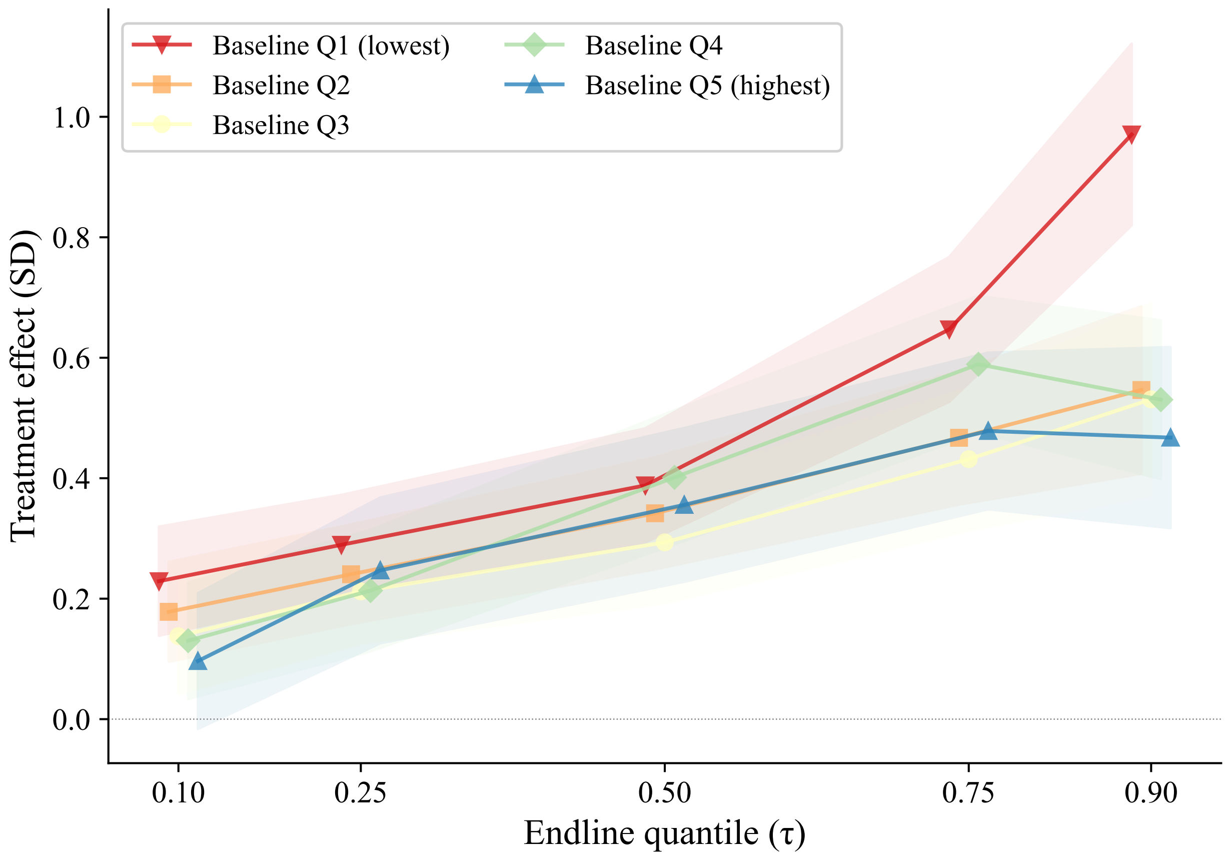

FIGURE I9. CONDITIONAL QUANTILE TREATMENT EFFECTS BY BASELINE ABILITY

Notes: Each line shows quantile treatment effects within a baseline ability quintile, addressing the rank-invariance violation in unconditional quantile regressions (Figure I8). Within every baseline quintile, effects increase from the 10th to the 75th percentile of the endline distribution, then flatten or decline at the 90th. Lines for lower-baseline quintiles sit above those for higher-baseline quintiles, consistent with the across-quintile gradient in Table 4. All regressions include controls, strata FE, and grade FE. Shaded bands show 95% confidence intervals.